

2026 Summer Chemicals, Agriculture & Packaging Outlook

Specialty, Commodity, and Agricultural Chemicals & Packaging

Jeffrey J. Zekauskas ^{AC}

1-(212)-622-6644

jeffrey.zekauskas@jpmorgan.com

Bloomberg Finance L.P.

J.P. Morgan Securities LLC

Silke Kueck

1-(212)-622-6503

Silke.x.kueck@jpmorgan.com

J.P. Morgan Securities LLC

Katie Zhang

1-(212)-622-3262

Katie.zhang@jpmorgan.com

J.P. Morgan Securities LLC

Lydia Huang

1-(212)-622-0086

Lydia.huang@jpmorgan.com

J.P. Morgan Securities LLC

See the end pages of this presentation for analyst certification and important disclosures, including non-US analyst disclosures.

J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Table of Contents

Pages 3-13: Equity Performances & Valuation

Pages 14-17: Macroeconomic trends

Pages 18-50: Commodity Pricing, Margin, Supply/Demand, Expansions, Exports

Pages 51-57: Housing & Construction Data

Pages 58-70: Global Automotive Sales, Production, Mileage, Collision & Vehicle Repair

Pages 71-74: Beverage Can Supply/Demand, Market Share

Pages 75-91: Agricultural Commodities / Grains

Pages 92-113: Fertilizers

Stock Performance, 2026 YTD

	Market Value (\$b)	Rating	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Name	7/10/2026		Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)	Incr/(Decr)
Compass Minerals	1.25	N	7.1%	8.5%	(13.3%)	4.1%	(7.8%)	(42.3%)	46.2%	1.2%	(17.2%)	(19.7%)	(38.2%)	(55.6%)	74.6%	51.9%
CF Industries Holdings	17.96	N	14.7%	16.9%	(25.1%)	(22.9%)	35.1%	2.3%	9.7%	(18.9%)	82.8%	20.4%	(6.7%)	7.3%	(9.4%)	51.2%
Chemours	2.68	N	NA	NA	NA	312.1%	126.6%	(43.6%)	(35.9%)	37.0%	35.4%	(8.8%)	3.0%	(46.4%)	(30.2%)	51.1%
Tronox	0.98	N	26.4%	3.5%	(83.6%)	163.7%	98.9%	(62.1%)	46.8%	28.0%	64.4%	(42.9%)	3.3%	(28.9%)	(58.6%)	47.2%
Bayer AG	49.61	OW	41.8%	10.8%	2.5%	(14.4%)	4.9%	(40.8%)	20.2%	(33.9%)	(2.4%)	4.5%	(30.4%)	(42.6%)	91.6%	36.4%
Valvoline	4.93	N	NA	NA	NA	NA	16.6%	(22.8%)	10.6%	8.1%	61.1%	(12.4%)	15.1%	(3.7%)	(19.7%)	33.0%
Cabot Corp.	4.51	UW	29.2%	(14.7%)	(6.8%)	23.6%	21.9%	(30.3%)	10.7%	(5.6%)	25.2%	18.9%	24.9%	9.4%	(27.4%)	31.7%
LyondellBasell	18.19	N	40.6%	(1.1%)	9.5%	(1.3%)	28.6%	(24.6%)	13.6%	(3.0%)	0.6%	(10.0%)	14.5%	(21.9%)	(41.7%)	30.1%
Corteva	57.52	N	NA	NA	NA	NA	NA	NA	NA	31.0%	22.1%	24.3%	(18.5%)	18.9%	17.7%	27.8%
Linde	245.08	OW	18.8%	(0.4%)	(21.0%)	14.4%	32.0%	0.9%	36.4%	23.8%	31.5%	(5.8%)	25.9%	1.9%	1.8%	24.3%
Dow	20.92	OW	NA	NA	NA	NA	NA	NA	NA	1.4%	2.2%	(11.2%)	8.8%	(26.8%)	(41.7%)	24.2%
Methanex	3.76	N	85.9%	(22.6%)	(28.0%)	32.7%	38.2%	(20.5%)	(19.5%)	19.3%	(14.2%)	(4.3%)	25.1%	5.4%	(20.5%)	22.5%
Air Products & Chemicals	66.70	OW	33.0%	29.0%	(9.8%)	19.6%	14.1%	(2.5%)	46.8%	16.3%	11.4%	1.3%	(11.2%)	5.9%	(14.8%)	21.3%
Ball Corp	16.42	OW	16.9%	34.2%	5.5%	2.4%	0.8%	18.4%	43.8%	44.1%	2.4%	(45.9%)	12.5%	(4.2%)	(3.9%)	16.5%
Intl. Flavors & Fragrances	19.79	OW	29.2%	17.9%	18.0%	(1.5%)	29.5%	(12.0%)	(3.9%)	(15.6%)	38.4%	(30.4%)	(22.8%)	4.4%	(20.3%)	15.0%
PPG Industries	26.03	OW	40.1%	21.9%	(14.5%)	(4.1%)	23.3%	(12.5%)	30.6%	8.0%	19.6%	(27.1%)	18.9%	(20.1%)	(14.2%)	14.0%
Ashland	3.06	OW	20.7%	23.4%	(14.2%)	6.4%	33.2%	(0.3%)	7.8%	3.5%	35.9%	(0.1%)	(21.6%)	(15.2%)	(17.9%)	13.8%
Scotts Miracle-Gro	3.81	N	41.2%	0.2%	3.5%	48.1%	12.0%	(42.6%)	72.8%	93.1%	(19.2%)	(69.8%)	31.2%	4.1%	(12.0%)	12.2%
DuPont	18.40	OW	37.3%	2.7%	12.9%	11.1%	24.5%	(24.9%)	(15.6%)	10.8%	13.6%	(15.0%)	12.1%	(0.9%)	26.1%	11.7%
Silgan	4.75	OW	15.6%	11.6%	0.2%	(4.7%)	14.8%	(19.6%)	31.6%	19.3%	15.8%	21.0%	(12.7%)	15.0%	(22.4%)	11.4%
Huntsman Corp.	1.94	N	54.7%	(7.4%)	(50.1%)	67.8%	74.5%	(42.1%)	25.2%	4.1%	38.7%	(21.2%)	(8.6%)	(28.3%)	(44.5%)	11.3%
Celanese	5.15	OW	24.2%	8.4%	12.3%	16.9%	36.0%	(16.0%)	36.8%	5.5%	29.3%	(39.2%)	52.0%	(55.5%)	(38.9%)	11.0%
Crown	12.46	OW	21.1%	14.2%	(0.4%)	3.7%	7.0%	(26.1%)	74.5%	38.1%	10.4%	(25.7%)	12.0%	(10.2%)	24.5%	8.3%
Orion	0.32	UW	NA	NA	(25.8%)	49.6%	35.8%	(1.3%)	(23.7%)	(11.2%)	7.1%	(3.0%)	55.7%	(43.1%)	(66.6%)	7.8%
Nutrien	31.36	OW	NA	NA	NA	NA	NA	NA	1.9%	0.5%	56.1%	(2.9%)	(22.9%)	(20.6%)	37.9%	6.0%
Eastman Chemical	7.73	OW	21.0%	(4.2%)	(11.5%)	10.9%	23.2%	(22.0%)	8.4%	26.5%	20.6%	(32.6%)	10.3%	1.7%	(30.1%)	5.9%
Ecolab	77.20	OW	45.0%	0.2%	9.4%	2.5%	14.5%	9.8%	31.0%	12.1%	8.4%	(38.0%)	36.3%	18.1%	12.0%	4.5%
Amcort	19.96	OW	NA	NA	NA	NA	NA	NA	(0.8%)	8.6%	2.0%	(0.8%)	(19.1%)	(2.4%)	(11.4%)	3.5%
Sherwin-Williams	82.37	OW	19.3%	43.3%	(1.3%)	3.5%	52.6%	(4.0%)	48.3%	25.9%	43.8%	(32.6%)	31.4%	9.0%	(4.7%)	3.1%
Westlake	9.67	N	53.9%	0.1%	(11.1%)	3.1%	90.3%	(37.9%)	6.0%	16.3%	19.0%	5.6%	36.5%	(18.1%)	(35.5%)	2.1%
RPM International	13.41	OW	41.4%	22.2%	(13.1%)	22.2%	(2.6%)	12.1%	30.6%	18.3%	11.3%	(3.5%)	14.6%	10.2%	(15.5%)	1.1%
Axalta	6.96	NR	NA	NA	2.4%	2.1%	19.0%	(27.6%)	29.8%	(6.1%)	16.0%	(23.1%)	33.4%	0.7%	(5.6%)	0.6%
Olin	2.36	N	33.6%	(21.1%)	(24.2%)	48.4%	38.9%	(43.5%)	(14.2%)	42.4%	134.2%	(8.0%)	1.9%	(37.3%)	(38.4%)	(0.7%)
H.B. Fuller	3.01	OW	49.5%	(14.4%)	(18.1%)	32.5%	11.5%	(20.8%)	20.9%	0.6%	56.1%	(11.6%)	13.7%	(17.1%)	(11.9%)	(6.0%)
Mosaic Company	6.84	UW	(16.5%)	(3.4%)	(39.6%)	6.3%	(12.5%)	13.8%	(25.9%)	6.3%	70.8%	11.7%	(18.6%)	(31.2%)	(2.0%)	(10.7%)
Albemarle Corp.	14.87	N	2.0%	(5.1%)	(6.9%)	53.7%	48.6%	(39.7%)	(5.2%)	102.0%	58.5%	(7.2%)	(33.4%)	(40.4%)	64.3%	(10.9%)
Avery Dennison	12.29	OW	43.7%	3.4%	20.8%	12.1%	63.6%	(21.8%)	45.6%	18.6%	39.6%	(16.4%)	11.7%	(7.4%)	(2.8%)	(11.6%)
FMC	1.36	N	26.6%	(22.8%)	(30.3%)	48.9%	66.0%	(23.5%)	52.5%	14.2%	(3.3%)	13.3%	(49.5%)	(22.9%)	(71.5%)	(21.3%)
S&P 500			29.6%	11.4%	(0.7%)	9.5%	19.4%	(6.2%)	28.9%	16.3%	26.9%	(19.4%)	24.2%	23.3%	16.4%	10.7%

NOTE: Companies at or above the shaded line represent companies that outperformed the S&P 500. *Bayer is co-covered with Richard Vossler. DuPont is covered by Chigusa Katoku.

Source: Bloomberg Finance L.P. & J.P. Morgan Estimates.

We have suspended our rating for Axalta (AXTA) due to J.P. Morgan acting as financial advisor to Axalta Coating Systems Ltd. regarding a merger with Akzo Nobel N.V.

J.P.Morgan

Quarter-to-Date Performance

Ticker	6/30/26	7/10/26	% Change
CF	108.26	116.92	8.0%
LYB	52.65	56.35	7.0%
DOW	27.36	29.03	6.1%
MEOH	46.15	48.66	5.4%
HUN	10.62	11.13	4.8%
BAYN GR	€ 48.21	€ 50.50	4.8%
OLN	19.82	20.68	4.3%
NTR	62.95	65.43	3.9%
WLK	73.00	75.49	3.4%
APD	293.18	299.53	2.2%
LIN	518.94	529.79	2.1%
CE	46.00	46.92	2.0%
MOS	21.19	21.51	1.5%
ASH	65.89	66.75	1.3%
CTVA	84.69	85.68	1.2%
EMN	66.98	67.57	0.9%
CCK	111.82	111.47	(0.3%)
AMCR	43.35	43.18	(0.4%)
DD	135.64	134.68	(0.7%)
AVY	162.35	160.71	(1.0%)
BALL	62.40	61.69	(1.1%)
ECL	278.61	274.31	(1.5%)
IFF	79.22	77.53	(2.1%)
VVV	39.54	38.64	(2.3%)
TROX	6.30	6.14	(2.5%)
SHW	344.32	333.99	(3.0%)
SLGN	46.39	44.98	(3.0%)
PPG	121.29	116.76	(3.7%)
CBT	90.82	87.32	(3.9%)
SMG	68.11	65.48	(3.9%)
FUL	58.29	55.91	(4.1%)
CMP	31.13	29.84	(4.1%)
AXTA	34.22	32.51	(5.0%)
FMC	11.50	10.91	(5.1%)
RPM	111.15	105.10	(5.4%)
ALB	135.03	126.05	(6.7%)
CC	20.52	17.81	(13.2%)
OEC	6.63	5.69	(14.2%)
RIY	4093.86	4125	0.8%
RDG	2362.63	2299	(2.7%)
XLB	50.83	50.89	0.1%
SPX	7499.36	7575	1.0%

Source: Bloomberg Finance L.P.

NOTE: Companies at and above the shaded line represent companies that outperformed the S&P 500

Quarterly Stock Performance

Ticker	3/31/26	6/30/26	% Change	Ticker	12/31/25	3/31/26	% Change	Ticker	9/30/25	12/31/25	% Change	Ticker	6/30/25	9/30/25	% Change
CMP	23.35	31.13	33.3%	TROX	4.17	9.77	134.3%	ALB	81.08	141.44	74.4%	CC	11.45	15.84	38.3%
AXTA	27.70	34.22	23.5%	CC	11.79	22.03	86.9%	BAYN GR	€ 28.18	€ 37.01	31.3%	ALB	62.67	81.08	29.4%
BAYN GR	€ 39.74	€ 48.21	21.3%	LYB	43.30	80.56	86.1%	DD	32.58	40.20	23.4%	OLN	20.09	24.99	24.4%
CBT	75.31	90.82	20.6%	DOW	23.38	41.65	78.1%	ASH	47.91	58.67	22.5%	MEOH	33.10	39.76	20.1%
SLGN	38.80	46.39	19.6%	CF	77.34	129.84	67.9%	AXTA	28.62	32.31	12.9%	BAYN GR	€25.55	€28.18	10.3%
ASH	55.61	65.89	18.5%	WLK	73.94	116.82	58.0%	AVY	162.17	181.88	12.2%	RPM	109.84	117.88	7.3%
VVV	33.68	39.54	17.4%	CE	42.28	65.77	55.6%	HUN	8.98	10.00	11.4%	ECL	269.44	273.86	1.6%
PPG	106.88	121.29	13.5%	MEOH	39.72	59.54	49.9%	IFF	61.54	67.39	9.5%	WLK	75.93	77.06	1.5%
SMG	60.81	68.11	12.0%	OLN	20.83	29.73	42.7%	CCK	96.59	102.97	6.6%	CBT	75.00	76.05	1.4%
RPM	99.40	111.15	11.8%	HUN	10.00	13.31	33.1%	NTR	58.71	61.72	5.1%	LIN	469.18	475.00	1.2%
CCK	100.25	111.82	11.5%	ALB	141.44	179.53	26.9%	BALL	50.42	52.97	5.1%	SHW	343.36	346.26	0.8%
IFF	72.55	79.22	9.2%	CTVA	67.03	83.71	24.9%	TROX	4.02	4.17	3.7%	NTR	58.24	58.71	0.8%
AMCR	39.75	43.35	9.1%	FMC	13.87	17.22	24.2%	SMG	56.95	58.35	2.5%	FUL	60.15	59.28	(1.4%)
SHW	320.55	344.32	7.4%	OEC	5.28	6.50	23.1%	CMP	19.20	19.64	2.3%	CF	92.00	89.70	(2.5%)
BALL	59.11	62.40	5.6%	NTR	61.72	75.46	22.3%	DOW	22.93	23.38	2.0%	APD	282.06	272.72	(3.3%)
ECL	266.02	278.61	4.7%	EMN	63.83	76.32	19.6%	AMCR	40.90	41.70	2.0%	AXTA	29.69	28.62	(3.6%)
LIN	495.76	518.94	4.7%	CMP	19.64	23.35	18.9%	EMN	63.05	63.83	1.2%	CMP	20.09	19.20	(4.4%)
OEC	6.50	6.63	2.0%	APD	247.02	290.49	17.6%	CE	42.08	42.28	0.5%	ASH	50.28	47.91	(4.7%)
CTVA	83.71	84.69	1.2%	LIN	426.39	495.76	16.3%	FUL	59.28	59.46	0.3%	MOS	36.48	34.68	(4.9%)
APD	290.49	293.18	0.9%	VVV	29.06	33.68	15.9%	MEOH	39.76	39.72	(0.1%)	VVV	37.87	35.91	(5.2%)
DD	137.40	135.64	(1.3%)	DD	40.20	45.80	13.9%	CTVA	67.63	67.03	(0.9%)	CCK	102.98	96.59	(6.2%)
FUL	61.68	58.29	(5.5%)	CBT	66.28	75.31	13.6%	PPG	105.11	102.46	(2.5%)	AVY	175.47	162.17	(7.6%)
AVY	172.68	162.35	(6.0%)	BALL	52.97	59.11	11.6%	WLK	77.06	73.94	(4.0%)	PPG	113.75	105.11	(7.6%)
CC	22.03	20.52	(6.9%)	IFF	67.39	72.55	7.7%	ECL	273.86	262.52	(4.1%)	CTVA	74.53	67.63	(9.3%)
EMN	76.32	66.98	(12.2%)	BAYN GR	€ 37.01	€ 39.74	7.4%	SLGN	43.01	40.37	(6.1%)	BALL	56.09	50.42	(10.1%)
NTR	75.46	62.95	(16.6%)	MOS	24.09	25.50	5.9%	SHW	346.26	324.03	(6.4%)	AMCR	9.19	8.18	(11.0%)
CF	129.84	108.26	(16.6%)	PPG	102.46	106.88	4.3%	APD	272.72	247.02	(9.4%)	DOW	26.48	22.93	(13.4%)
MOS	25.50	21.19	(16.9%)	SMG	58.35	60.81	4.2%	LIN	475.00	426.39	(10.2%)	SMG	65.96	56.95	(13.7%)
HUN	13.31	10.62	(20.2%)	FUL	59.46	61.68	3.7%	LYB	49.04	43.30	(11.7%)	HUN	10.42	8.98	(13.8%)
MEOH	59.54	46.15	(22.5%)	ECL	262.52	266.02	1.3%	RPM	117.88	104.00	(11.8%)	LYB	57.86	49.04	(15.2%)
ALB	179.53	135.03	(24.8%)	SHW	324.03	320.55	(1.1%)	CBT	76.05	66.28	(12.8%)	EMN	74.66	63.05	(15.6%)
CE	65.77	46.00	(30.1%)	CCK	102.97	100.25	(2.6%)	CF	89.70	77.34	(13.8%)	IFF	73.55	61.54	(16.3%)
FMC	17.22	11.50	(33.2%)	SLGN	40.37	38.80	(3.9%)	OLN	24.99	20.83	(16.6%)	FMC	41.75	33.63	(19.4%)
OLN	29.73	19.82	(33.3%)	RPM	104.00	99.40	(4.4%)	VVV	35.91	29.06	(19.1%)	SLGN	54.18	43.01	(20.6%)
DOW	41.65	27.36	(34.3%)	AMCR	41.70	39.75	(4.7%)	CC	15.84	11.79	(25.6%)	TROX	5.07	4.02	(20.7%)
LYB	80.56	52.65	(34.6%)	AVY	181.88	172.68	(5.1%)	OEC	7.58	5.28	(30.3%)	CE	55.33	42.08	(23.9%)
TROX	9.77	6.30	(35.5%)	ASH	58.67	55.61	(5.2%)	MOS	34.68	24.09	(30.5%)	OEC	10.49	7.58	(27.7%)
WLK	116.82	73.00	(37.5%)	AXTA	32.31	27.70	(14.3%)	FMC	33.63	13.87	(58.8%)				
RIY	3565.72	4094	14.8%	RIY	3733	3566	(4.5%)	RIY	3656	3733	2.1%	RIY	3395.61	3655.86	7.7%
RDG	2065.95	2363	14.4%	RDG	2210	2066	(6.5%)	RDG	2298	2210	(3.9%)	RDG	2239.39	2298.29	2.6%
XLB	49.97	50.83	1.7%	XLB	45.35	49.97	10.2%	XLB	44.81	45.35	1.2%	SPX	6204.95	6688.46	7.8%
SPX	6528.52	7499	14.9%	SPX	6846	6529	(4.6%)	SPX	6688	6846	2.3%				

Source: Bloomberg Finance L.P.

Pre-Conflict and Since-Conflict Stock Performance

Dec 31, 2025 – Feb 27, 2026

Ticker	12/31/25	2/27/26	% Change
TROX	4.17	7.48	79.4%
CC	11.79	18.24	54.7%
WLK	73.94	105.38	42.5%
LYB	43.30	57.52	32.8%
DOW	23.38	30.73	31.4%
VVV	29.06	37.80	30.1%
CF	77.34	99.54	28.7%
CMP	19.64	25.20	28.3%
MEOH	39.72	50.53	27.2%
BALL	52.97	67.13	26.7%
HUN	10.00	12.65	26.5%
ALB	141.44	178.67	26.3%
DD	120.60	150.12	24.5%
IFF	67.39	82.23	22.0%
OLN	20.83	25.37	21.8%
NTR	61.72	75.07	21.6%
PPG	102.46	123.27	20.3%
SMG	58.35	70.12	20.2%
CTVA	67.03	80.12	19.5%
LIN	426.39	508.08	19.2%
SLGN	40.37	48.05	19.0%
EMN	63.83	75.51	18.3%
CE	42.28	49.94	18.1%
ECL	262.52	308.35	17.5%
AMCR	41.70	48.43	16.1%
MOS	24.09	27.84	15.6%
CBT	66.28	76.14	14.9%
BAYN GR	€ 37.01	€ 41.80	12.9%
SHW	324.03	362.59	11.9%
APD	247.02	275.67	11.6%
CCK	102.97	114.60	11.3%
FUL	59.46	65.72	10.5%
RPM	104.00	114.12	9.7%
AVY	181.88	196.35	8.0%
OEC	5.28	5.69	7.8%
ASH	58.67	62.36	6.3%
FMC	13.87	14.74	6.3%
AXTA	32.31	33.41	3.4%
RIY	3732.87	3757	0.6%
RDG	2209.78	2206	(0.2%)
XLB	45.35	53.41	17.8%
SPX	6845.50	6879	0.5%

Feb 27, 2026 – Jul 10, 2026

Ticker	2/27/26	7/10/26	% Change
BAYN GR	€ 41.80	€ 50.50	20.8%
CMP	25.20	29.84	18.4%
CF	99.54	116.92	17.5%
CBT	76.14	87.32	14.7%
APD	275.67	299.53	8.7%
ASH	62.36	66.75	7.0%
CTVA	80.12	85.68	6.9%
LIN	508.08	529.79	4.3%
VVV	37.80	38.64	2.2%
OEC	5.69	5.69	0.0%
LYB	57.52	56.35	(2.0%)
CC	18.24	17.81	(2.4%)
AXTA	33.41	32.51	(2.7%)
CCK	114.60	111.47	(2.7%)
MEOH	50.53	48.66	(3.7%)
PPG	123.27	116.76	(5.3%)
DOW	30.73	29.03	(5.5%)
IFF	82.23	77.53	(5.7%)
CE	49.94	46.92	(6.0%)
SLGN	48.05	44.98	(6.4%)
SMG	70.12	65.48	(6.6%)
SHW	362.59	333.99	(7.9%)
RPM	114.12	105.10	(7.9%)
BALL	67.13	61.69	(8.1%)
DD	150.12	134.68	(10.3%)
EMN	75.51	67.57	(10.5%)
AMCR	48.43	43.18	(10.8%)
ECL	308.35	274.31	(11.0%)
HUN	12.65	11.13	(12.0%)
NTR	75.07	65.43	(12.8%)
FUL	65.72	55.91	(14.9%)
TROX	7.48	6.14	(17.9%)
AVY	196.35	160.71	(18.2%)
OLN	25.37	20.68	(18.5%)
MOS	27.84	21.51	(22.7%)
FMC	14.74	10.91	(26.0%)
WLK	105.38	75.49	(28.4%)
ALB	178.67	126.05	(29.5%)
RIY	3756.96	4125	9.8%
RDG	2205.95	2299	4.2%
XLB	53.41	50.89	(4.7%)
SPX	6878.88	7575	10.1%

Chemical Stock Performance: Trailing 12 months

Ticker	7/10/25	7/10/26	% Change
BAYN GR	€ 27.72	€ 50.50	82.2%
ALB	74.27	126.05	69.7%
MEOH	33.98	48.66	43.2%
DD	95.52	134.68	41.0%
CMP	22.44	29.84	33.0%
CC	14.05	17.81	26.8%
ASH	54.47	66.75	22.5%
CF	95.76	116.92	22.1%
CTVA	74.83	85.68	14.5%
LIN	470.34	529.79	12.6%
CBT	79.06	87.32	10.4%
NTR	60.80	65.43	7.6%
TROX	5.72	6.14	7.3%
BALL	58.40	61.69	5.6%
AXTA	30.94	32.51	5.1%
CCK	107.45	111.47	3.7%
ECL	266.79	274.31	2.8%
APD	294.10	299.53	1.8%
IFF	76.62	77.53	1.2%
VVV	39.05	38.64	(1.0%)
PPG	119.07	116.76	(1.9%)
DOW	30.23	29.03	(4.0%)
SHW	351.83	333.99	(5.1%)
SMG	69.41	65.48	(5.7%)
HUN	11.88	11.13	(6.3%)
RPM	114.30	105.10	(8.0%)
OLN	22.64	20.68	(8.7%)
AMCR	49.15	43.18	(12.1%)
WLK	86.48	75.49	(12.7%)
FUL	64.06	55.91	(12.7%)
AVY	184.63	160.71	(13.0%)
LYB	65.32	56.35	(13.7%)
EMN	81.56	67.57	(17.2%)
SLGN	55.33	44.98	(18.7%)
CE	62.39	46.92	(24.8%)
MOS	35.94	21.51	(40.2%)
OEC	11.32	5.69	(49.7%)
FMC	42.66	10.91	(74.4%)
RIY	3437.82	4125	20.0%
RDG	2243.30	2299	2.5%
XLB	46.14	50.89	10.3%
SPX	6280.46	7575	20.6%

Source: Bloomberg Finance L.P.

NOTE: Companies at or above the shaded line represent companies that outperformed the S&P 500

Five-Year Stock Performance

Ticker	7/9/21	7/10/26	% Change
CF	50.66	116.92	130.8%
CTVA	43.45	85.68	97.2%
LIN	292.71	529.79	81.0%
CBT	58.41	87.32	49.5%
MEOH	34.10	48.66	42.7%
DD	98.47	134.68	36.8%
ECL	212.76	274.31	28.9%
SHW	276.22	333.99	20.9%
VVV	32.02	38.64	20.7%
RPM	90.52	105.10	16.1%
SLGN	41.64	44.98	8.0%
AXTA	30.37	32.51	7.0%
NTR	61.21	65.43	6.9%
CCK	105.57	111.47	5.6%
APD	291.67	299.53	2.7%
BAYN GR	€ 51.04	€ 50.50	(1.1%)
FUL	64.58	55.91	(13.4%)
WLK	89.63	75.49	(15.8%)
AVY	210.35	160.71	(23.6%)
ASH	87.92	66.75	(24.1%)
AMCR	57.55	43.18	(25.0%)
BALL	84.79	61.69	(27.2%)
ALB	175.51	126.05	(28.2%)
PPG	171.41	116.76	(31.9%)
MOS	31.79	21.51	(32.3%)
EMN	114.95	67.57	(41.2%)
LYB	102.73	56.35	(45.1%)
IFF	147.70	77.53	(47.5%)
CC	34.57	17.81	(48.5%)
CMP	60.82	29.84	(50.9%)
OLN	44.59	20.68	(53.6%)
DOW	63.08	29.03	(54.0%)
HUN	27.01	11.13	(58.8%)
SMG	183.36	65.48	(64.3%)
CE	153.53	46.92	(69.4%)
TROX	20.23	6.14	(69.6%)
OEC	18.91	5.69	(69.9%)
FMC	107.66	10.91	(89.9%)
RIY	2457.65	4125	67.9%
RDG	1837.11	2299	25.2%
XLB	41.49	50.89	22.7%
SPX	4369.55	7575	73.4%

Source: Bloomberg Finance L.P.

NOTE: Companies at and above the shaded line represent companies that outperformed the S&P 500

Historical and Forward EV/EBITDA Multiples

	SHW	PPG	LIN	APD	ECL	IFF	RPM	AXTA	DD	ASH	VVV	10Y treasury
2020	19.9	13.5	16.5	16.3	23.2	11.6	14.6	10.0	12.7	10.3	9.6	0.91%
2021	25.0	14.9	17.5	16.9	22.9	17.7	17.4	10.4	11.2	11.9	11.9	1.51%
2022	17.3	12.6	15.5	16.3	16.4	14.0	14.3	9.7	11.8	10.2	15.7	3.87%
2023	20.0	12.8	16.6	13.9	18.8	14.1	14.5	9.9	12.0	11.7	13.5	3.88%
2024	19.3	10.4	15.9	15.0	18.9	13.4	14.6	8.8	10.9	9.8	12.1	4.57%
2025	18.6	9.7	15.5	13.5	19.0	10.9	12.4	8.1	11.8	8.9	9.7	4.17%
2026E	19.4	10.9	19.1	14.7	21.1	12.0	13.0	8.1	12.0	10.9	12.1	4.56%
2027E	18.0	10.4	18.3	14.1	18.9	11.3	13.0	7.4	11.3	9.8	10.6	
	CE	EMN	FUL	HUN	ALB	CMP						
2020	11.0	9.6	9.6	7.3	22.5	9.1						
2021	9.0	9.5	10.9	6.1	25.0	11.0						
2022	8.3	7.6	9.2	6.8	6.1	10.6						
2023	10.4	8.2	9.3	9.4	9.1	7.7						
2024	7.9	7.8	8.1	8.5	14.0	7.3						
2025	8.0	7.4	7.8	9.9	17.0	7.0						
2026E	6.9	7.5	7.6	9.3	5.8	9.2						
2027E	6.8	7.3	7.8	8.1	5.1	8.4						
	CF	FMC	NTR	MOS	CTVA	SMG						
2020	10.5	12.4	9.1	6.9	11.9	15.1						
2021	5.1	11.7	5.8	3.7	10.9	13.0						
2022	4.0	12.1	4.2	3.1	11.2	9.1						
2023	7.2	10.8	6.4	6.1	9.4	10.9						
2024	8.2	8.9	6.1	5.3	10.4	10.2						
2025	6.8	6.6	6.7	4.8	10.9	8.9						
2026E	4.4	7.1	6.3	7.0	13.5	10.4						
2027E	5.8	6.5	6.5	6.6	12.6	10.3						
	DOW	LYB	WLK	OLN	MEOH	CC	TROX	CBT	OEC			
2020	8.2	8.7	9.4	8.2	8.2	6.7	6.5	7.2	7.3			
2021	5.3	5.2	4.8	4.5	4.8	5.5	5.8	6.3	5.7			
2022	6.4	6.7	5.2	4.9	5.8	5.3	5.7	6.4	5.3			
2023	7.8	7.0	7.5	6.9	6.4	6.2	7.4	7.4	6.4			
2024	6.7	6.5	6.3	6.0	5.2	6.1	6.4	7.0	5.4			
2025	8.5	7.6	8.3	6.3	6.2	5.7	8.7	6.2	5.4			
2026E	4.7	4.3	7.1	9.4	3.3	7.3	13.0	7.7	6.2			
2027E	4.6	4.5	6.2	7.4	3.8	6.2	10.0	7.3	6.1			
	BALL	CCK	SLGN	AVY	AMCR							
2020	17.1	11.1	8.7	13.4	11.8							
2021	15.8	10.3	8.5	14.4	11.2							
2022	11.0	8.9	8.5	11.7	11.0							
2023	12.2	8.9	8.1	13.5	10.0							
2024	10.4	8.1	8.8	11.5	9.7							
2025	9.4	8.2	7.4	11.1	8.5							
2026E	10.5	8.7	7.8	10.2	9.1							
2027E	9.8	8.1	7.2	9.6	8.4							

Note: Historical EV/EBITDA is based on Bloomberg Finance L.P. Consensus

5-Year Average Prices

Ticker	5 year average	7/10/2026	% Change
CTVA	59.23	85.68	44.7%
CF	84.20	116.92	38.9%
DD	97.17	134.68	38.6%
LIN	395.85	529.79	33.8%
ECL	216.56	274.31	26.7%
BAYN GR	€ 40.56	€ 50.50	24.5%
CCK	93.98	111.47	18.6%
CBT	76.13	87.32	14.7%
MEOH	43.53	48.66	11.8%
SHW	301.16	333.99	10.9%
VVV	35.23	38.64	9.7%
APD	276.20	299.53	8.4%
AXTA	30.55	32.51	6.4%
RPM	102.13	105.10	2.9%
NTR	65.27	65.43	0.2%
BALL	62.81	61.69	(1.8%)
SLGN	46.07	44.98	(2.4%)
CMP	30.84	29.84	(3.3%)
PPG	128.17	116.76	(8.9%)
AVY	188.14	160.71	(14.6%)
AMCR	51.38	43.18	(16.0%)
SMG	78.48	65.48	(16.6%)
FUL	68.00	55.91	(17.8%)
IFF	94.73	77.53	(18.2%)
ASH	82.42	66.75	(19.0%)
ALB	165.50	126.05	(23.8%)
EMN	88.89	67.57	(24.0%)
CC	25.41	17.81	(29.9%)
WLK	110.16	75.49	(31.5%)
LYB	83.52	56.35	(32.5%)
DOW	47.41	29.03	(38.8%)
MOS	36.42	21.51	(40.9%)
HUN	22.92	11.13	(51.4%)
OLN	43.11	20.68	(52.0%)
TROX	13.09	6.14	(53.1%)
CE	106.83	46.92	(56.1%)
OEC	16.56	5.69	(65.6%)
FMC	72.65	10.91	(85.0%)
RIY	2,838	4,125	45.4%
RDG	1,798	2,299	27.9%
SPX	5,173	7,575	46.4%

Source: Bloomberg Finance L.P.

NOTE: Companies at and above the shaded line represent companies that outperformed the S&P 500

2026E EV/EBITDA Turn Per Share

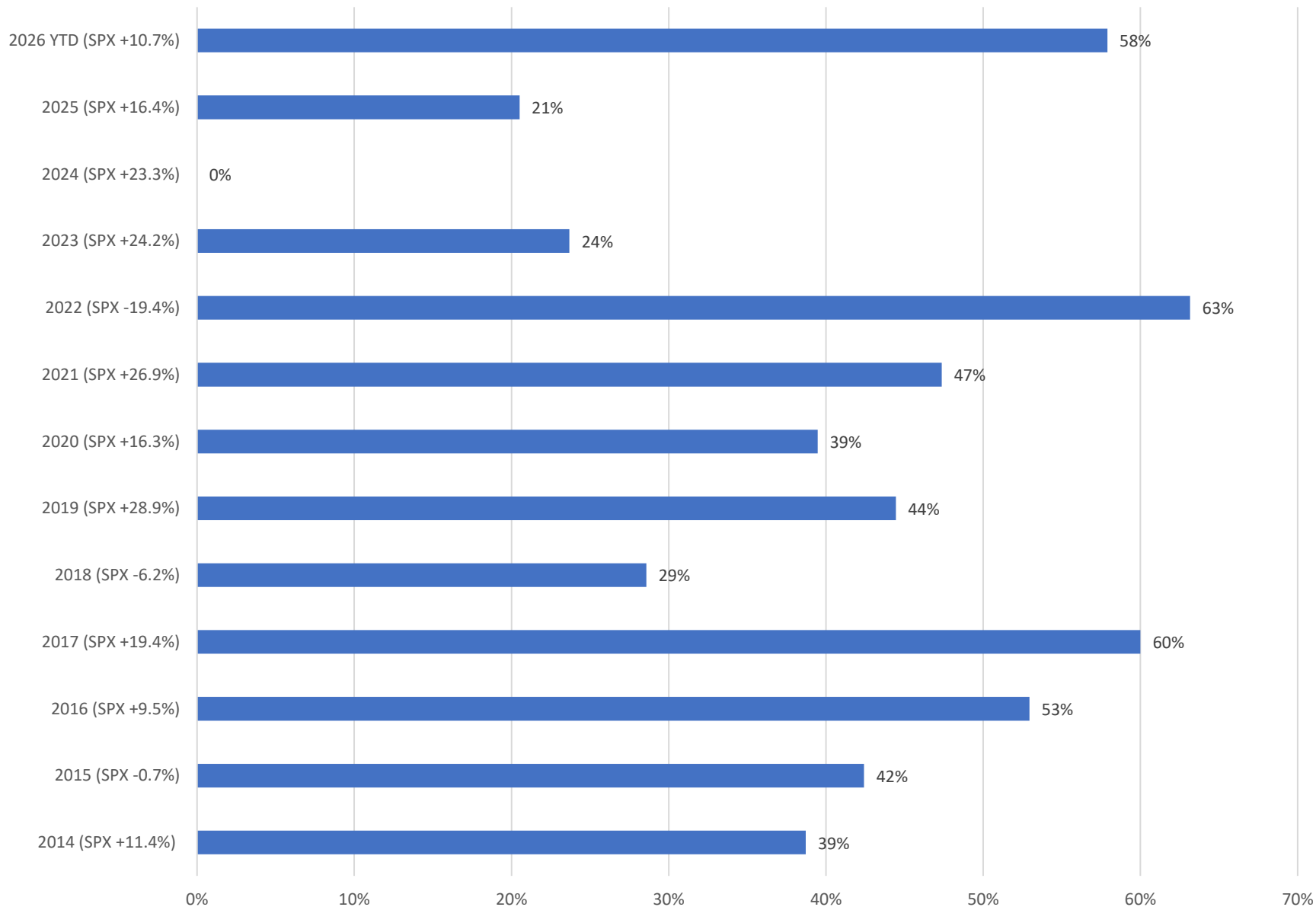
Ticker	1x 2026E EBITDA Turn
LIN	\$32
CF	\$27
MEOH	\$25
APD	\$24
ALB	\$24
LYB	\$21
AVY	\$20
SHW	\$20
CCK	\$19
CE	\$19
ECL	\$16
WLK	\$15
CBT	\$14
EMN	\$14
NTR	\$14
PPG	\$13
FUL	\$12
DOW	\$10
BAYN GR	\$10
SLGN	\$10
SMG	\$9
RPM	\$9
BALL	\$9
IFF	\$8
AMCR	\$8
ASH	\$8
MOS	\$6
FMC	\$6
CC	\$6
CTVA	\$6
CMP	\$5
AXTA	\$5
OLN	\$5
VVV	\$4
OEC	\$3
TROX	\$2
HUN	\$2

Source: J.P. Morgan estimates

Large Cap Stock Performance, Volume & Price Growth

LIN	2019	2020	2021	2022	2023	2024	2025
Volume	2.0%	(2.0%)	7.3%	1.7%	(0.7%)	(0.2%)	(0.3%)
Price	2.0%	2.0%	2.6%	6.7%	5.7%	2.1%	2.0%
EBITDA growth	7.6%	5.7%	17.7%	6.8%	11.6%	5.7%	4.2%
Share Perf.	36.4%	23.8%	31.5%	(5.8%)	25.9%	1.9%	1.8%
APD	F2019	F2020	F2021	F2022	F2023	F2024	F2025
Volume	2.0%	2.0%	4.4%	5.2%	3.1%	0.6%	(4.0%)
Price	3.0%	3.0%	2.3%	6.7%	5.6%	1.1%	1.0%
EBITDA growth	11.2%	4.4%	7.3%	9.4%	10.7%	7.3%	0.8%
Share Perf.	30.9%	34.3%	(14.0%)	(9.1%)	21.8%	6.7%	(14.8%)
DOW	2019E	2020	2021	2022	2023	2024	2025
Volume	(1.4%)	(2.7%)	1.0%	(2.4%)	(6.6%)	0.4%	(0.1%)
Price	(10.8%)	(7.7%)	39.6%	9.8%	(14.9%)	(4.5%)	(6.6%)
EBITDA growth	(20.3%)	(23.3%)	121.4%	(24.5%)	(42.4%)	1.7%	(40.6%)
Share Perf.	9.9%	1.4%	2.2%	(11.2%)	8.8%	(26.8%)	(41.7%)
LYB	2019	2020	2021	2022	2023	2024	2025
Volume	(1.4%)	(1.6%)	0.2%	(2.1%)	1.1%	(0.3%)	(3.7%)
Price	(9.5%)	(18.5%)	66.2%	11.4%	(19.6%)	(1.6%)	(21.5%)
EBITDA growth	(14.9%)	(35.7%)	145.7%	(24.9%)	(20.0%)	(14.9%)	(45.8%)
Share Perf.	13.6%	(3.0%)	0.6%	(10.0%)	14.5%	(21.9%)	(41.7%)
ECL	2019	2020	2021	2022	2023	2024	2025
Volume	(21.5%)	(8.4%)	3.0%	2.5%	0.2%	1.5%	0.8%
Price	2.3%	1.2%	2.1%	9.8%	8.3%	2.5%	2.1%
EBITDA growth	7.1%	(24.7%)	8.4%	2.6%	12.5%	16.5%	9.1%
Share Perf.	31.0%	12.1%	8.4%	(38.0%)	36.3%	18.1%	12.0%
IFF	2019E	2020	2021	2022	2023	2024	2025
Volume	0.6%	(0.0%)	5.8%	(0.9%)	(6.9%)	6.5%	1.3%
Price	1.6%	0.0%	2.0%	9.9%	5.5%	0.0%	0.2%
EBITDA growth			3.5%	1.2%	(19.3%)	11.4%	(5.4%)
Share Perf.	(3.9%)	(15.6%)	38.4%	(30.4%)	(22.8%)	4.4%	(20.3%)
SHW	2019	2020	2021	2022	2023	2024	2025
Volume	1.0%	2.3%	2.7%	(0.1%)	(0.7%)	0.1%	(1.6%)
Price	2.2%	1.7%	5.1%	11.1%	3.8%	0.4%	2.0%
EBITDA growth	10.0%	12.2%	(6.9%)	12.0%	19.5%	1.6%	4.8%
Share Perf.	48.3%	25.9%	43.8%	(32.6%)	31.4%	9.0%	(4.7%)
CTVA	2019	2020	2021	2022	2023	2024	2025
Volume	0.0%	5.7%	5.1%	4.7%	(10.1%)	1.6%	3.2%
Price	0.0%	2.7%	3.9%	10.3%	7.5%	(0.5%)	0.7%
EBITDA growth	(4.1%)	5.0%	23.4%	25.2%	4.9%	(0.1%)	14.0%
Share Perf.	1.9%	31.0%	22.1%	24.3%	(18.5%)	18.9%	17.7%
PPG	2019	2020	2021	2022	2023	2024	2025
Volume	(2.9%)	(9.6%)	6.2%	(3.5%)	(1.2%)	(0.9%)	1.1%
Price	2.1%	1.5%	4.7%	11.3%	3.4%	(0.0%)	0.9%
EBITDA growth	4.0%	(7.1%)	6.1%	(5.0%)	19.1%	2.8%	(7.3%)
Share Perf.	30.6%	8.0%	19.6%	(27.1%)	18.9%	(20.1%)	(14.2%)
BALL	2019	2020	2021	2022	2023	2024	2025
Volume			7.4%	1.9%	(5.9%)	1.0%	3.8%
Price			9.8%	10.0%	(3.2%)	(2.1%)	5.2%
EBITDA growth	1.5%	4.2%	10.3%	(8.3%)	7.9%	(10.1%)	6.0%
Share Perf.	40.7%	44.1%	3.7%	(46.4%)	12.5%	(4.2%)	(3.9%)

Percentage of Coverage Universe Outperformance vs. S&P 500

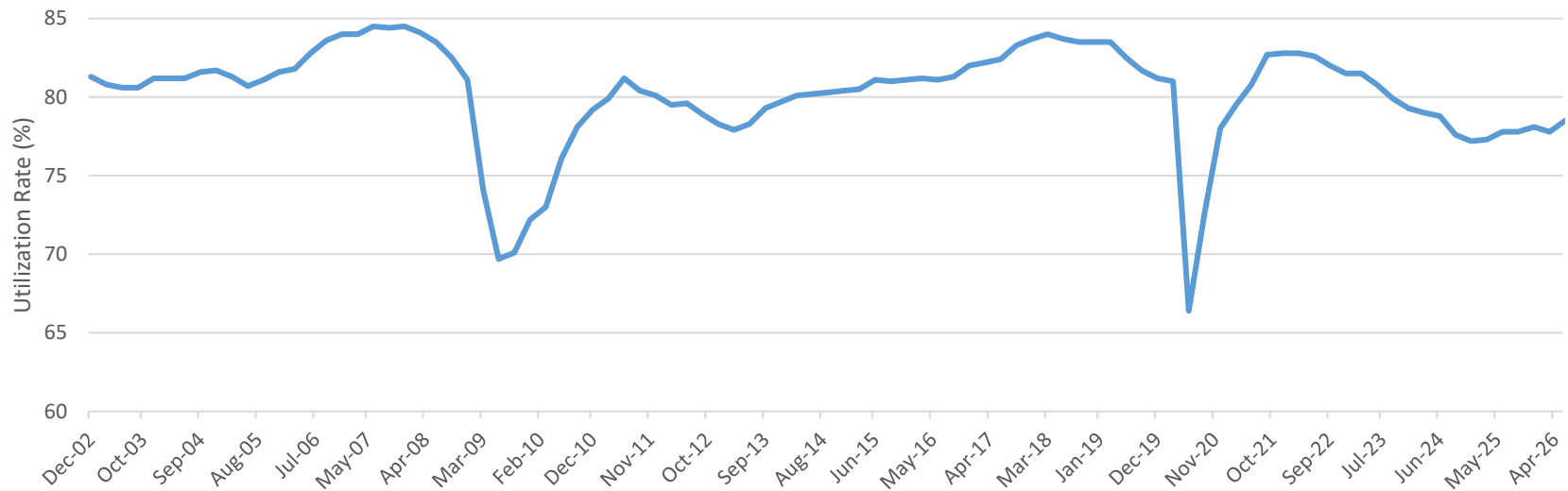


Source: Bloomberg Finance L.P.

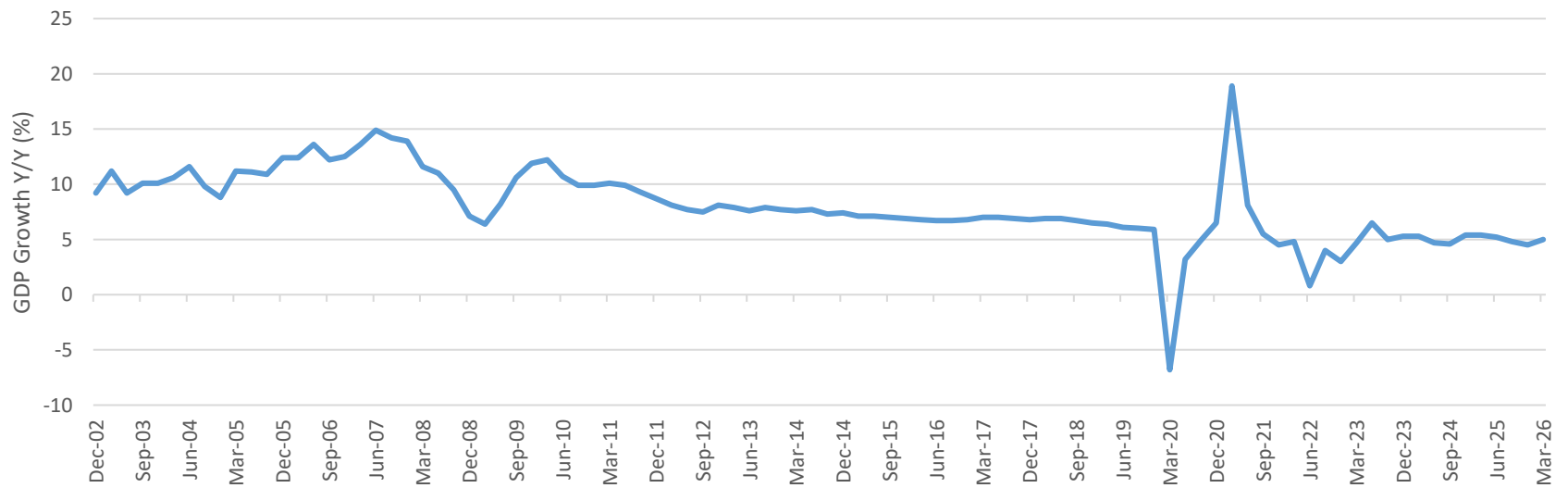
*Chemicals & Packaging stocks covered by Jeff Zekauskas, J.P. Morgan

Industrial Capacity Utilization

European Utilization Rates



China GDP Growth Y/Y



U.S. Chemical Railcar Data

(Total railcar shipments, monthly data represents average weekly railcar shipments.)

	Freight Carloads		Y-O-Y % Change	Monthly U.S. Railcar Shipments		
As of	7/3/26	31,608	1.0%			
Year Ago Week		31,280				
Week Prior	6/26/26	33,933				
YTD Total		882,733	2.8%			
Year-Ago YTD Total		858,452				
Weekly Average	2025	2026	Y-O-Y % Change			
January	31,443	32,675	3.9%			
February	33,476	34,606	3.4%			
March	33,725	35,451	5.1%			
April	33,745	34,683	2.8%			
May	32,728	33,593	2.6%			
June	32,744	32,911	0.5%			
July	32,978	31,608	(4.2%)			
August	33,303					
September	32,244					
October	32,172					
November	31,852					
December	31,652					

	Total shipments	% change
2025	1,764,979	4.7%
2024	1,685,683	4.1%
2023	1,619,355	(0.3%)
2022	1,624,544	(4.1%)
2021	1,694,133	3.6%
2020	1,634,822	(1.6%)
2019	1,661,742	(0.6%)
2018	1,672,126	3.8%
2017	1,610,781	1.2%

Major Currency Changes

Global Currencies (Spot Price)	Price	Week	Month	Quarterly Average			% Y/Y	Annual Average		% Change	Since 2005	
USD Denominated	10-Jul	Ago	Ago	QTD	2Q:26	3Q:25	QTD	YTD	2025	YTD	Low	High
Euro	0.876	0.874	0.864	0.875	0.860	0.856	2.3%	0.858	0.885	(2.97%)	0.625	1.042
Chinese Renminbi	6.777	6.781	6.762	6.788	6.806	7.157	(5.2%)	6.861	7.187	(4.53%)	6.041	8.283
Brazilian Real	5.110	5.168	5.058	5.157	5.047	5.447	(5.3%)	5.152	5.584	(7.73%)	1.540	6.293
Canadian Dollar	1.415	1.420	1.399	1.419	1.384	1.377	3.0%	1.380	1.397	(1.20%)	0.920	1.458
Japanese Yen	161.681	161.342	160.256	161.854	159.399	147.481	9.7%	158.336	149.525	5.89%	75.827	161.786
Mexican Peso	17.470	17.470	17.229	17.493	17.387	18.622	(6.1%)	17.478	19.164	(8.80%)	9.858	25.361
Australian Dollar	1.438	1.441	1.419	1.442	1.409	1.528	(5.6%)	1.425	1.551	(8.10%)	0.907	1.741
South Korea Won*	14.988	15.298	15.172	15.231	15.011	13.870	9.8%	14.853	14.209	4.53%	9.022	15.838
Global Currencies (Spot Price)	Price	Week	Month	Quarterly Average			% Y/Y	Annual Average		% Change	Since 2005	
Offshore Denominated	10-Jul	Ago	Ago	QTD	2Q:26	3Q:25	QTD	YTD	2025	YTD	Low	High
Euro	1.142	1.144	1.157	1.142	1.162	1.169	(2.3%)	1.165	1.130	3.06%	0.959	1.599
Chinese Renminbi	0.148	0.147	0.148	0.147	0.147	0.140	5.4%	0.146	0.139	4.74%	0.121	0.166
Brazilian Real	0.196	0.194	0.198	0.194	0.198	0.184	5.6%	0.194	0.179	8.38%	0.159	0.649
Canadian Dollar	0.707	0.704	0.715	0.705	0.722	0.726	(3.0%)	0.725	0.716	1.21%	0.686	1.087
Japanese Yen	0.006	0.006	0.006	0.006	0.006	0.007	(8.9%)	0.006	0.007	(5.56%)	0.006	0.013
Mexican Peso	0.057	0.057	0.058	0.057	0.058	0.054	6.5%	0.057	0.052	9.64%	0.039	0.101
Australian Dollar	0.695	0.694	0.705	0.693	0.710	0.654	6.0%	0.702	0.645	8.81%	0.574	1.102
South Korea Won*	0.067	0.065	0.066	0.066	0.067	0.072	(8.9%)	0.067	0.070	(4.34%)	0.063	0.111

Note: KRWUSD represents 100 KRW in USD

Source: Bloomberg Finance L.P. and J.P. Morgan estimates.

Geographic Exposure

Sales Exposure	North America	EMEA	APAC	LATAM & Other
Valvoline	96%	4%		
Compass Minerals	94%	5%	1%	
Scotts Miracle-Gro	92%	8%		
CF Industries	85%	5%	10%	
Sherwin-Williams	80%	10%	10%	
Westlake	80%	11%	4%	5%
RPM	78%	14%	2%	6%
Nutrien	71%	2%	16%	11%
Silgan	70%	24%	6%	
Olin	68%	10%	23%	
Ecolab	56%	23%	13%	7%
LyondellBasell	55%	27%	18%	
Corteva	52%	18%	8%	23%
Ball Corp	51%	32%	0%	17%
H.B. Fuller	51%	30%	19%	0%
DuPont	50%	21%	24%	5%
Arcor	48%	29%	11%	12%
Linde	48%	27%	21%	4%
Chemours	46%	20%	21%	13%

Sales Exposure	North America	EMEA	APAC	LATAM & Other
Air Products	45%	26%	28%	1%
Crown Holdings	45%	25%	12%	18%
Eastman	44%	26%	24%	6%
Dow	40%	31%	18%	11%
Mosaic	39%	2%	14%	45%
Huntsman	39%	26%	29%	6%
Cabot	37%	24%	39%	0%
Bayer AG	37%	30%	16%	17%
Avery Dennison	35%	28%	31%	6%
Axalta	34%	36%	18%	12%
Orion	34%	44%	22%	0%
PPG	34%	34%	19%	13%
FMC	32%	25%	4%	39%
Celanese	31%	35%	33%	1%
IFF	29%	34%	23%	14%
Ashland	27%	37%	26%	10%
Tronox	26%	40%	27%	7%
Methanex	25%	26%	35%	14%
Albemarle	24%	7%	69%	0%

Source: Company 10-K and J.P. Morgan estimates. Data based on F2025 reporting.

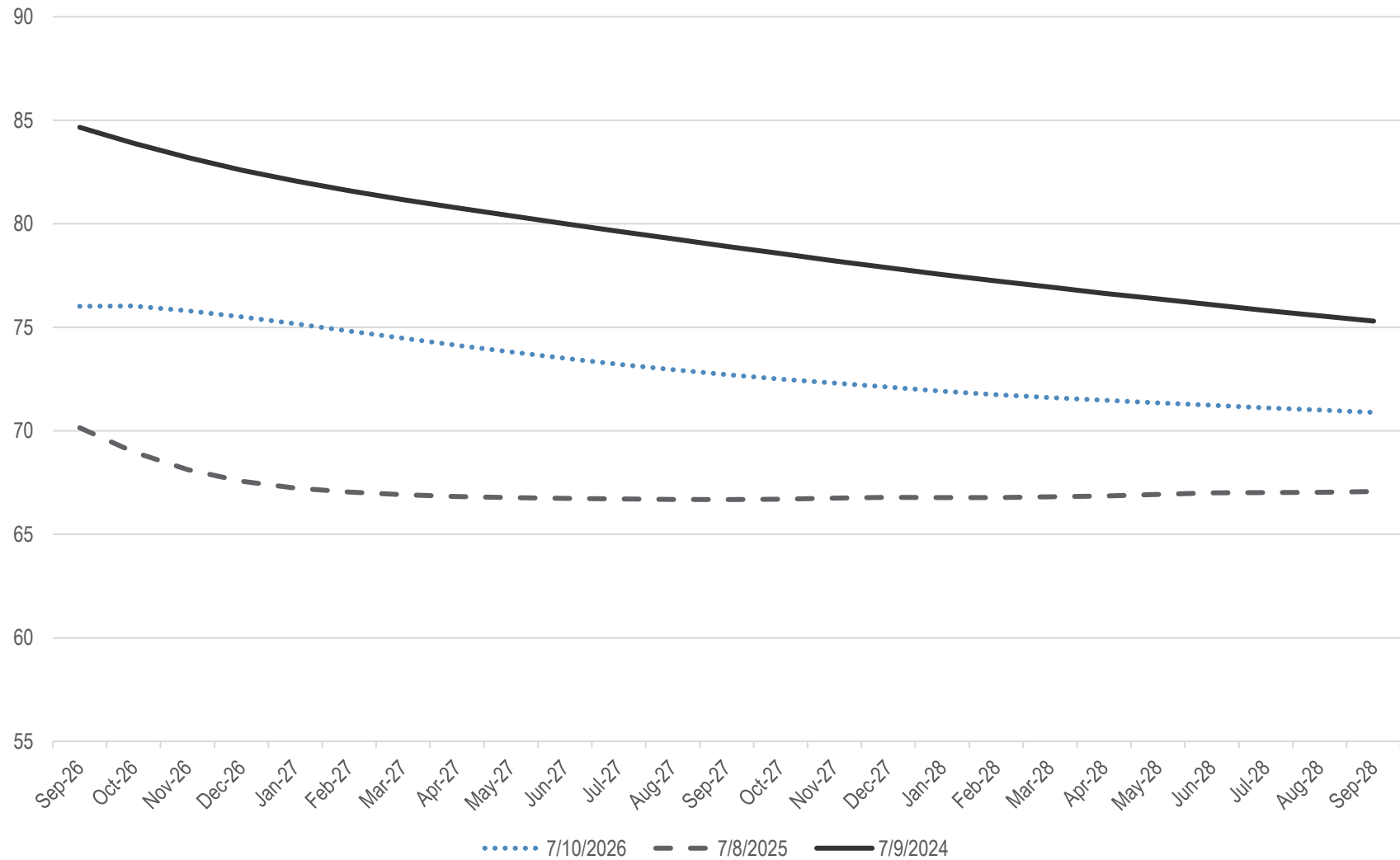
Commodity Trends:

	7/10/26	3/31/26	12/31/25	9/30/25	6/30/25	3/31/25	12/30/24	3/28/24	12/29/23	12/30/22	12/31/21	% Change y/y*	% Change vs 1Q:26*
Brent Oil \$/bbl	76.0	85.9	60.3	65.3	64.9	73.3	73.6	85.4	76.8	85.3	73.8	26.2%	(11.5%)
Natural Gas \$/mmBTU	2.77	3.14	3.99	3.12	3.26	4.11	3.40	1.54	2.58	3.52	3.65	(30.5%)	(11.7%)
HDPE Houston Spot \$/mt	880	1,365	795	805	840	895	870	850	850	890	1,440	10.7%	(35.5%)
HDPE Far East \$/mt	1,030	1,300	850	900	920	925	940	970	970	970	1,125	21.2%	(20.8%)
Propylene Gulf Coast Polymer Grade Spot c/lb	37.5	55.5	26.5	31.5	35.5	38.5	39.0	44.0	44.0	33.0	49.0	41.5%	(32.4%)
Polypropylene Houston Spot c/lb	55.1	68.5	37.4	42.4	47.4	47.4	48.1	49.9	49.9	43.3	90.5	47.4%	(19.5%)
PVC US Export c/lb	30.6	45.6	26.8	25.6	29.7	30.8	31.3	32.0	32.0	35.2	76.7	14.2%	(32.8%)
Chlorine \$/mt	608	624	679	755	755	756	809	819	906	1,262	810	(10.5%)	(2.6%)
China TiO2 \$/mt	2,413	2,234	2,041	1,941	1,939	2,119	2,041	2,366	2,315	2,283	3,252	18.2%	8.0%
China MDI \$/mt	2,287	2,538	2,478	2,148	2,200	2,372	2,478	2,266	2,215	2,229	2,983	(7.7%)	(9.9%)
Corn \$/bu	4.60	4.84	4.48	4.32	4.26	4.42	4.69	4.78	4.84	6.78	5.46	2.7%	(5.0%)
Soy \$/bu	11.97	11.71	10.31	10.02	10.24	10.15	9.98	11.92	12.94	15.19	13.29	16.1%	2.2%
US DAP NOLA \$/st	770	693	616	778	725	618	578	600	575	613	750	25.0%	11.2%
US Potash NOLA \$/st	338	305	310	338	336	315	255	313	323	485	685	8.9%	10.7%
US Urea NOLA \$/st	387	688	369	383	419	364	332	380	300	453	780	4.7%	(43.8%)
EURO/USD	1.14	1.16	1.17	1.17	1.18	1.08	1.04	1.08	1.10	1.07	1.14	(2.4%)	(1.2%)
USD/REAL	5.11	5.18	5.50	5.32	5.43	5.71	6.17	5.01	4.85	5.29	5.57	(7.1%)	(1.4%)
USD/RMB	6.78	6.89	6.99	7.12	7.16	7.26	7.30	7.23	7.10	6.90	6.36	(3.0%)	(1.7%)

Source: Bloomberg Finance L.P.

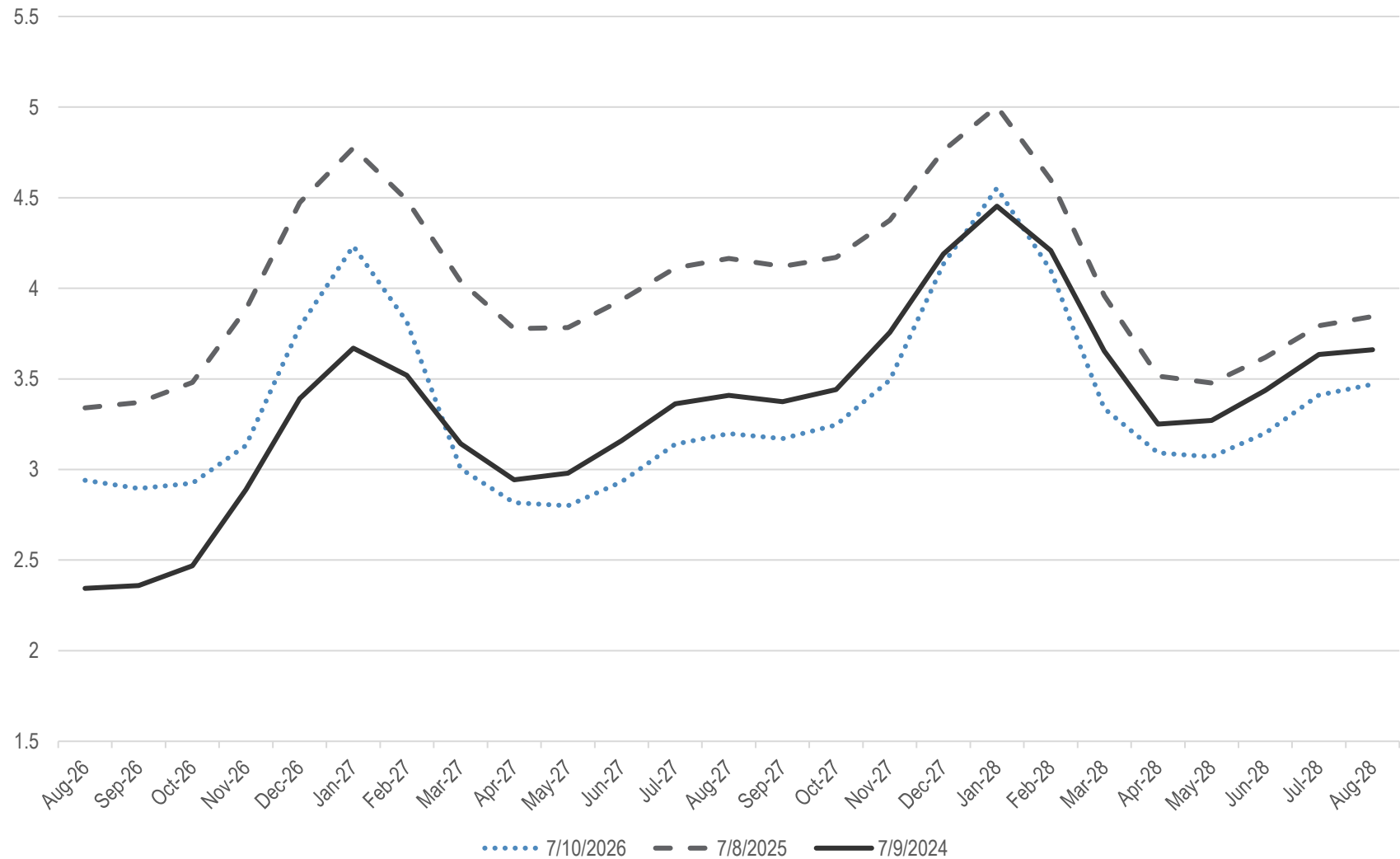
Brent Futures

(In \$ per barrel.)



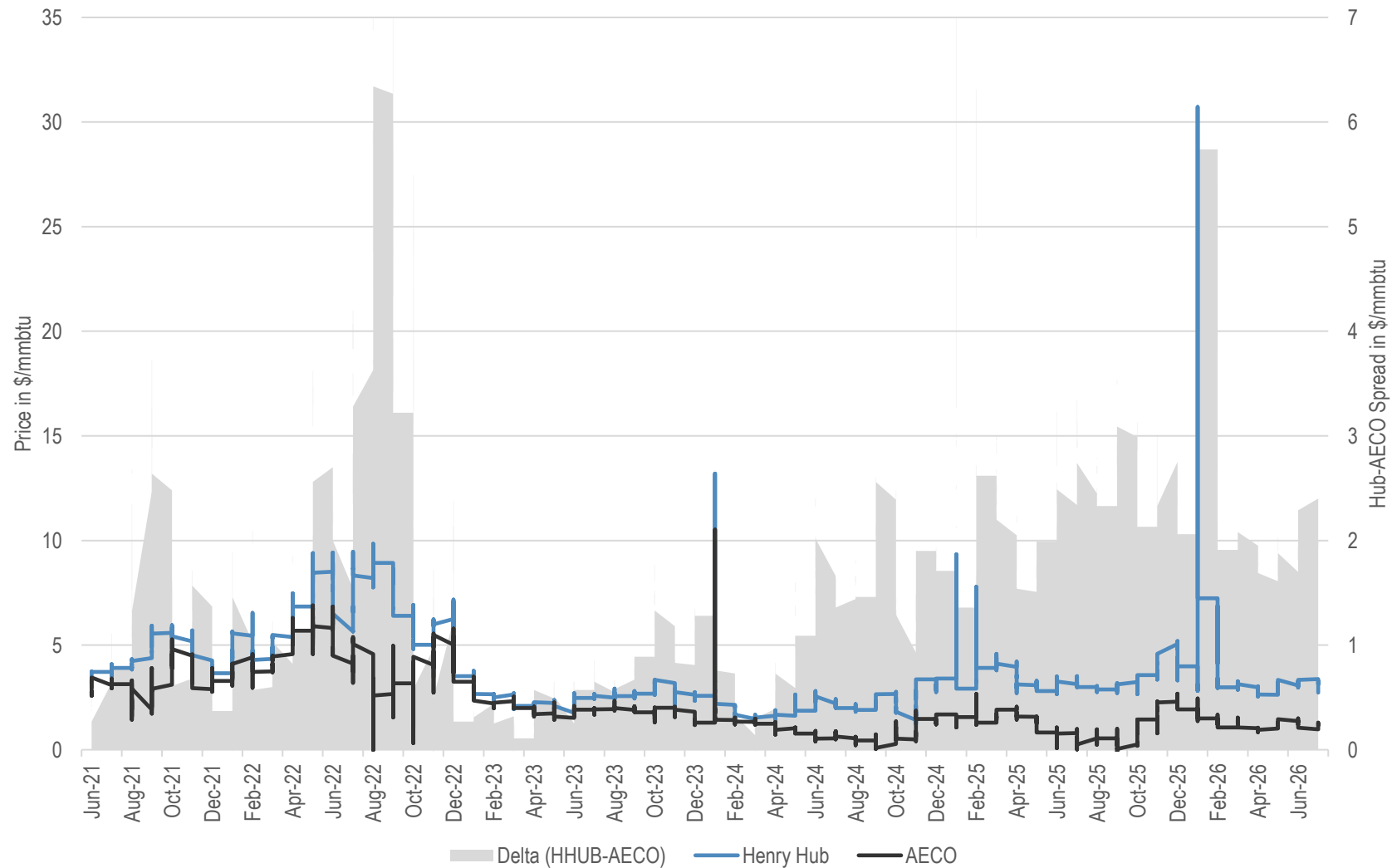
U.S. Henry Hub Natural Gas Futures

(In \$ per mmBTU.)



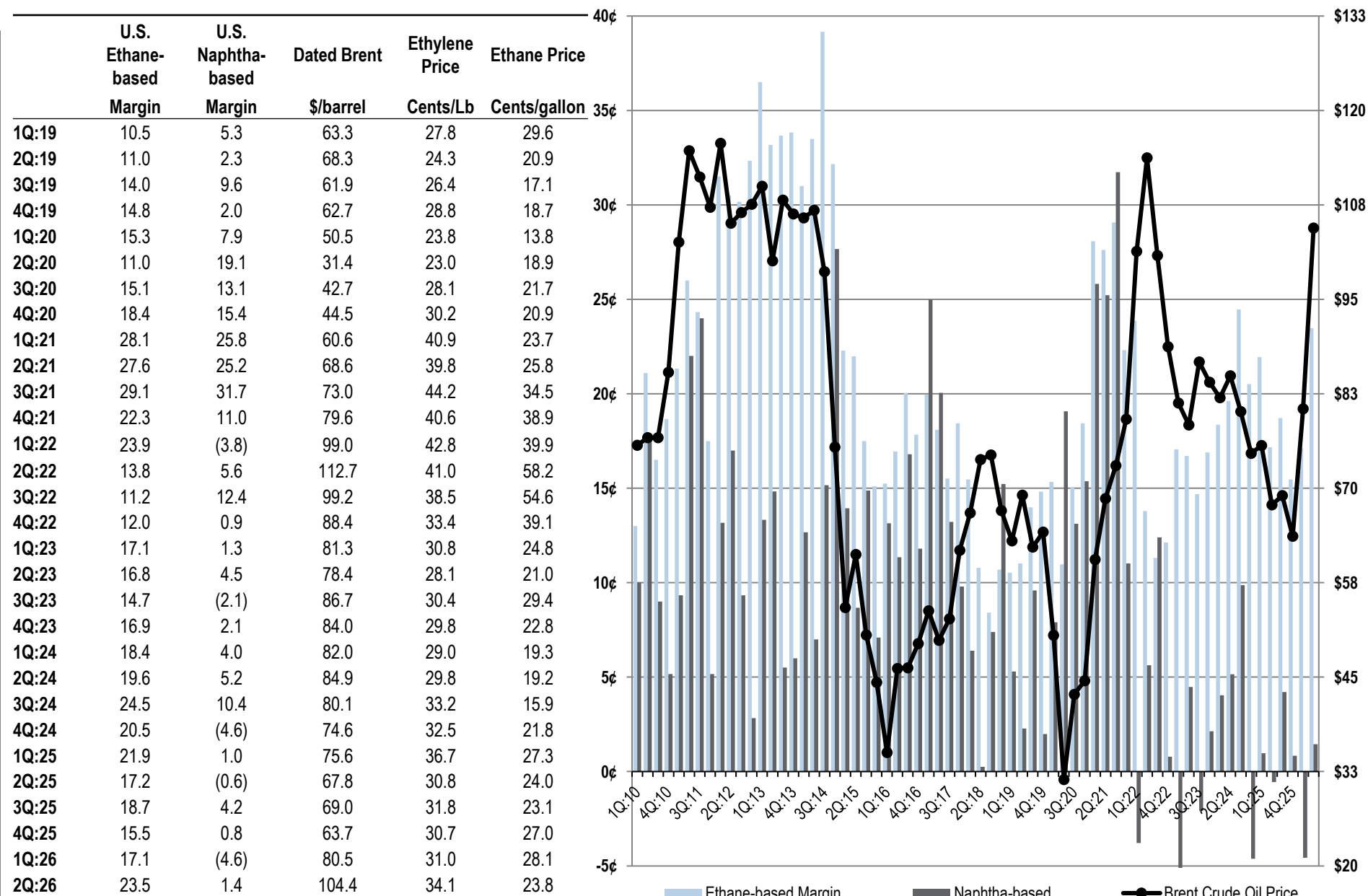
U.S. Henry Hub vs AECO Gas

The average spread between Henry Hub and AECO gas is ~\$1.65/mmBTU (mid-2021-2026TD), implying a 13 cents/gal ethane price differential (all else equal)



North American Ethylene Margins

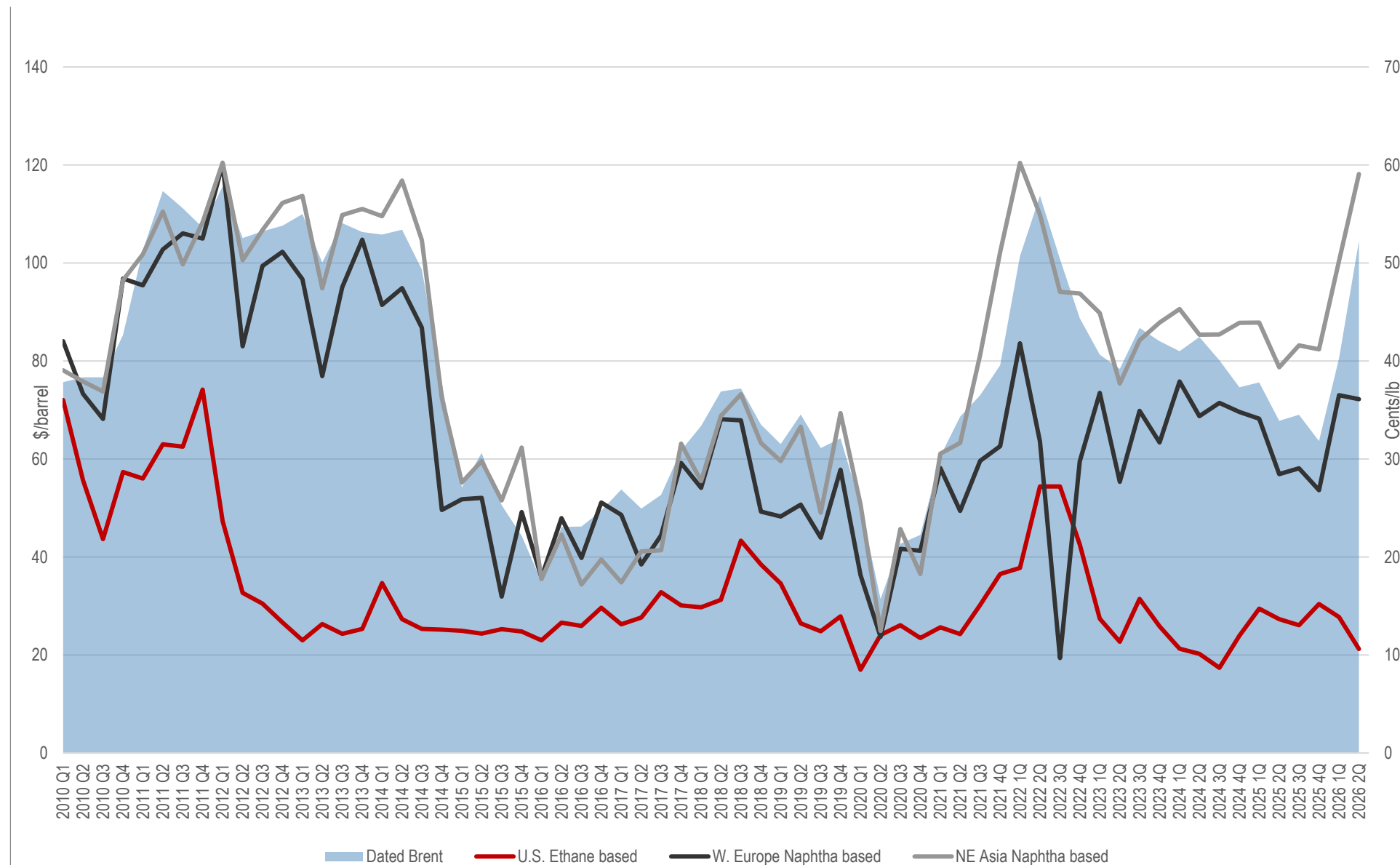
(Contract ethylene margins in cents per pound; European Brent crude oil price in \$ per barrel)



Note: Ethane and naphtha margins are net of co-product credits and additional operational costs

Regional Ethylene Cash Costs

(European Brent crude oil price in \$ per barrel; Rt Axis Ethylene costs in cents per pound)

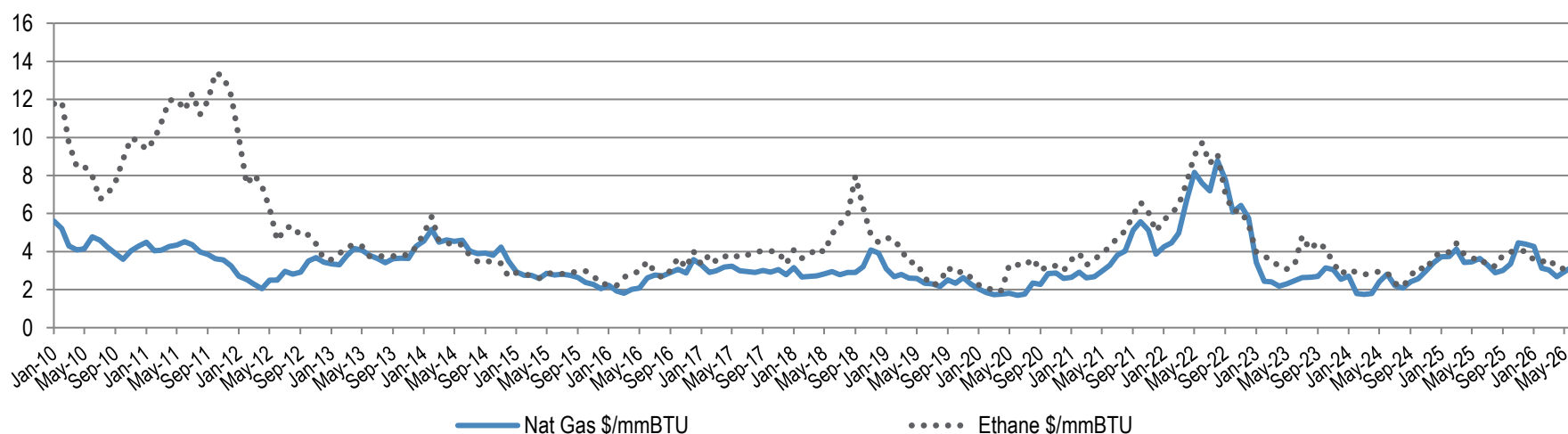


Source: Chemical Market Analytics by OPIS, a Dow Jones Company, Bloomberg Finance L.P., and J.P. Morgan estimates

U.S. Natural Gas-Ethane Spread

(Prices are quoted as a monthly average)

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Average
2022	Nat Gas \$/mmBTU	4.26	4.46	4.98	6.71	8.16	7.60	7.19	8.78	7.76	6.09	6.43	5.77	6.51
	Ethane cents/gal	37.33	39.33	42.70	50.14	59.77	63.94	56.24	60.36	46.68	40.25	40.65	35.52	47.74
	Ethane \$/mmBTU	5.65	5.96	6.47	7.60	9.05	9.69	8.52	9.15	7.07	6.10	6.16	5.38	7.23
	Ethane-Nat Gas Spread \$/mmBTU	1.40	1.50	1.49	0.89	0.89	2.09	1.33	0.37	(0.69)	0.01	(0.27)	(0.39)	0.72
2023	Nat Gas \$/mmBTU	3.42	2.44	2.41	2.19	2.30	2.48	2.64	2.65	2.70	3.15	3.06	2.54	2.66
	Ethane cents/gal	26.14	24.57	23.80	21.00	20.36	21.58	31.80	27.07	29.71	27.38	22.55	18.61	24.55
	Ethane \$/mmBTU	3.96	3.72	3.61	3.18	3.09	3.27	4.82	4.10	4.50	4.15	3.42	2.82	3.72
	Ethane-Nat Gas Spread \$/mmBTU	0.54	1.29	1.20	0.99	0.79	0.79	2.18	1.46	1.81	1.00	0.36	0.28	1.06
2024	Nat Gas \$/mmBTU	2.72	1.80	1.75	1.79	2.42	2.81	2.21	2.09	2.41	2.58	2.98	3.41	2.41
	Ethane cents/gal	20.36	19.07	18.40	19.07	19.47	19.13	15.36	14.07	17.94	21.08	19.98	24.52	19.04
	Ethane \$/mmBTU	3.08	2.89	2.79	2.89	2.95	2.90	2.33	2.13	2.72	3.19	3.03	3.72	2.88
	Ethane-Nat Gas Spread \$/mmBTU	0.37	1.09	1.04	1.10	0.53	0.09	0.12	0.05	0.31	0.62	0.05	0.56	0.49
2025	Nat Gas \$/mmBTU	3.72	3.74	4.14	3.43	3.46	3.64	3.30	2.89	3.01	3.35	4.47	4.39	3.63
	Ethane cents/gal	26.33	26.22	29.30	25.45	24.57	22.25	23.11	21.14	25.15	26.31	26.74	26.46	25.25
	Ethane \$/mmBTU	3.99	3.97	4.44	3.86	3.72	3.37	3.50	3.20	3.81	3.99	4.05	4.01	3.83
	Ethane-Nat Gas Spread \$/mmBTU	0.27	0.23	0.30	0.43	0.26	(0.27)	0.20	0.31	0.80	0.63	(0.42)	(0.39)	0.20
2026	Nat Gas \$/mmBTU	4.27	3.13	3.04	2.69	2.94	3.20							3.21
	Ethane cents/gal	23.46	22.74	23.90	20.72	20.46	22.86							22.36
	Ethane \$/mmBTU	3.55	3.45	3.62	3.14	3.10	3.46							3.39
	Ethane-Nat Gas Spread \$/mmBTU	(0.71)	0.32	0.58	0.45	0.16	0.26							0.18



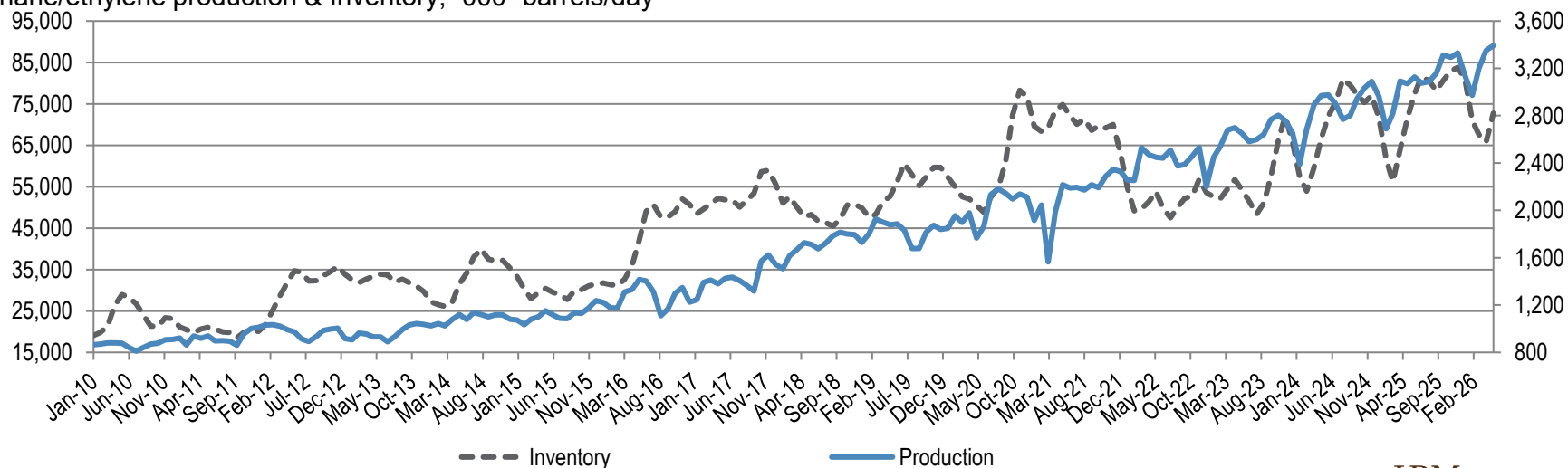
Note: Bloomberg Finance L.P. ticker for natural gas: NG1 Comdty; for ethane: LPGSMBPE Index

U.S. Daily-Average Ethane-Ethylene Inventory

(Production and consumption in thousands of barrels per day. Inventories in thousands of barrels.)

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Average	% growth
2022	Production	2,035	1,556	1,980	2,203	2,175	2,184	2,162	2,217	2,191	2,290	2,347	2,330	2,139	6.1%
	Consumption	2,168	2,053	2,084	2,065	1,982	2,118	2,180	1,849	1,932	2,016	1,963	1,826	2,020	12.8%
	Inventories	54,590	49,136	49,643	51,324	53,751	49,873	47,519	50,063	52,159	52,713	56,797	53,546	51,760	(23.0%)
2023	Production	2,367	2,485	2,617	2,702	2,625	2,613	2,570	2,598	2,734	2,781	2,746	2,612	2,621	9.0%
	Consumption	1,922	2,028	2,019	2,136	2,246	2,194	2,203	2,008	2,003	2,098	2,162	2,459	2,123	5.1%
	Inventories	52,518	52,141	54,299	56,724	54,244	51,534	48,347	51,042	58,045	66,185	72,043	65,796	56,910	10.0%
2024	Production	2,369	2,689	2,840	2,938	2,953	2,862	2,732	2,753	2,911	2,969	3,034	2,907	2,830	6.7%
	Consumption	2,118	2,364	2,259	2,207	2,355	2,273	2,080	2,387	2,432	2,537	2,463	2,500	2,331	8.2%
	Inventories	58,280	53,491	58,251	66,539	72,078	75,277	80,889	79,499	77,227	75,301	76,967	71,553	70,446	23.9%
2025	Production	2,689	2,818	3,092	3,070	3,127	3,076	3,088	3,158	3,313	3,294	3,329	3,140	3,100	8.1%
	Consumption	2,472	2,385	2,325	2,325	2,409	2,485	2,557	2,728	2,561	2,601	2,582	2,632	2,505	5.5%
	Inventories	61,716	56,200	63,933	71,395	77,409	81,568	80,576	78,162	80,666	82,968	83,795	80,916	74,942	6.3%
2026	Production	2,973	3,201	3,353	3,353									3,220	3.9%
	Consumption	2,631	2,590	2,668	2,630									2,630	5.0%
	Inventories	70,962	67,594	65,780	72,705									69,260	(7.6%)

Ethane/ethylene production & Inventory, -000- barrels/day



Source: DOE & J.P. Morgan estimates

Ethane Exports

(Ethane in barrels per day)

Ethane export capacity

Projects	Export	Ethane capacity (bpd)	Location	Start-up
Cochin (EP)	Pipeline	8,000	Conway, KS to Sarnia	1979
Mariner West	Pipeline	50,000	Houston, PA to Sarnia	4Q:13
Vantage	Pipeline	68,000	Tioga, ND to Alberta	2Q:14
Mariner East 1	Marine	40,000	Marcus Hook, PA	1Q:16
Morgan's Point (Enterprise)*	Marine	240,000	Morgan's Point, Texas	3Q:16
Utopia East	Pipeline	50,000	OH to Ontario	1Q:18
Mariner East 2	Marine	30,000	Marcus Hook, PA	1Q:19
Energy Transfer Partners and Satellite Petrochemical USA	Marine	170,000	Nederland, Texas	1Q:21
Neches River (Enterprise)	Marine	120,000	Orange County, Texas	3Q:25
Energy Transfer Flexport Expansion	Marine	250,000*	Nederland, Texas	4Q:25
Neches River Phase II (Enterprise)	Marine	180,000**	Orange County, Texas	2H:26

Total

1,206,000+ est.

Average Ethane Daily Exports in 2026 (Average YTD): 677,000 bpd

Average Ethane Daily Exports in December 2025: 613,000 bpd

Average Ethane Daily Exports in December 2024: 521,000 bpd

Average Ethane Daily Exports in December 2023: 375,000 bpd

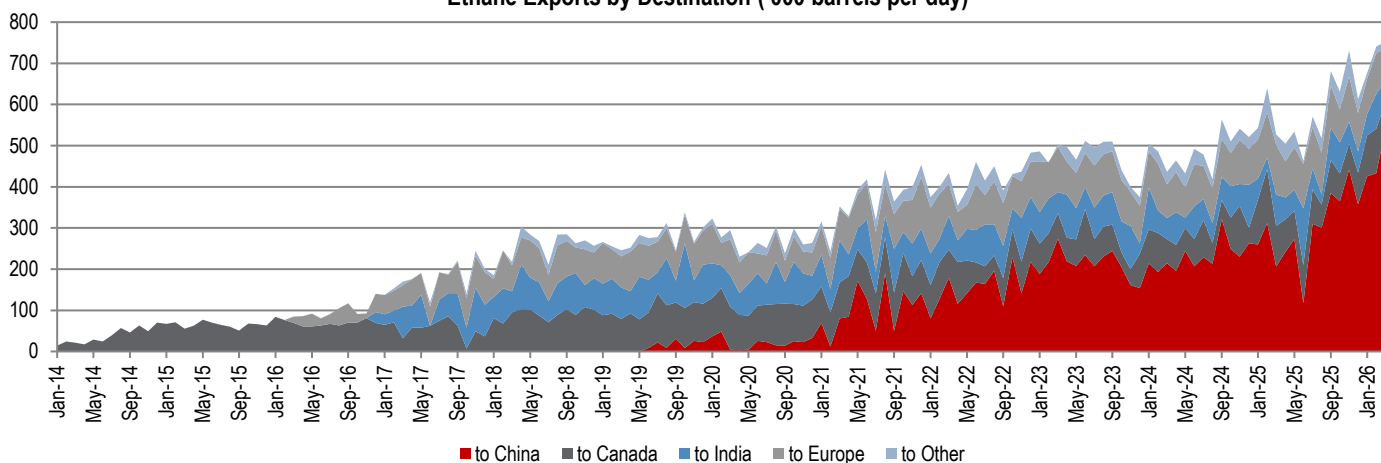
Average Ethane Daily Exports in December 2022: 483,000 bpd

Average Ethane Daily Exports in December 2021: 453,000 bpd

Average Ethane Exports

('000 barrels per day)

Ethane Exports by Destination ('000 barrels per day)



2014	38
2015	65
2016	95
2017	178
2018	256
2019	277
2020	271
2021	369
2022	419
2023	471
2024	487
2025	580

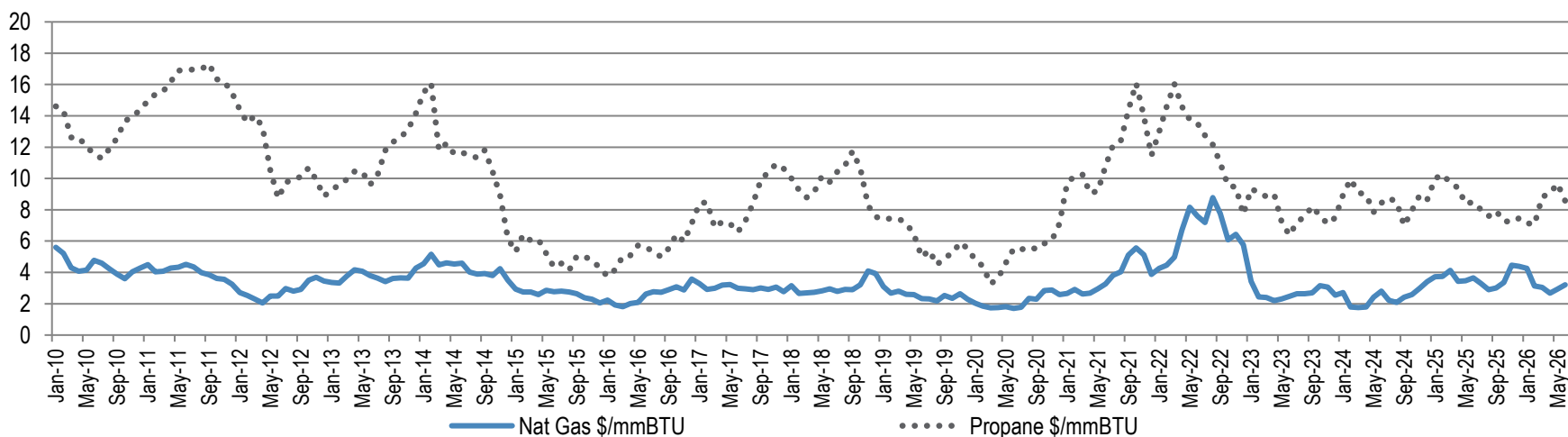
Source: ONEOK, EIA, company reports and J.P. Morgan Estimates

*Flexport is 250MBPD of NGL export capacity, with the flexibility of loading ethylene beginning 4Q:25

U.S. Natural Gas-Propane Spreads

(Prices are quoted as a monthly average)

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Average
2022	Nat Gas \$/mmBTU	4.26	4.46	4.98	6.71	8.16	7.60	7.19	8.78	7.76	6.09	6.43	5.77	6.51
	Propane cents/gal	116.84	131.27	144.30	131.25	123.88	121.73	114.67	109.36	98.88	85.93	85.10	69.38	111.05
	Propane \$/mmBTU	12.98	14.58	16.03	14.58	13.76	13.52	12.74	12.15	10.99	9.55	9.45	7.71	12.34
	Propane-Nat Gas Spread \$/mmBTU	8.73	10.12	11.05	7.88	5.60	5.93	5.55	3.37	3.23	3.46	3.02	1.94	5.82
2023	Nat Gas \$/mmBTU	3.42	2.44	2.41	2.19	2.30	2.48	2.64	2.65	2.70	3.15	3.06	2.54	2.66
	Propane cents/gal	84.06	81.94	79.59	80.90	65.00	57.44	64.80	68.20	74.05	68.54	63.92	67.44	71.32
	Propane \$/mmBTU	9.34	9.10	8.84	8.99	7.22	6.38	7.20	7.58	8.23	7.61	7.10	7.49	7.92
	Propane-Nat Gas Spread \$/mmBTU	5.92	6.67	6.43	6.80	4.92	3.91	4.56	4.93	5.53	4.47	4.04	4.95	5.26
2024	Nat Gas \$/mmBTU	2.72	1.80	1.75	1.79	2.42	2.81	2.21	2.09	2.41	2.58	2.98	3.41	2.41
	Propane cents/gal	80.59	89.60	81.88	80.67	70.71	76.01	78.81	76.44	62.81	71.69	80.48	77.76	77.29
	Propane \$/mmBTU	8.95	9.95	9.10	8.96	7.86	8.44	8.76	8.49	6.98	7.96	8.94	8.64	8.59
	Propane-Nat Gas Spread \$/mmBTU	6.24	8.16	7.35	7.17	5.44	5.63	6.55	6.41	4.57	5.39	5.96	5.23	6.17
2025	Nat Gas \$/mmBTU	3.72	3.74	4.14	3.43	3.46	3.64	3.30	2.89	3.01	3.35	4.47	4.39	3.63
	Propane cents/gal	89.82	92.40	87.63	84.59	75.42	76.51	72.09	68.33	71.28	65.54	65.08	67.10	76.32
	Propane \$/mmBTU	9.98	10.27	9.74	9.40	8.38	8.50	8.01	7.59	7.92	7.28	7.23	7.45	8.48
	Propane-Nat Gas Spread \$/mmBTU	6.26	6.52	5.60	5.97	4.92	4.86	4.71	4.70	4.91	3.93	2.76	3.06	4.85
2026	Nat Gas \$/mmBTU	4.27	3.13	3.04	2.69	2.94	3.20							3.21
	Propane cents/gal	64.35	63.60	79.40	81.40	87.14	77.16							75.50
	Propane \$/mmBTU	7.15	7.07	8.82	9.04	9.68	8.57							8.39
	Propane-Nat Gas Spread \$/mmBTU	2.88	3.94	5.78	6.36	6.74	5.37							5.18



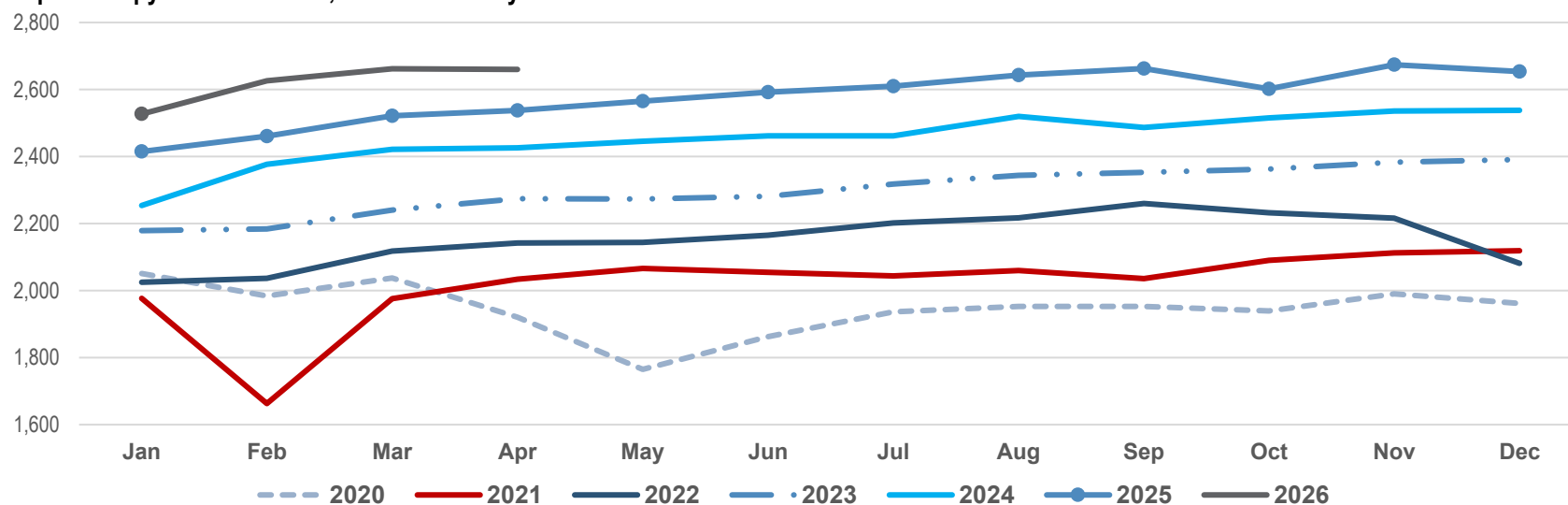
Note: Bloomberg Finance L.P. ticker for natural gas: NG1 Comdty; for propane LPGSMBPP Index

U.S. Propane-Propylene Inventories

(Production and consumption in thousands of barrels per day. Inventories in thousands of barrels.)

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Average	% growth
2022	Production	2,025	2,037	2,118	2,142	2,144	2,165	2,202	2,217	2,260	2,232	2,216	2,081	2,153	6.6%
	Consumption	1,294	1,239	941	681	540	565	613	563	746	758	986	1,104	836	0.7%
	Inventories	49,223	38,912	37,337	41,176	50,761	55,327	65,390	73,929	83,063	87,751	88,840	77,980	62,474	3.1%
2023	Production	2,179	2,184	2,240	2,274	2,273	2,281	2,318	2,344	2,353	2,363	2,383	2,391	2,299	6.7%
	Consumption	1,147	1,066	742	649	474	550	595	629	631	863	979	1,052	781	(6.5%)
	Inventories	69,507	61,297	56,342	61,880	72,195	80,330	87,909	97,558	102,622	99,253	91,648	80,549	80,091	28.2%
2024	Production	2,254	2,377	2,422	2,426	2,446	2,462	2,462	2,520	2,487	2,515	2,536	2,538	2,454	6.8%
	Consumption	1,331	1,020	789	631	560	529	511	573	644	843	875	1,197	792	1.3%
	Inventories	60,728	50,310	52,479	58,164	67,637	76,198	88,338	95,120	98,639	98,782	93,871	82,024	76,858	(4.0%)
2025	Production	2,415	2,461	2,522	2,538	2,565	2,592	2,610	2,643	2,663	2,602	2,674	2,654	2,578	5.1%
	Consumption	1,484	1,273	866	648	547	525	586	730	641	784	819	1,173	840	6.0%
	Inventories	60,461	47,445	45,263	49,504	63,652	76,400	83,565	95,448	101,330	105,145	104,609	98,077	77,575	0.9%
2026	Production	2,527	2,626	2,662	2,660									2,619	1.6%
	Consumption	1,404	1,259	668	666									999	19.0%
	Inventories	81,166	73,857	79,467	79,735									78,556	1.3%

Propane/Propylene Production, -000- barrels/day



Major North American Ethylene Expansions

(In millions of pounds)

Company	Location	Expansion	Type	Start up Date
Post 2024:				
CPCChem/Qatar	Orange, TX	4,410	Greenfield	Early 2027
Dow (Phase I)	Alberta, Canada	2,830	Greenfield	~2029
Dow (Phase II)	Alberta, Canada	1,320	Expansion	~2030
Shintech	Plaquemine, LA	1,380	Expansion	~2031
Completed				Start up Date
Nova Chemical	Sarnia, Canada	950	Expansion	2Q:23
Shell	Monaca, PA	3,310	Greenfield	4Q:22*
Total/Borealis ("Bayport Polymers")	Port Arthur, TX	2,210	Greenfield	3Q:22
Exxon/SABIC ("Gulf Coast Growth Ventures")	Corpus Christi, TX	3,970	Greenfield	Jan-22
Dow	Alberta, Canada	290	Expansion	1H:21
Dow	Freeport, TX	1,100	Expansion	May-20
Shintech	Plaquemine, LA	1,100	Greenfield	Mar-20
Formosa	Point Comfort, TX	2,650	Greenfield	Feb-20
Indorama	Lake Charles, LA	925	Restart	Jan-20
Sasol	Lake Charles, LA	3,310	Greenfield	Dec-19
Westlake/Lotte	Lake Charles, LA	2,210	Greenfield	Aug-19
ExxonMobil Chemical	Baytown, TX	3,310	Greenfield	Jul-18
Chevron Phillips Chemical	Cedar Bayou, TX	3,310	Greenfield	Mar-18
Dow	Freeport, TX	3,310	Greenfield	Jan-18
Westlake	Calvert City, KY	70	Expansion	2Q:17
LyondellBasell	Corpus Christi, TX	800	Expansion	Mar-17
Occidental Chemical/MexiChem	Ingleside, TX	1,210	Greenfield	Mar-17
Dow	Plaquemine, LA	550	Expansion	Dec-16
Westlake	Lake Charles, LA	250	Expansion	Jul-16
Braskem/Idesa	Coatzacoalcas, Mexico	2,210	Greenfield	Mar-16
N. American Completed Total		37,045		
North American Ethylene Capacity (YE 2015)		79,860		
North American Ethylene Capacity (YE 2020)		106,465		
North American Ethylene Capacity (YE 2024)		118,960		

Note: We define "start up" as the achievement of run rate production levels; Shell is limited by PE startups

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, J.P. Morgan Estimates **29**

Global Ethylene Supply/Demand Balance

(In -000- metric tons)

Capacity by Region	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E	2028E
North America	33,010	33,700	33,740	34,640	34,750	35,440	36,920	38,450	41,650	44,110	48,010	48,490	51,060	52,940	53,960	53,960	53,440	56,040	56,040
South America	5,320	5,470	5,450	5,450	5,420	5,400	5,400	5,400	5,400	5,380	5,380	5,380	5,380	5,440	5,440	5,440	5,440	5,440	5,440
West Europe	24,370	24,120	23,800	23,610	23,020	22,820	22,570	22,570	22,570	22,570	22,570	22,570	22,300	22,110	21,700	19,710	18,680	18,730	18,160
Central Europe	2,410	2,410	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320	2,320
CIS & Baltic States	3,660	3,880	3,940	3,640	3,670	3,740	4,140	4,440	4,610	4,910	6,350	6,350	6,350	6,410	6,410	7,190	7,660	9,020	9,880
Middle East	25,260	27,110	28,130	29,330	30,100	30,990	32,290	33,910	33,980	33,980	34,080	35,420	35,460	35,500	36,500	36,500	37,720	40,860	43,650
Africa	1,850	1,850	1,420	1,330	1,420	1,430	1,540	1,890	1,890	1,890	1,890	1,890	1,890	1,920	2,220	2,220	2,220	2,220	2,220
Indian Subcontinent	3,730	3,980	3,980	3,980	3,980	3,980	4,350	6,390	7,470	7,470	7,470	7,470	7,470	8,070	8,670	8,670	8,670	10,060	10,060
Northeast Asia ex-China	19,650	19,800	20,180	20,620	20,740	20,510	19,700	19,740	20,040	20,580	20,850	21,910	23,490	23,570	23,590	23,770	22,020	21,680	20,800
China	15,220	15,890	16,520	17,680	19,340	21,050	22,190	23,900	25,360	26,820	31,980	39,630	45,510	51,640	53,310	59,750	65,880	68,470	73,400
Southeast Asia	9,550	9,990	9,990	10,660	11,140	11,630	11,940	12,040	12,130	12,560	13,430	14,240	14,310	14,310	15,140	14,660	14,720	14,280	13,990
World Capacity	144,030	148,200	149,470	153,260	155,900	159,310	163,360	171,050	177,420	182,590	194,330	205,670	215,540	224,230	229,260	234,190	238,770	249,120	255,960
% change	8.6%	2.9%	0.9%	2.5%	1.7%	2.2%	2.5%	4.7%	3.7%	2.9%	6.4%	5.8%	4.8%	4.0%	2.2%	2.2%	2.0%	4.3%	2.7%
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021E	2022E	2023E	2024E	2025E	2026E	2027E	2028E
World Demand (Consultants)	121,520	125,630	127,960	131,970	135,550	140,890	146,290	152,570	159,910	163,940	169,710	177,870	177,380	181,830	185,710	188,950	195,250	201,690	208,080
% change	9.3%	3.4%	1.9%	3.1%	2.7%	3.9%	3.8%	4.3%	4.8%	2.5%	3.5%	4.8%	-0.3%	2.5%	2.1%	1.7%	3.3%	3.3%	3.2%
Operating Rate	84.4%	84.8%	85.6%	86.1%	86.9%	88.4%	89.6%	89.2%	90.1%	89.8%	87.3%	86.5%	82.3%	81.1%	81.0%	80.7%	81.8%	81.0%	81.3%
World Demand (JPM)	121,520	125,630	127,960	131,970	135,550	140,890	146,290	152,570	159,910	163,940	169,710	177,870	177,380	181,830	185,710	188,950	191,790	196,580	202,480
% change	9.3%	3.4%	1.9%	3.1%	2.7%	3.9%	3.8%	4.3%	4.8%	2.5%	3.5%	4.8%	-0.3%	2.5%	2.1%	1.7%	1.5%	2.5%	3.0%
Operating Rate	84.4%	84.8%	85.6%	86.1%	86.9%	88.4%	89.6%	89.2%	90.1%	89.8%	87.3%	86.5%	82.3%	81.1%	81.0%	80.7%	80.3%	78.9%	79.1%

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, J.P. Morgan Estimates

Global Polyethylene Supply/Demand

(In -000- metric tons)

Capacity by Region	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E	2028E
North America	19,590	19,790	20,000	20,380	20,560	20,750	21,530	23,160	25,850	26,990	28,110	29,050	30,230	32,840	33,470	33,770	34,820	36,070	36,070
South America	4,230	4,250	4,250	4,250	4,240	4,230	4,230	4,230	4,230	4,290	4,310	4,310	4,310	4,310	4,310	4,310	4,310	4,310	4,310
West Europe	14,870	15,060	15,010	14,750	14,370	14,070	14,020	14,020	14,020	14,020	14,020	14,020	14,020	14,020	14,020	12,590	11,480	10,880	10,880
Central Europe	1,800	1,830	1,700	1,710	1,710	1,710	1,760	1,760	1,760	1,760	1,960	2,030	2,030	2,030	2,030	2,030	2,030	2,030	2,030
CIS & Baltic States	2,460	2,480	2,370	2,370	2,370	2,370	2,800	2,800	2,870	3,340	4,350	4,810	4,810	4,810	4,810	5,130	5,760	7,560	7,920
Middle East	14,470	16,010	16,390	17,910	18,300	19,780	20,840	22,110	22,280	22,280	22,340	23,360	23,580	23,650	23,700	23,700	24,080	26,630	29,000
Africa	1,400	1,400	1,400	1,400	1,400	1,400	1,500	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800
Indian Subcontinent	2,560	2,930	2,930	2,930	2,930	2,930	3,260	4,700	5,620	5,620	5,620	5,620	5,660	6,490	6,870	6,870	6,870	7,710	7,810
Northeast Asia ex-China	9,530	9,610	9,730	9,690	9,980	9,970	10,030	10,080	10,030	10,100	10,450	11,670	12,830	12,830	12,830	12,810	12,780	13,490	12,920
China	11,110	11,470	12,010	13,340	14,610	15,640	16,540	17,610	18,810	19,820	22,210	25,960	30,280	33,120	34,130	39,080	42,790	48,680	53,100
Southeast Asia	7,190	7,570	7,900	8,900	8,970	9,000	9,110	9,340	9,750	9,650	10,250	10,220	10,320	11,340	11,570	11,430	11,430	10,580	10,190
World Capacity	89,210	92,410	93,690	97,630	99,420	101,850	105,610	111,610	117,030	119,670	125,420	132,850	139,870	147,240	149,540	153,520	158,150	169,740	176,030
% change	9.8%	3.6%	1.4%	4.2%	1.8%	2.4%	3.7%	5.7%	4.9%	2.3%	4.8%	5.9%	5.3%	5.3%	1.6%	2.7%	3.0%	7.3%	3.7%
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024E	2025E	2026E	2027E	2028E
World Demand (Consultants)	73,840	76,630	78,460	81,720	84,640	88,090	92,040	96,220	100,750	103,940	109,940	114,540	115,070	115,320	118,390	120,710	124,410	128,360	132,150
% change	8.7%	3.8%	2.4%	4.2%	3.6%	4.1%	4.5%	4.5%	4.7%	3.2%	5.8%	4.2%	0.5%	0.2%	2.7%	2.0%	3.1%	3.2%	3.0%
Operating Rate	82.8%	82.9%	83.7%	83.7%	85.1%	86.5%	87.2%	86.2%	86.1%	86.9%	87.7%	86.2%	82.3%	78.3%	79.2%	78.6%	78.7%	75.6%	75.1%
World Demand (JPM)	73,840	76,630	78,460	81,720	84,640	88,090	92,040	96,220	100,750	103,940	109,940	114,540	115,070	115,320	118,390	120,710	121,910	125,570	129,340
% change	8.7%	3.8%	2.4%	4.2%	3.6%	4.1%	4.5%	4.5%	4.7%	3.2%	5.8%	4.2%	0.5%	0.2%	2.7%	2.0%	1.0%	3.0%	3.0%
Operating Rate	82.8%	82.9%	83.7%	83.7%	85.1%	86.5%	87.2%	86.2%	86.1%	86.9%	87.7%	86.2%	82.3%	78.3%	79.2%	78.6%	77.1%	74.0%	73.5%

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, J.P. Morgan estimates

U.S. and Canada Polyethylene Capacity Expansions

(In millions of pounds)

Company	Location	Capacity	Type	Expected Start up
Post 2024:				
CPChem/Qatar	Orange, TX	4,410	HDPE	4Q:26
Dow (Phase I)	Alberta, Canada	2,830	HDPE, LLDPE	~2029
Dow (Phase II)	Alberta, Canada	1,320	TBD	~2030
Completed				Start up Date
Dow	Freeport, TX	1,300	LLDPE	3Q:25
Bayport Polymers (Total/Borealis)	Bayport, TX	1,380	HDPE	4Q:23
Shell (last train)	Monaca, PA	1,100	LLDPE, HDPE	4Q:23-2Q:24
Novachem	Sarnia, Canada	950	LLDPE	4Q:23- Aug. 2024
Shell (first two trains)	Monaca, PA	2,200	LLDPE, HDPE	4Q:22
Gulf Coast Growth Ventures (Exxon/SABIC)	Corpus Christi, TX	2,865	LLDPE	4Q:21 – 1H:22
Formosa	Point Comfort, TX	1,765	HD/LLDPE, LDPE	3Q:19 HD/LLDPE; 4Q:20 LDPE
SASOL	Lake Charles, LA	1,960	LDPE, LLDPE	2Q:19 LLDPE; 4Q:20 LDPE
LyondellBasell	La Porte, TX	1,100	HDPE	1Q:20 - Completed
ExxonMobil	Beaumont, TX	1,435	LLDPE	3Q:19 - Completed
Dow	Taft, LA	275	HDPE	4Q:18 - Completed
Dow	Plaquemine, LA	770	LDPE	1Q:18 - Completed
Ineos-Sasol	Deer Park, TX	1,035	HDPE	4Q:17 - Completed
ExxonMobil	Mont Belvieu, TX	2,865	LLDPE	4Q:17 - Completed
Dow	Freeport, TX	880	LLDPE	4Q:17 - Completed
Chevron Phillips	Sweeny, TX	2,205	HDPE, LLDPE	3Q:17 - Completed
Novachem	Joffre, Canada	950	LLDPE	4Q:16 - Completed
U.S. and Canada Completed Total		23,735		
U.S. and Canada Polyethylene Capacity (YE 2015)		44,100		
U.S. and Canada Polyethylene Capacity (YE 2020)		59,720		
U.S. and Canada Polyethylene Capacity (YE 2024)		69,720		

Note: We define “start up” as the achievement of run rate production levels

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, J.P. Morgan Estimates

Major Ethylene/Polyethylene Reductions

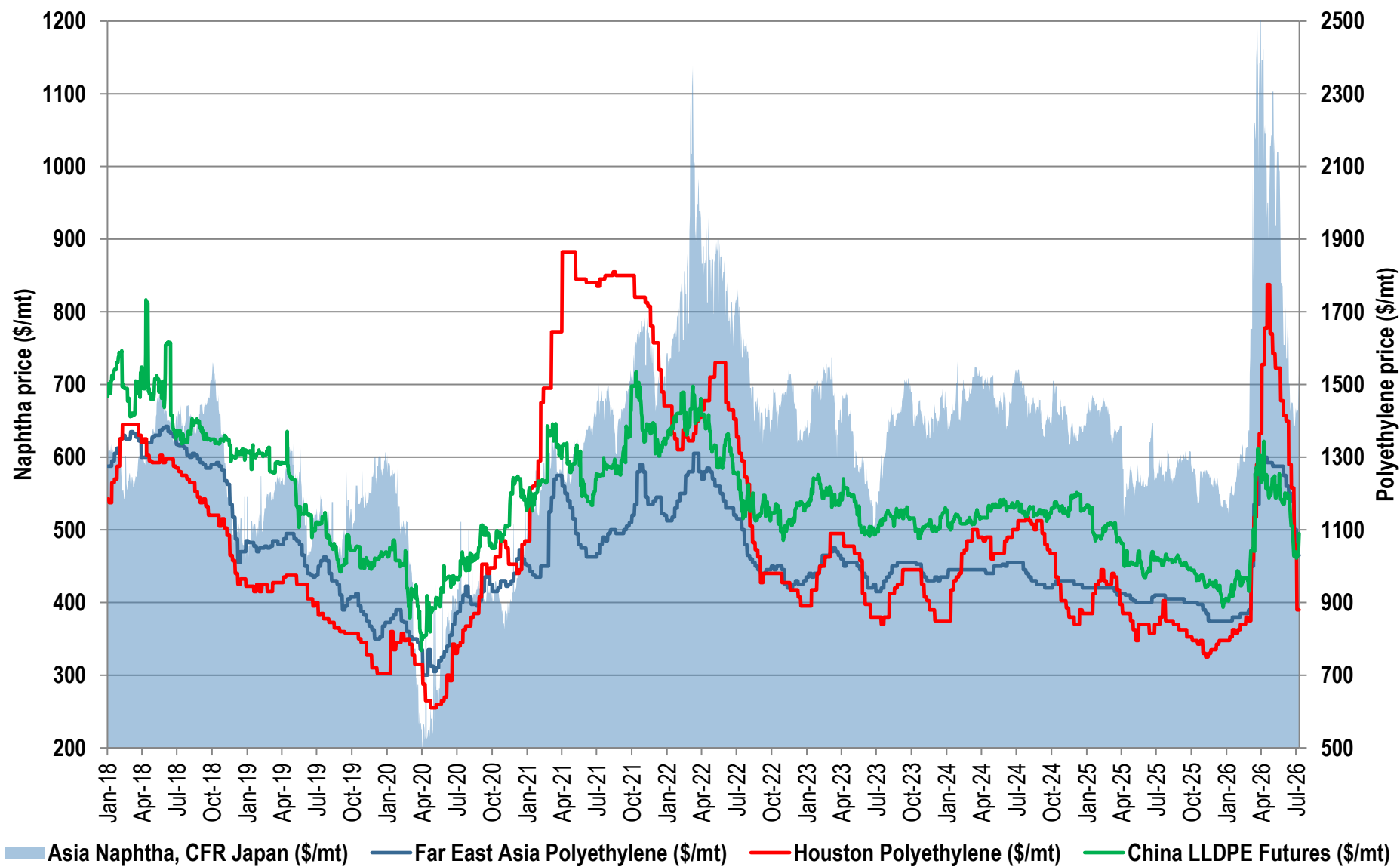
(In -000- metric tons)

Company	Location	Capacity	Product	Timing
TotalEnergies	Antwerp, Belgium	570	Ethylene	End-2027
Dow	Boehlen, Germany	610	Ethylene	4Q:27
Eni/Versalis	Ragusa, Italy	160	LDPE	2026
ExxonMobil	Fife, Scotland	830	Ethylene	Feb 2026
Eni/Versalis	Brindisi, Italy	490	Ethylene	March 2025
Eni/Versalis	Priolo, Italy	450	Ethylene	July 2025
SABIC	Wilton, UK	865	Ethylene	2025
SABIC	Geleen, Netherlands	575	Ethylene	April 2024
ExxonMobil	Gravenchon, France	400/425	Ethylene/LLDPE	2024
Europe Subtotal		4,790	Ethylene	
		585	Polyethylene	
Idemitsu Kosan	Chiba, Japan	415	Ethylene	2028
ENEOS	Kawasaki, Japan	445	Ethylene	End-2027
Maruzen Petrochemical	Chiba, Japan	525	Ethylene	1Q:26
YNCC (part of planned reduction)	Yeosu, S.Korea	470	Ethylene	1Q:26
Lotte Chemical	Pasir Gudang, Malaysia	285	Ethylene	End-2024
Qenos	Botany Bay, Australia	265/215	Ethylene/LLDPE, LLDPE	April 2024
Sinopec Zhongyuan PC	Puyang, China	290	LLDPE	Oct 2025
Asia Subtotal		2,405	Ethylene	
		505	Polyethylene	
Potential:				
South Korea planned reduction	Various	~2,700-3,700	Ethylene	TBD
China capacity over 20 y.o.	Various	~7,000	Ethylene	TBD
		~5,560	Polyethylene	TBD

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, ICIS News, J.P. Morgan Estimates

Polyethylene: Global Prices

(In \$ per metric ton, prices as of July 9th, 2026)



Middle East Polyethylene, Polypropylene, Ethylene

(in billions of pounds)

Global Capacity	PE	% total	PP	% total	Ethylene	% total
Dow	19.5	6%			36.0	7%
Lyondell	13.7	4%	11.7	4%	18.0	3%
Westlake	2.6	1%			4.8	1%
Middle East	53.1	15%	24.2	9%	83.2	16%
Gulf Cooperation Council Inside Strait of Hormuz	31.4	9%	16.7	6%		
Iran	11.2	3%	2.5	1%	19.6	4%
N America	76.8	22%	23.8	9%	117.8	22%
NE Asia	122.5	35%	152.1	55%	193.8	37%
Steam Cracker					171.9	33%
Other					21.9	4%
W Europe	25.3	7%	19.7	7%	41.2	8%
RoW	71.0	20%	59.1	21%	90.5	17%
Global (2026E)	348.7		278.9		526.4	

Approximately 15% of global PE and ethylene capacity sits in the Middle East, and about 9% of global capacity is inside the Strait of Hormuz.

North American Propane Dehydrogenation (PDH) Capacity

(In -000- metric tons)

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E	Start-Up
INVISTA (Koch) Houston, TX	658	658	658	742	825	825	825	825	825	825	825	825	825	825	2010
Dow Freeport, TX	-	188	750	750	750	750	750	750	750	750	750	750	750	750	2015
Dow (FCDh tech) Plaquemine, TX	-	-	-	-	-	-	-	-	-	115	150	150	150	150	1Q:23
Enterprise Mont Belvieu, TX	-	-	-	125	750	750	750	750	750	750	750	750	750	750	4Q:17
Inter Pipeline Redwater, Alberta	-	-	-	-	-	-	-	-	260	525	525	525	525	525	2H:22
Enterprise Mont Belvieu, TX	-	-	-	-	-	-	-	-	-	565	750	750	750	750	2Q:23
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Total N. American PDH Capacity (-000 mt)	658	846	1,408	1,617	2,325	2,325	2,325	2,325	2,585	3,530	3,750	3,750	3,750	3,750	3,750
Total N. American PDH Capacity (M lbs)	1,451	1,865	3,104	3,565	5,126	5,126	5,126	5,126	5,700	7,783	8,267	8,267	8,267	8,267	8,267
Total N.A. Est. PDH Propane Consumed (-000 bpd)	27	35	58	66	95	95	95	95	106	144	153	153	153	153	153

Propylene Production via PDH using Propane

Propane gallon/lb conversion

4.22 lbs of propane per gallon of propane

If propane is 70.4 cents/gallon => 16.7 cents/lb

Need 1.2 lb of propane per 1 lb of propylene

Cash RM costs: 16.7 cents/lb * 1.2 => 20.0 cents/lb

Add 8 cents in operating costs

Total op costs: 20 cents/lb + 9 cents/lb => 29.0 cents/lb

Propylene Price (Polymer Grade): 43.5 cents/lb

Propane/propylene cash margin => 14.5 cents/lb

Propylene Dehydrogenation Economics

Propylene Capacity 1,550 M lbs

Investment Costs:

Onsite (Reactors & P/P Tower) \$835

Offsite \$315

Fixed Capital \$1,150

Working Capital \$95

Total Capital Cost \$1,245 million

Capital Cost per Lb \$0.80 per lb of Propylene Capacity

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, and J.P. Morgan estimates

U.S. and Canada Polypropylene Expansions

(In millions of pounds)

Company	Location	Capacity	Start up
Formosa Plastics	Point Comfort, TX	550	2H:25
INVISTA	Longview, TX	100	4Q:23
ExxonMobil	Baton Rouge, LA	990	4Q:22
Heartland Polymers	Alberta, Canada	1,160	3Q:22
Braskem	La Porte, TX	990	3Q:20
U.S. and Canada Completed Total		3,140	
U.S. and Canada Polypropylene Capacity (YE 2020)		19,740	
U.S. and Canada Polypropylene Capacity (YE 2025)		22,880	

Global Polypropylene Supply/Demand

(In -000- metric tons)

	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Global Capacity (CMA by OPIS)	82,600	85,530	90,980	96,220	102,100	108,210	112,610	120,880	124,750	133,770
<i>Global incremental capacity</i>	2,720	2,930	5,450	5,240	5,880	6,110	4,400	8,270	3,870	9,020
<i>Capacity growth</i>	3.4%	3.5%	6.4%	5.8%	6.1%	6.0%	4.1%	7.3%	3.2%	7.2%
Global Demand (CMA by OPIS)	73,840	76,200	80,120	83,910	83,280	83,920	86,580	88,580	88,020	94,940
<i>Growth rate</i>	4.3%	3.2%	5.1%	4.7%	-0.8%	0.8%	3.2%	2.3%	-0.6%	7.9%
<i>Implied Operating Rate</i>	89.4%	89.1%	88.1%	87.2%	81.6%	77.6%	76.9%	73.3%	70.6%	71.0%

Note: We define "start up" as the achievement of run rate production levels

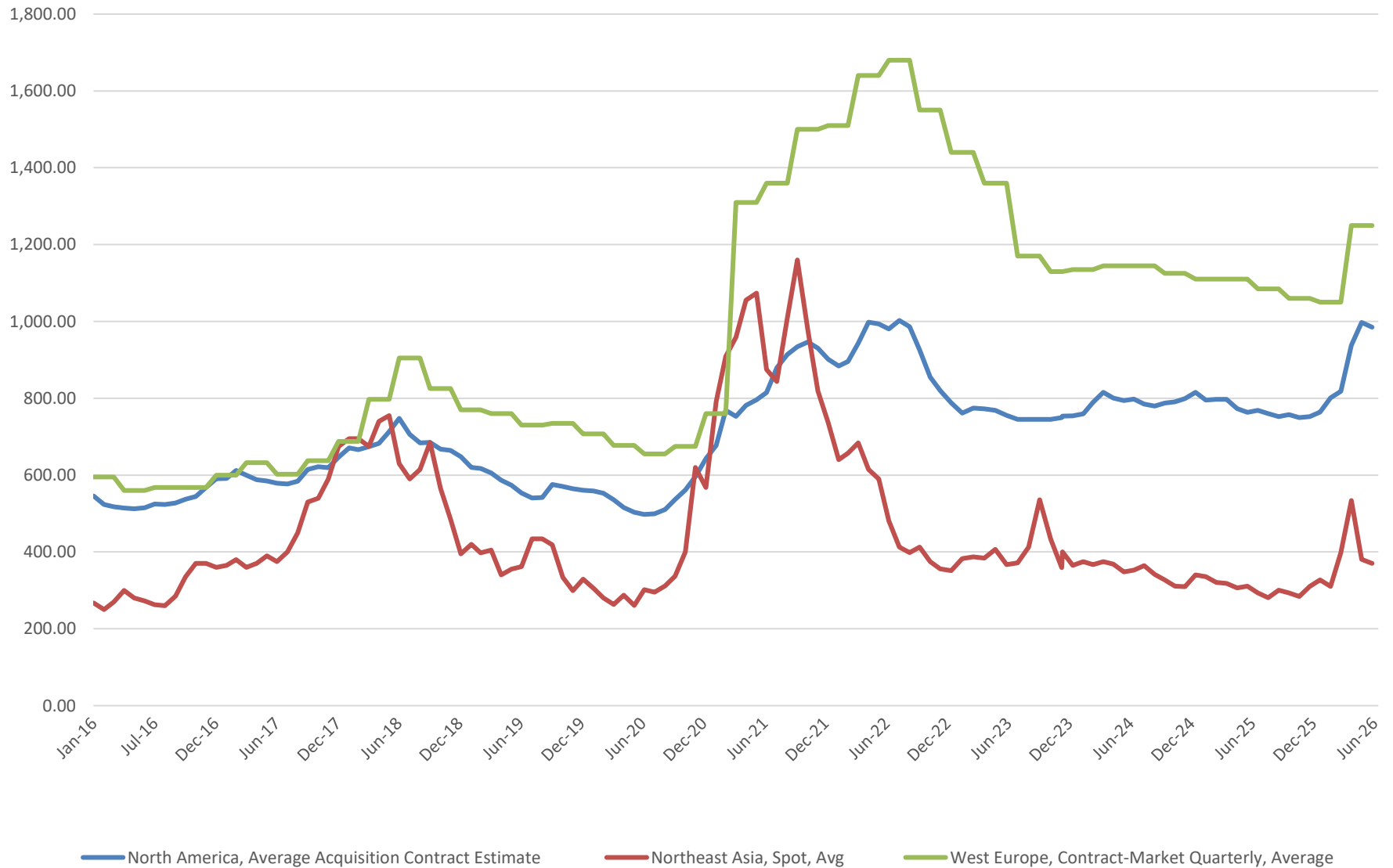
Source: Chemical Market Analytics by OPIS, a Dow Jones Company, J.P. Morgan Estimates

Global Methanol Supply/Demand

Capacity by Region	2019	2020	2021	2022	2023	2024E	2025E	2026E	2027E	2028E
United States	7,920	8,090	8,760	9,970	9,970	11,240	11,670	11,670	11,670	11,670
Canada	600	600	600	600	600	600	600	600	600	600
Mexico	200	200	200	200	200	200	200	200	200	200
Mainland China	89,100	92,360	98,590	102,180	113,000	113,000	115,400	115,400	115,400	115,400
Malaysia	2,490	2,490	2,490	2,490	2,490	3,370	4,240	4,240	4,240	4,240
New Zealand	2,420	2,420	2,420	2,420	2,420	2,420	2,420	2,420	2,420	2,420
India	850	850	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Other Southeast Asia	850	850	850	850	850	850	850	850	850	1,850
Indonesia	720	720	720	720	720	720	720	720	720	720
Australia	70	70	70	70	70	70	70	70	70	70
Europe	3,960	4,080	4,170	4,260	4,260	4,260	4,260	4,260	4,260	4,260
Russia	6,260	6,260	6,260	6,760	6,760	6,760	6,760	6,760	6,760	6,760
Africa	3,210	3,210	3,210	3,210	3,210	3,210	3,210	3,210	3,210	3,210
Saudi Arabia	7,640	7,640	7,640	7,640	7,640	7,640	7,640	7,640	7,640	7,640
Malaysia	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300	1,300
Iran	6,990	9,490	12,110	13,340	13,340	13,340	13,340	13,340	13,340	13,340
Oman	2,400	2,400	2,400	2,400	2,400	2,400	2,400	2,400	2,400	2,400
Other Middle East	440	440	440	440	440	440	440	440	440	440
South America	11,250	11,750	12,250	12,250	12,250	12,250	12,250	12,250	12,250	12,250
World Capacity	148,670	155,220	165,480	172,100	182,920	185,070	188,770	188,770	188,770	189,770
% growth		4.4%	6.6%	4.0%	6.3%	1.2%	2.0%	0.0%	0.0%	0.5%
World Demand	98,300	102,150	106,950	106,000	108,550	112,000	115,750	119,650	123,550	127,500
utilization rate	66.1%	65.8%	64.6%	61.6%	59.3%	60.5%	61.3%	63.4%	65.5%	67.2%
% growth		3.9%	4.7%	(0.9%)	2.4%	3.2%	3.3%	3.4%	3.3%	3.2%
Exports (mm tons)	Iran	Saudi Arabia	Qatar	Global						
2025E	8.9	8.5	0.9	37.0						
% of total	24%	23%	2%							
Imports (mm tons)	China	South Korea	India	Malaysia						
2025E	14.9	2.7	4.5	1.1						

Acetic Acid: Global Pricing

(Prices as of Jul 10, 2026)

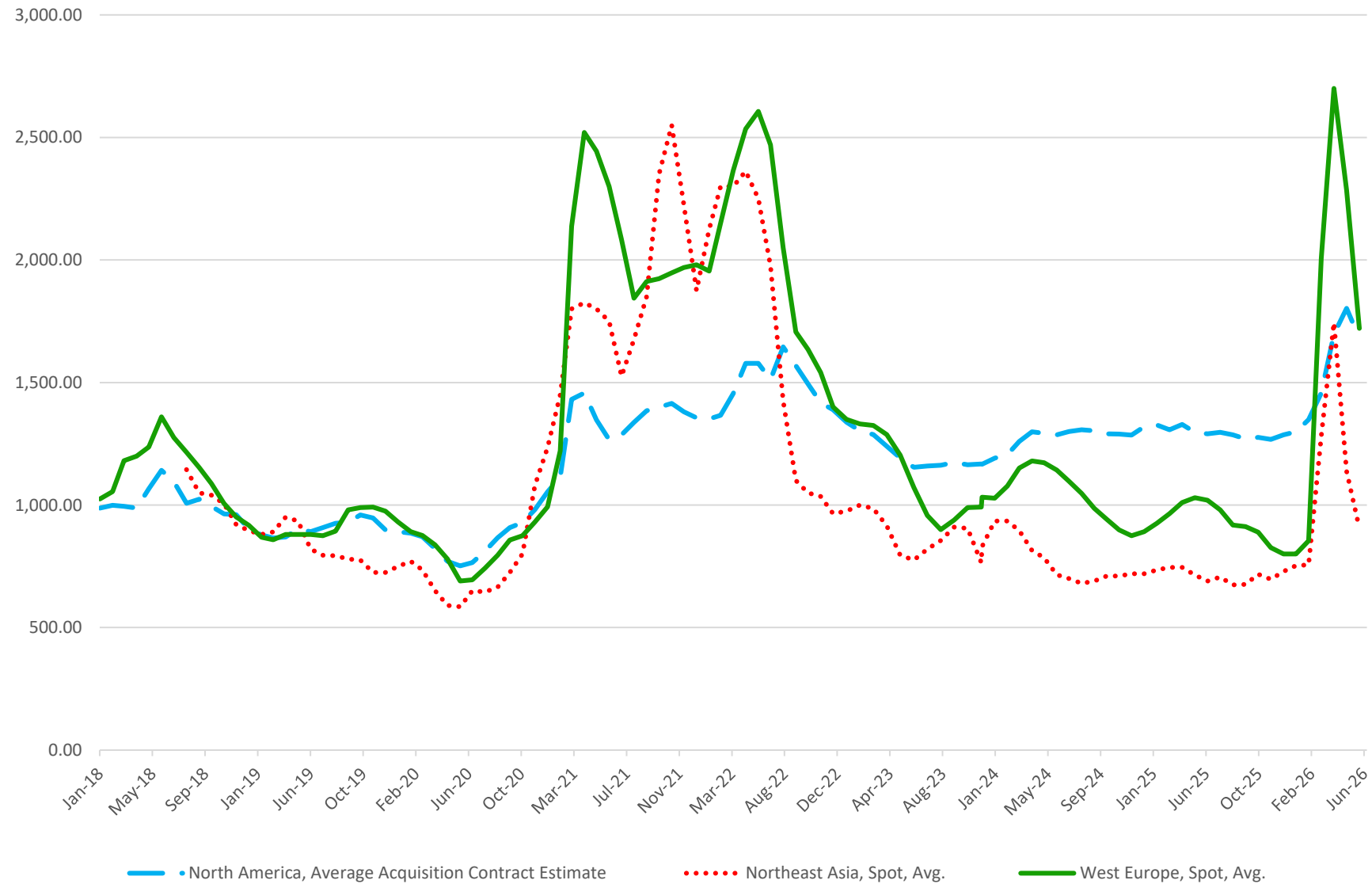


Source: Chemical Market Analytics by OPIS, a Dow Jones Company & J.P. Morgan Estimates. North America June 2026 Acetic Acid prices are estimated.

North America and Northeast Asia prices are in USD per metric ton; West Europe prices are in euro per metric ton

VAM: Global Pricing

(Prices as of Jul 10, 2026)



Source: Chemical Market Analytics by OPIS, a Dow Jones Company & J.P. Morgan Estimates. North America and Europe June 2026 VAM prices are estimated.
North America and Northeast Asia prices are in USD per metric ton; West Europe prices are in euro per metric ton.

Carbon Black Capacity

World Carbon Black Capacity												
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Unit: 000 tons per year												
World Capacity	15,710	15,845	16,525	17,050	17,770	18,125	18,375	18,990	19,375	19,920	21,025	21,530
Amt of New Capacity	50	135	680	525	720	355	250	615	385	545	1,105	505
		0.9%	4.3%	3.2%	4.2%	2.0%	1.4%	3.3%	2.0%	2.8%	5.5%	2.4%
World excluding China	8,700	8,775	9,000	9,375	9,500	9,700	9,800	10,050	10,250	10,625	11,325	11,525
North America	2,100	2,100	2,075	2,155	2,155	2,155	2,155	2,065	2,140	2,160	2,160	2,160
% growth	0.0%	0.0%	(1.2%)	3.9%	0.0%	0.0%	0.0%	(4.2%)	3.6%	0.9%	0.0%	0.0%
South America	740	740	745	755	755	755	755	755	755	755	755	755
% growth	(1.3%)	0.0%	0.7%	1.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Europe (Total)	2,230	2,255	2,365	2,425	2,500	2,525	2,525	2,535	2,620	2,720	2,780	2,780
% growth	(3.0%)	1.1%	4.9%	2.5%	3.1%	1.0%	0.0%	0.4%	3.4%	3.8%	2.2%	0.0%
European Union & UK	1,115	1,120	1,160	1,185	1,205	1,230	1,230	1,235	1,310	1,350	1,350	1,350
% growth	(7.1%)	0.4%	3.6%	2.2%	1.7%	2.1%	0.0%	0.4%	6.1%	3.1%	0.0%	0.0%
Eastern Europe	1,115	1,135	1,205	1,240	1,300	1,300	1,300	1,300	1,310	1,370	1,430	1,430
% growth	1.4%	1.8%	6.2%	2.9%	4.8%	0.0%	0.0%	0.0%	0.8%	4.6%	4.4%	0.0%
Asia	10,105	10,175	10,740	11,115	11,755	12,085	12,335	13,035	13,255	13,685	14,550	15,055
% growth	1.6%	0.7%	5.6%	3.5%	5.8%	2.8%	2.1%	5.7%	1.7%	3.2%	6.3%	3.5%
China	7,010	7,080	7,520	7,685	8,270	8,430	8,570	8,950	9,125	9,300	9,700	10,000
% growth	3.9%	1.0%	6.2%	2.2%	7.6%	1.9%	1.7%	4.4%	2.0%	1.9%	4.3%	3.1%
All Other Asia	3,095	3,095	3,220	3,425	3,485	3,655	3,765	4,085	4,130	4,385	4,850	5,055
% growth	(3.3%)	0.0%	4.0%	6.4%	1.8%	4.9%	3.0%	8.5%	1.1%	6.2%	10.6%	4.2%
Africa/Mideast	540	580	605	605	605	605	605	605	605	605	785	785
% growth	(1.8%)	7.4%	4.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	29.8%	0.0%

Carbon Black Production

World Carbon Black Production by Country/Region												
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Unit: 000 tons												
World Production	12,375	12,900	13,300	13,250	12,200	13,500	13,650	13,700	14,525	14,975	15,225	15,700
% growth		4.2%	3.1%	-0.4%	-7.9%	10.7%	1.1%	0.4%	6.0%	3.1%	1.7%	3.1%
Utilization Rate	79%	81%	80%	78%	69%	74%	74%	72%	75%	75%	72%	73%
World excl. China	7,150	7,375	7,575	7,500	6,700	7,700	7,700	7,775	7,850	8,000	8,225	8,600
North America	1,750	1,825	1,900	1,825	1,550	1,850	1,850	1,775	1,775	1,775	1,775	1,825
utilization rate	83.3%	86.9%	91.6%	84.7%	71.9%	85.8%	85.8%	86.0%	82.9%	82.2%	82.2%	84.5%
growth rate	0.0%	4.3%	4.1%	(3.9%)	(15.1%)	19.4%	0.0%	(4.1%)	0.0%	0.0%	0.0%	2.8%
South America	575	625	600	575	475	550	575	525	525	500	525	550
utilization rate	77.7%	84.5%	80.5%	76.2%	62.9%	72.8%	76.2%	69.5%	69.5%	66.2%	69.5%	72.8%
growth rate	(4.2%)	8.7%	(4.0%)	(4.2%)	(17.4%)	15.8%	4.5%	(8.7%)	0.0%	(4.8%)	5.0%	4.8%
Europe (Total)	1,925	2,000	2,125	2,100	1,875	2,175	1,950	2,025	2,125	2,125	2,125	2,200
utilization rate	86.3%	88.7%	89.9%	86.6%	75.0%	86.1%	77.2%	79.9%	81.1%	78.1%	76.4%	79.1%
growth rate	4.1%	3.9%	6.3%	(1.2%)	(10.7%)	16.0%	(10.3%)	3.8%	4.9%	0.0%	0.0%	3.5%
European Union & UK	950	975	1,000	975	850	975	925	1,025	1,125	1,125	1,100	1,125
utilization rate	85.2%	87.1%	86.2%	82.3%	70.5%	79.3%	75.2%	83.0%	85.9%	83.3%	81.5%	83.3%
growth rate	(5.0%)	2.6%	2.6%	(2.5%)	(12.8%)	14.7%	(5.1%)	10.8%	9.8%	0.0%	(2.2%)	2.3%
Eastern Europe	975	1,025	1,125	1,125	1,025	1,200	1,025	1,000	1,000	1,000	1,025	1,075
utilization rate	87.4%	90.3%	93.4%	90.7%	78.8%	92.3%	78.8%	76.9%	76.3%	73.0%	71.7%	75.2%
growth rate	8.3%	5.1%	9.8%	0.0%	(8.9%)	17.1%	(14.6%)	(2.4%)	0.0%	0.0%	2.5%	4.9%
Asia (Total)	7,725	8,050	8,200	8,325	7,925	8,550	8,850	8,950	9,725	10,200	10,300	10,550
utilization rate	76.4%	79.1%	76.4%	74.9%	67.4%	70.7%	71.7%	68.7%	73.4%	74.5%	70.8%	70.1%
growth rate	3.0%	4.2%	1.9%	1.5%	(4.8%)	7.9%	3.5%	1.1%	8.7%	4.9%	1.0%	2.4%
China	5,225	5,525	5,725	5,750	5,500	5,800	5,950	5,925	6,675	6,975	7,000	7,100
utilization rate	74.5%	78.0%	76.1%	74.8%	66.5%	68.8%	69.4%	66.2%	73.2%	75.0%	72.2%	71.0%
growth rate	4.5%	5.7%	3.6%	0.4%	(4.3%)	5.5%	2.6%	(0.4%)	12.7%	4.5%	0.4%	1.4%
All Other Asia	2,500	2,525	2,475	2,575	2,425	2,750	2,900	3,025	3,050	3,225	3,300	3,450
utilization rate	80.8%	81.6%	76.9%	75.2%	69.6%	75.2%	77.0%	74.1%	73.8%	73.5%	68.0%	68.2%
growth rate	0.0%	1.0%	(2.0%)	4.0%	(5.8%)	13.4%	5.5%	4.3%	0.8%	5.7%	2.3%	4.5%
Africa/Mideast	400	400	450	425	375	400	425	425	400	400	475	550
utilization rate	74.1%	69.0%	74.4%	70.2%	62.0%	66.1%	70.2%	70.2%	66.1%	66.1%	60.5%	70.1%
growth rate	0.0%	0.0%	12.5%	(5.6%)	(11.8%)	6.7%	6.3%	0.0%	(5.9%)	0.0%	18.8%	15.8%

Carbon Black Demand

World Carbon Black Demand by Country/Region

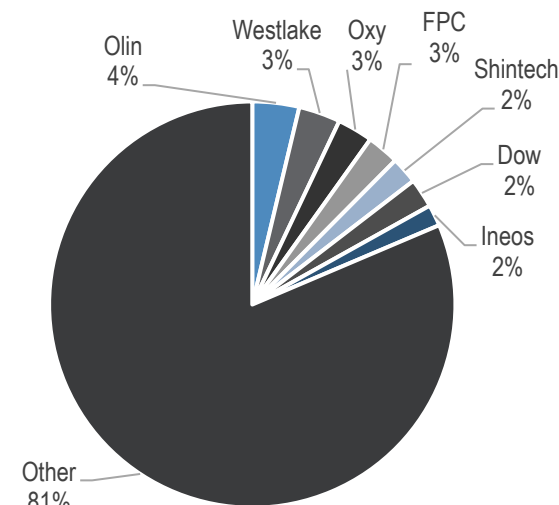
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Unit: 000 tons												
World Consumption	12,725	13,275	13,650	13,600	12,550	13,850	13,950	14,025	14,725	14,900	15,250	15,700
% growth		4.3%	2.8%	(0.4%)	(7.7%)	10.4%	0.7%	0.5%	5.0%	1.2%	2.3%	3.0%
World excl. China	8,150	8,375	8,725	8,600	7,625	8,700	8,700	8,550	8,925	8,900	9,075	11,050
North America	1,775	1,825	1,875	1,850	1,550	1,825	1,850	1,750	1,725	1,700	1,725	1,750
growth rate	(1.4%)	2.8%	2.7%	(1.3%)	(16.2%)	17.7%	1.4%	(5.4%)	(1.4%)	(1.4%)	1.5%	1.4%
South America	625	650	625	650	525	625	625	575	575	600	600	625
growth rate	(3.8%)	4.0%	(3.8%)	4.0%	(19.2%)	19.0%	0.0%	(8.0%)	0.0%	4.3%	0.0%	4.2%
Europe Total	2,000	2,050	2,150	2,050	1,800	2,025	1,975	1,900	1,975	1,900	1,900	1,925
growth rate	2.6%	2.5%	4.9%	(4.7%)	(12.2%)	12.5%	(2.5%)	(3.8%)	3.9%	(3.8%)	0.0%	1.3%
European Union & UK	1,600	1,650	1,750	1,650	1,425	1,625	1,600	1,475	1,500	1,450	1,425	1,450
growth rate	0.0%	3.1%	6.1%	(5.7%)	(13.6%)	14.0%	(1.5%)	(7.8%)	1.7%	(3.3%)	(1.7%)	1.8%
Eastern Europe	400	400	400	400	375	400	375	425	475	475	475	475
growth rate	14.3%	0.0%	0.0%	0.0%	(6.3%)	6.7%	(6.3%)	13.3%	11.8%	0.0%	0.0%	0.0%
Asia Total	7,950	8,375	8,600	8,650	8,300	8,950	9,050	9,350	10,025	10,275	10,575	10,925
growth rate	3.9%	5.3%	2.7%	0.6%	(4.0%)	7.8%	1.1%	3.3%	7.2%	2.5%	2.9%	3.3%
China	4,575	4,900	4,925	5,000	4,925	5,175	5,250	5,400	5,800	6,000	6,200	6,400
growth rate	5.2%	7.1%	0.5%	1.5%	(1.5%)	5.1%	1.4%	2.9%	7.4%	3.4%	3.3%	3.2%
All other Asia	3,375	3,475	3,650	3,650	3,400	3,800	3,800	3,950	4,225	4,275	4,375	4,525
growth rate	2.3%	3.0%	5.0%	0.0%	(6.8%)	11.8%	0.0%	3.9%	7.0%	1.2%	2.3%	3.4%
Africa/Mideast	375	375	400	400	375	425	450	450	425	425	450	475
growth rate	7.1%	0.0%	6.7%	0.0%	(6.3%)	13.3%	5.9%	0.0%	(5.6%)	0.0%	5.9%	5.6%

Caustic Soda Capacity Expansions – Growth Offshore

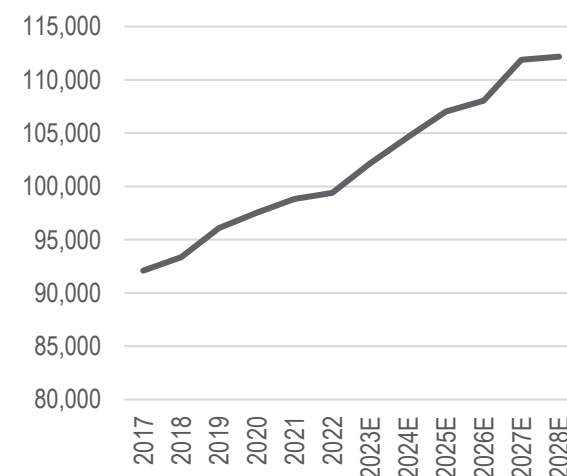
(Capacity in -000- metric tons.)

Region/Company	2020	2021	2022	2023	2024E	2025E	2026E	2027E	2028E
Canada	560	560	560	560	560	560	560	560	560
Mexico	485	485	485	525	610	620	620	620	525
United States									
Olin	5,610	5,235	4,800	4,530	4,530	4,530	4,030	4,030	4,030
Oxy	3,175	3,105	2,985	2,985	2,985	2,985	3,105	3,450	3,450
Westlake	2,865	2,865	2,865	2,865	3,085	3,140	2,730	2,730	2,730
Shintech	1,260	1,305	1,530	1,530	1,595	1,920	1,920	1,920	1,920
FPC USA	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Other	1,230	1,230	1,230	1,230	1,230	1,230	1,185	1,170	1,170
Total - United States	15,140	14,740	14,410	14,140	14,425	14,805	13,970	14,300	14,300
U.S. Incremental capacity	0	-400	-330	-270	285	380	-835	330	0
Capacity Growth	0.0%	-2.6%	-2.2%	-1.9%	2.0%	2.6%	-5.6%	2.4%	0.0%
South America	2,555	2,535	2,535	2,565	2,585	2,645	2,275	2,280	2,295
West Europe	10,525	10,690	10,695	10,810	10,835	10,680	10,635	10,635	10,635
Central Europe	1,225	1,225	1,225	1,225	1,225	1,225	1,280	1,280	1,280
CIS & Baltic States	1,960	1,960	1,960	1,975	2,050	2,050	2,145	2,145	2,145
Middle East	3,330	3,320	3,325	3,275	3,255	3,210	3,210	3,225	3,415
Africa	1,060	1,060	1,060	1,060	1,060	1,060	1,060	1,060	1,060
Indian Subcontinent	5,125	5,425	5,730	6,200	6,940	7,195	7,195	8,970	9,110
N.E. Asia (China, Japan, Korea, Taiwan)	52,660	53,885	54,465	56,780	58,000	59,855	61,720	63,035	63,035
N.E. Asia incremental capacity	935	1,225	580	2,315	1,220	1,855	1,865	1,315	0
Capacity growth	1.8%	2.3%	1.1%	4.3%	2.1%	3.2%	3.1%	2.1%	0.0%
Southeast Asia	2,915	2,945	2,945	3,025	3,070	3,125	3,370	3,770	3,810
TOTAL – World (CMA by OPIS/JPM)	97,540	98,830	99,395	102,140	104,615	107,030	108,040	111,880	112,170
Global incremental capacity	1,465	1,290	565	2,745	2,475	2,415	1,010	3,840	290
Capacity growth	1.5%	1.3%	0.6%	2.8%	2.4%	2.3%	0.9%	3.6%	0.3%
Global Demand (CMA by OPIS/JPM)	77,870	80,565	79,010	79,915	82,200	83,895	84,605	86,435	87,855
Growth rate	0.0%	3.5%	-1.9%	1.1%	2.9%	2.1%	0.8%	2.2%	1.6%
Implied Operating Rate	79.8%	81.5%	79.5%	78.2%	78.6%	78.4%	78.3%	77.3%	78.3%

% of Global Caustic Soda Capacity (2026E)



Global Caustic Soda Capacity

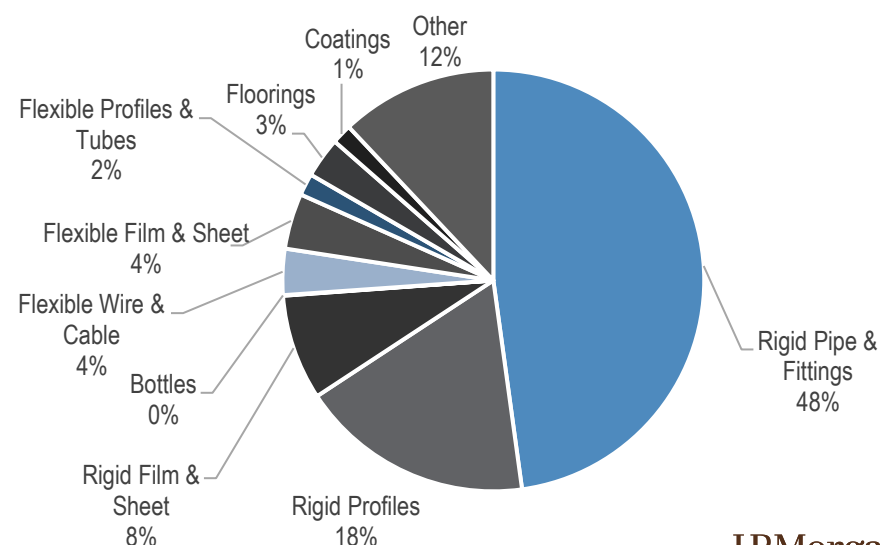
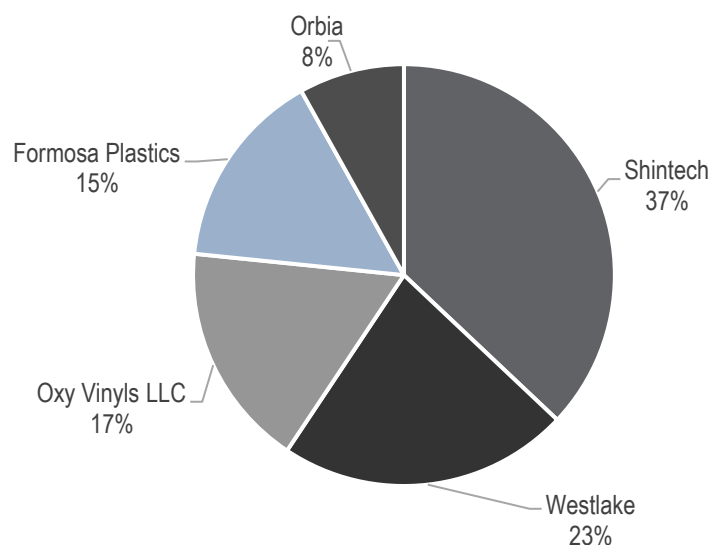


Source: Chemical Market Analytics by OPIS, a Dow Jones Company and J.P. Morgan estimates.
Note: Westlake and Olin capacity are pro forma for the transactions.

North America PVC Supply/Demand

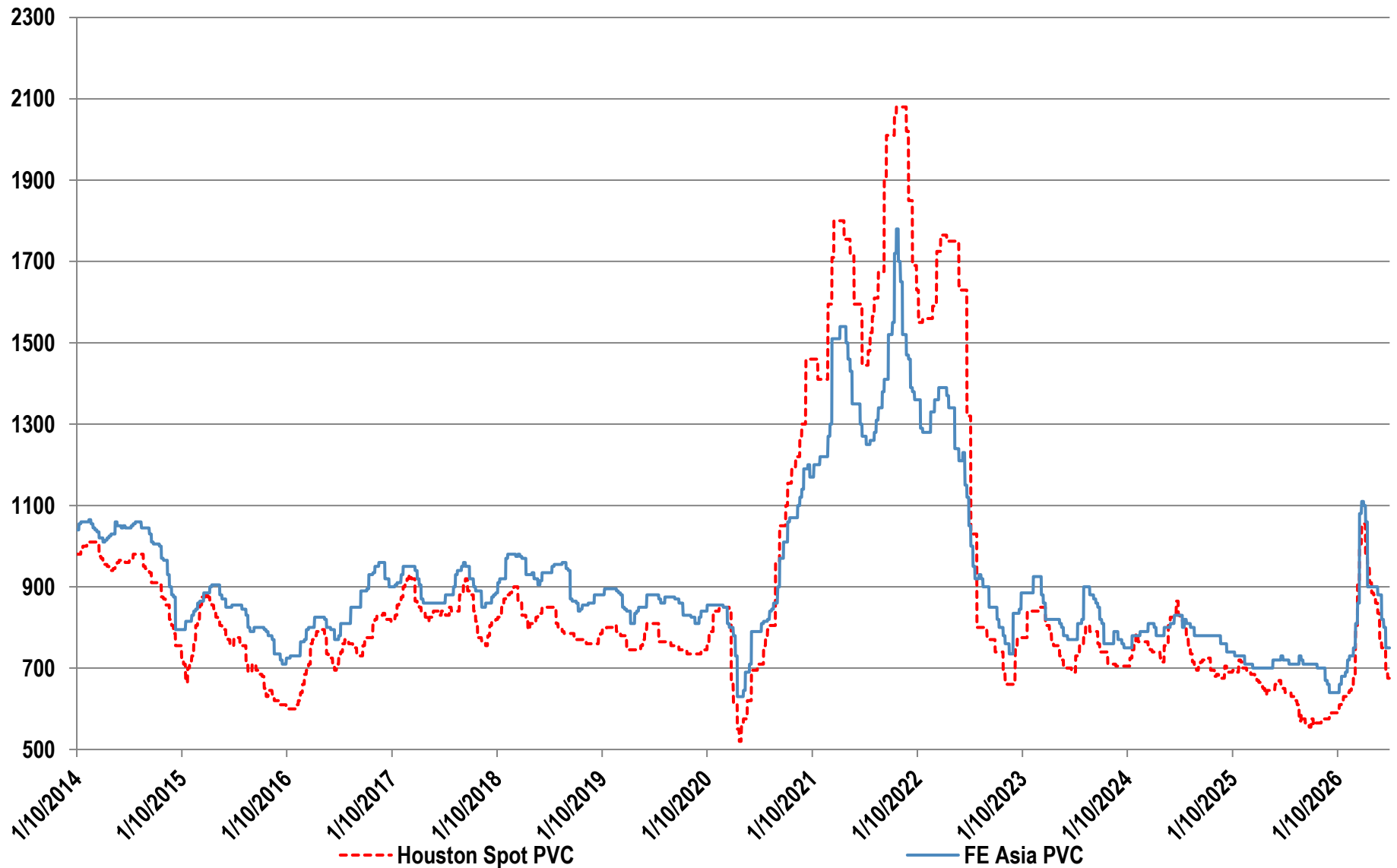
(In millions of pounds.)

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E	2028E
Certain-Teed	480	480	480	480	480	480	480										
Formosa Plastics	3,135	3,135	3,185	3,185	3,185	3,185	3,150	3,040	3,040	3,040	3,040	3,040	3,140	3,285	3,285	3,285	3,285
Axiall	2,655	2,655	2,655	2,655	1,770												
Orbia	1,200	1,350	1,910	1,980	1,980	1,980	1,980	1,980	1,980	1,980	1,980	1,980	1,980	1,740	1,740	1,740	1,740
Oxy Vinyls LLC	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710	3,710
PolyOne	255	105															
Shintech	5,820	5,820	5,820	5,820	6,070	6,485	6,485	6,485	6,485	6,485	7,120	7,120	7,225	7,960	7,960	7,960	7,960
Westlake	1,700	1,700	1,795	1,895	2,780	4,650	5,125	5,125	5,540	5,645	5,795	5,795	5,795	5,795	4,795	4,795	4,795
Total NAM Nameplate Capacity	18,955	18,955	19,555	19,725	19,975	20,490	20,930	20,340	20,755	20,860	21,645	21,645	21,850	22,490	21,490	21,490	21,490
<i>Growth</i>	<i>0.2%</i>	<i>0.0%</i>	<i>3.2%</i>	<i>0.9%</i>	<i>1.3%</i>	<i>2.6%</i>	<i>2.1%</i>	<i>-2.8%</i>	<i>2.0%</i>	<i>0.5%</i>	<i>3.8%</i>	<i>0.0%</i>	<i>0.9%</i>	<i>2.9%</i>	<i>-4.4%</i>	<i>0.0%</i>	<i>0.0%</i>
Domestic Consumption	10,825	11,085	11,360	11,515	11,900	12,630	12,540	12,090	12,560	14,020	13,170	11,530	12,595	12,470	12,350	12,635	12,890
Exports	7,740	7,540	6,950	6,825	7,180	7,090	7,715	7,805	6,615	5,350	6,220	7,790	7,845	8,345	7,915	8,075	8,235
Total NAM Demand	18,565	18,625	18,310	18,340	19,080	19,720	20,260	19,895	19,175	19,370	19,390	19,320	20,405	20,815	20,265	20,710	21,125
<i>Growth</i>	<i>6.0%</i>	<i>0.3%</i>	<i>(1.7%)</i>	<i>0.2%</i>	<i>4.0%</i>	<i>3.4%</i>	<i>2.7%</i>	<i>(1.8%)</i>	<i>(3.6%)</i>	<i>1.0%</i>	<i>0.1%</i>	<i>(0.4%)</i>	<i>5.6%</i>	<i>2.0%</i>	<i>(2.6%)</i>	<i>2.2%</i>	<i>2.0%</i>
Imports	2,045	2,085	2,070	2,070	2,090	2,140	2,240	2,145	2,190	2,500	2,495	2,350	2,330	2,660	2,585	2,650	2,650
Operating Rate	87.2%	87.3%	83.0%	82.5%	85.1%	85.8%	86.1%	87.3%	81.8%	80.9%	78.1%	78.4%	82.7%	80.7%	82.3%	84.0%	86.0%



PVC: U.S. vs. Far East Asia Spot Pricing

(In \$ per metric ton, prices as of July 9th, 2026)



Source: Bloomberg Finance L.P. & J.P. Morgan Estimates

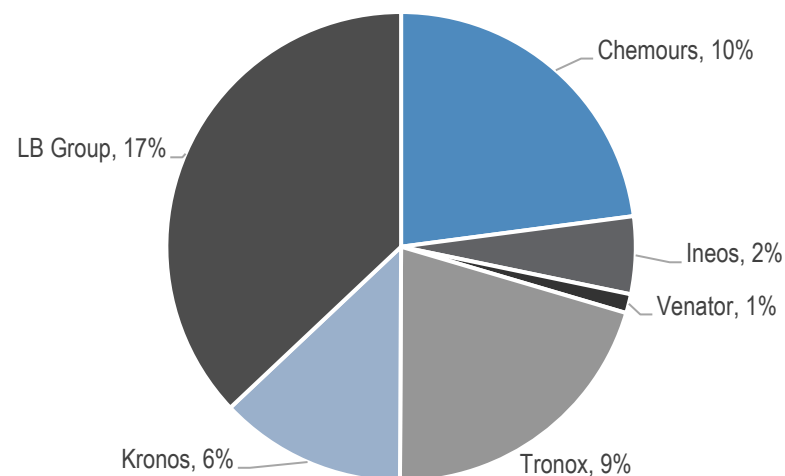
Global TiO2 Nameplate Capacity (2026E)

(In thousands of metric tons.)

Company	Capacity	%
Chemours	1,050	10%
Ineos	245	2%
Venator	60	1%
Tronox	940	9%
Kronos	590	6%
LB Group	1,700	17%
2026E Nameplate Capacity (large company)	4,590	47%
Others	945	9%
China ex LB Group	4,560	44%
2026E Nameplate Capacity	10,095	

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, Company reports and J.P. Morgan estimates

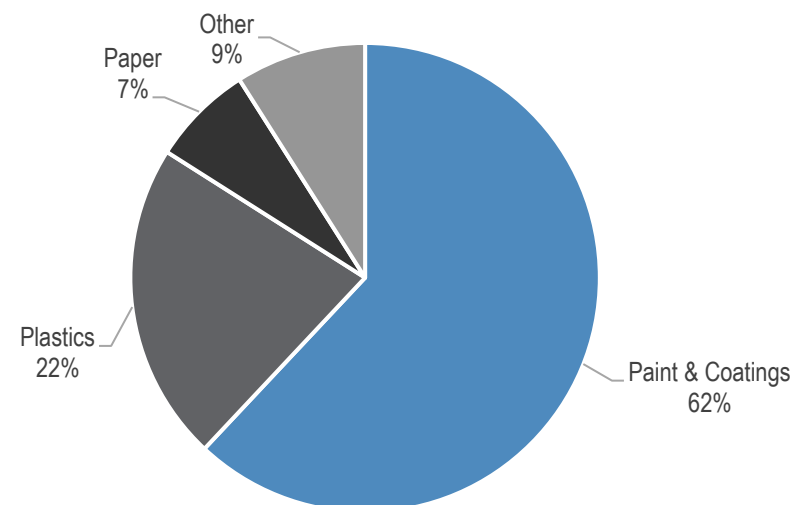
% of Global TiO2 Capacity Large Producers (2026E)



TiO2 End Markets

TiO2 End Markets	%
Paint & Coatings	62%
Plastics	22%
Paper	7%
Other	9%

Source: Chemical Market Analytics by OPIS, a Dow Jones Company.



TiO2 Supply/Demand Balance

(In thousands of metric tons.)

	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Chemours	1,230	1,210	1,210	1,210	1,210	1,210	1,210	1,210	1,130	1,050	1,050	1,050	1,050	1,050
Ineos	0	0	0	0	245	245	245	245	245	245	245	245	245	245
Venator*	820	820	800	675	675	650	600	600	580	485	395	60	0	0
Tronox	1,300	1,315	1,325	1,325	1,160	1,080	1,080	1,080	1,080	1,080	1,005	940	940	940
Kronos	545	560	560	560	560	560	555	555	555	565	590	590	590	590
LB Group	260	560	600	600	730	970	1,010	1,215	1,410	1,510	1,510	1,700	1,700	1,700
China Chloride-Based	210	210	350	350	600	640	680	1,030	1,130	1,130	1,130	1,170	1,170	1,170
China Sulfate-Based/other	3,095	3,135	3,300	3,300	3,310	3,360	3,535	3,770	4,030	4,115	4,405	4,940	5,330	5,330
Total China Capacity	3,305	3,345	3,650	3,650	3,910	4,000	4,215	4,800	5,160	5,245	5,535	6,110	6,500	6,500
Other Producers	700	700	685	685	685	685	685	685	710	750	770	945	895	935
Total Nameplate Capacity (JPMe)	7,905	7,955	8,225	8,105	8,445	8,430	8,590	9,175	9,460	9,420	9,590	10,095	10,375	10,415
Capacity Growth %		0.6%	3.4%	(1.5%)	4.2%	(0.2%)	1.9%	6.8%	3.1%	(0.4%)	1.8%	5.3%	2.8%	0.4%
Demand (JPMe)	5,500	5,700	6,100	5,900	6,100	6,400	7,300	6,800	6,800	7,300	7,300	7,300	7,445	7,595
Growth %		3.6%	7.0%	(3.3%)	3.4%	4.9%	14.1%	(6.8%)	0.0%	7.4%	0.0%	0.0%	2.0%	2.0%
<i>Global Utilization Rate</i>	69.6%	71.7%	74.2%	72.8%	72.2%	75.9%	85.0%	74.1%	71.9%	77.5%	76.1%	72.3%	71.8%	72.9%

Source: Chemical Market Analytics by OPIS, a Dow Jones Company, S&P Global Commodity Insights, company presentations & J.P. Morgan Estimates.

Note: Venator's historical capacity includes acquired Rockwood assets. Assumes the closure of Malaysian asset and sale of other assets

China Titanium Dioxide Trade Balance

(In thousands of metric tons.)

Exports	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
January	48.2	52.0	59.3	69.1	69.9	102.9	106.7	141.4	131.6	159.7	158.5	183.8
February	42.1	44.7	52.1	70.1	64.6	71.2	94.4	101.4	154.2	133.6	157.5	151.5
March	46.9	61.0	68.7	103.6	98.1	157.1	126.9	145.0	151.3	196.0	185.0	201.5
April	43.6	55.7	64.8	87.0	77.0	88.1	107.2	118.6	128.9	157.4	148.0	193.5
May	47.3	71.2	72.8	79.9	88.5	66.5	104.8	115.7	144.8	149.3	135.6	152.8
June	44.2	63.9	78.5	83.0	83.6	77.3	100.5	124.4	125.6	176.4	131.9	
July	46.5	63.6	68.8	70.2	88.6	97.5	87.5	125.8	135.9	159.6	134.7	
August	38.4	65.4	71.2	75.7	87.3	117.1	102.6	104.1	137.5	160.7	138.7	
September	40.4	63.1	62.4	74.0	83.8	111.8	108.0	96.3	145.0	146.8	156.3	
October	45.5	54.6	67.6	63.0	79.4	110.2	115.8	99.1	118.8	154.3	146.4	
November	39.7	63.2	76.3	70.0	81.9	112.3	123.6	107.2	124.9	149.3	152.2	
December	55.6	62.0	88.1	62.4	100.7	102.0	133.7	127.0	143.3	158.4	171.8	
Total YTD	538.4	720.5	830.9	908.0	1003.4	1214.1	1311.6	1405.9	1641.8	1901.5	1816.9	883.1
% change	(3%)	34%	15%	9%	11%	21%	8%	7%	17%	16%	(4%)	13%

Imports	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
January	13.7	13.5	17.9	20.1	12.4	7.5	15.9	12.0	3.6	6.6	7.3	6.1
February	13.4	13.0	18.6	8.6	8.4	17.7	14.5	12.0	4.8	7.2	5.2	3.5
March	14.3	13.7	17.6	17.0	11.6	12.2	16.2	12.9	4.7	10.5	8.3	5.9
April	20.3	13.1	18.7	19.1	13.2	12.3	18.7	15.4	5.5	6.8	6.5	5.1
May	19.4	15.9	20.4	18.3	13.7	14.0	15.8	15.0	6.4	9.2	7.5	5.3
June	18.3	19.3	21.3	17.6	15.9	13.9	16.3	12.7	7.1	7.5	4.9	
July	22.1	19.9	17.9	17.4	15.1	14.6	17.5	11.1	8.0	9.1	7.3	
August	19.0	13.2	17.0	15.9	15.2	12.3	15.1	12.4	9.2	6.4	3.4	
September	21.1	14.8	15.6	15.9	11.2	14.6	17.0	3.9	6.8	8.3	6.4	
October	12.7	13.0	18.7	16.5	18.2	16.6	12.2	3.3	7.8	6.4	5.6	
November	16.0	18.5	16.6	18.2	14.8	16.1	17.0	5.7	9.3	6.7	5.3	
December	13.5	21.3	14.6	12.8	17.1	16.5	15.8	6.9	11.2	7.3	6.7	
Total YTD	203.7	189.3	215.0	197.5	166.9	168.2	191.9	123.1	84.5	92.0	74.5	25.9
% change	(6%)	(7%)	14%	(8%)	(16%)	1%	14%	(36%)	(31%)	9%	(19%)	(26%)

Imports	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Net exports/ (imports)	334.8	531.2	615.9	710.5	836.5	1045.8	1119.7	1282.8	1557.3	1809.5	1742.4	857.2
Net change	(2.1)	196.4	84.7	94.6	126.0	209.3	73.9	163.1	274.5	252.2	(67.1)	107.4
% change	(0.6%)	58.7%	16.0%	15.4%	17.7%	25.0%	7.1%	14.6%	21.4%	16.2%	(3.7%)	14.3%

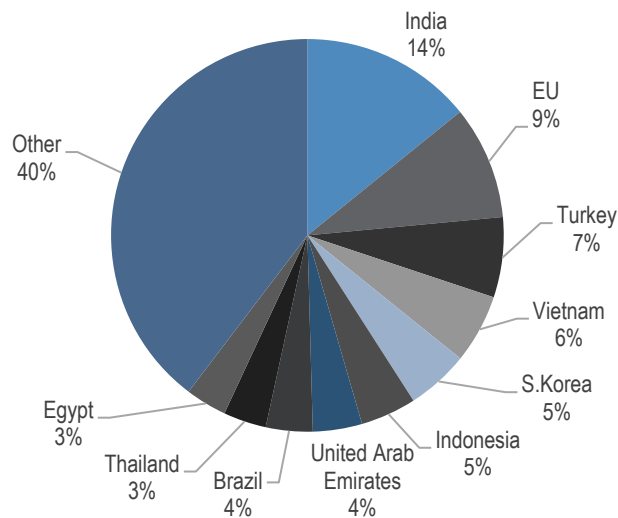
Source: China General Administration of Customs.

Anti-Dumping TiO2 Duties on Chinese Companies

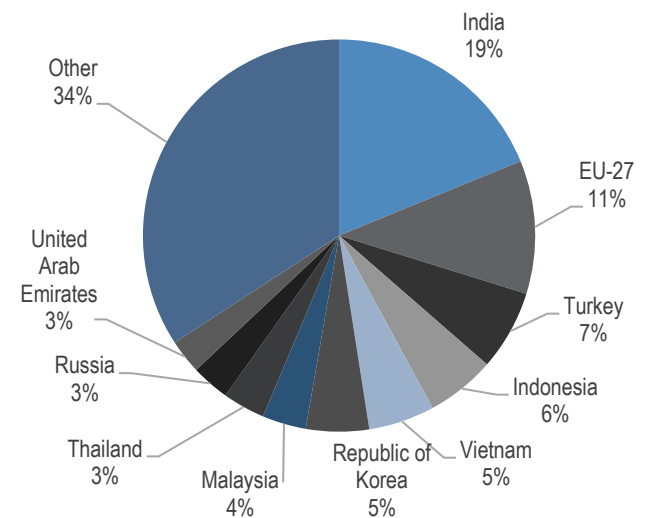
(\$ per metric ton, unless otherwise stated)

	EU Definitive	India Definitive	Brazil Preliminary	Saudi Arabia
<i>Effective:</i>	<i>Final Jan-25</i>	<i>Final May-25; halted Sep-25</i>	<i>Expired Apr-25, Final Oct-25</i>	<i>Final Oct-25</i>
Duties (per ton):				
LB Group	€ 740	\$460	\$1,159	31% (min \$773)
Anhui Gold Star Group	€ 250	\$609	\$1,149	29% (min \$667)
Shandong Lubei Group	€ 640	\$563	\$1,224	19% (min \$427)
Pangang	€ 640	\$510	\$1,224	37% (min \$827)
Yibin	€ 640	\$620	\$1,224	37% (min \$827)
Other Cooperating Cos.	€ 640	\$510	\$1,224	45% (min \$1,173)
Other Non-cooperating Cos.	€ 740	\$681	\$1,268	45% (min \$1,173)

2025 China Exports by Country:



2026YTD (May) China Exports by Country:



US Single-Family Housing Starts and Permits

(In million units, annualized)

U.S. Single-Family Housing Starts

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	0.36	0.51	0.43	0.52	0.60	0.57	0.70	0.75	0.79	0.87	0.92	0.97	1.13	1.17	0.83	1.21	1.00	0.89	(10.8%)
February	0.36	0.53	0.39	0.47	0.65	0.59	0.58	0.86	0.86	0.89	0.78	1.03	1.04	1.19	0.82	1.14	1.09	0.92	(5.6%)
March	0.35	0.54	0.43	0.47	0.62	0.65	0.63	0.75	0.84	0.88	0.84	0.88	1.21	1.17	0.83	1.03	0.94	1.02	8.3%
April	0.39	0.57	0.42	0.50	0.59	0.65	0.74	0.78	0.83	0.91	0.87	0.69	1.06	1.20	0.87	1.05	0.95	0.90	(5.7%)
May	0.41	0.46	0.41	0.52	0.61	0.65	0.71	0.75	0.83	0.97	0.84	0.74	1.09	1.05	1.00	1.00	0.95	0.88	(6.7%)
June	0.48	0.45	0.44	0.52	0.61	0.60	0.70	0.77	0.87	0.87	0.88	0.90	1.17	1.01	0.94	0.98	0.93	Release	7/17/26
July	0.51	0.43	0.43	0.52	0.59	0.66	0.76	0.77	0.84	0.87	0.89	1.01	1.15	0.91	1.01	0.88	0.96		
August	0.49	0.42	0.42	0.54	0.63	0.64	0.73	0.73	0.87	0.90	0.91	1.04	1.11	0.93	0.96	1.02	0.87		
September	0.51	0.45	0.42	0.59	0.58	0.66	0.75	0.78	0.84	0.88	0.93	1.11	1.10	0.89	0.98	1.05	0.84		
October	0.48	0.44	0.44	0.61	0.60	0.70	0.71	0.86	0.88	0.87	0.89	1.19	1.11	0.88	0.99	0.96	0.89		
November	0.50	0.45	0.47	0.57	0.70	0.66	0.77	0.82	0.92	0.78	0.92	1.17	1.21	0.79	1.12	1.01	0.93		
December	0.48	0.43	0.52	0.62	0.65	0.72	0.77	0.80	0.82	0.78	1.00	1.27	1.17	0.85	1.02	1.06	0.95		
Average	0.44	0.47	0.43	0.54	0.62	0.65	0.71	0.79	0.85	0.87	0.89	1.00	1.13	1.00	0.95	1.02	0.94	0.92 YTD	
y/y Δ	(28.2%)	6.6%	(7.9%)	23.6%	15.3%	4.4%	10.2%	10.2%	8.1%	2.7%	2.0%	12.6%	12.9%	(11.1%)	(5.6%)	7.2%	(7.3%)	(6.4%) YTD	

U.S. Single-Family Housing Permits

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	0.55	0.64	0.58	0.71	0.94	0.98	1.05	1.17	1.30	1.32	1.25	1.49	1.86	1.92	1.42	1.55	1.46	1.39	(4.8%)
February	0.56	0.65	0.54	0.74	0.98	1.04	1.13	1.17	1.23	1.32	1.29	1.46	1.70	1.85	1.61	1.57	1.45	1.54	6.5%
March	0.51	0.69	0.58	0.81	0.94	1.07	1.07	1.12	1.27	1.39	1.30	1.35	1.76	1.90	1.47	1.48	1.49	1.36	(8.6%)
April	0.52	0.64	0.58	0.73	1.01	1.09	1.17	1.16	1.25	1.38	1.32	1.08	1.71	1.85	1.48	1.46	1.45	1.42	(1.5%)
May	0.56	0.58	0.62	0.80	1.00	1.02	1.27	1.21	1.21	1.34	1.35	1.25	1.71	1.71	1.58	1.41	1.42	1.41	(0.4%)
June	0.60	0.59	0.64	0.79	0.94	1.01	1.38	1.21	1.35	1.35	1.30	1.33	1.66	1.73	1.45	1.46	1.40	Release	7/17/26
July	0.60	0.58	0.62	0.85	1.00	1.08	1.14	1.20	1.26	1.33	1.38	1.54	1.67	1.71	1.53	1.45	1.40		
August	0.62	0.58	0.65	0.84	0.96	1.05	1.17	1.20	1.32	1.27	1.48	1.53	1.74	1.57	1.60	1.49	1.35		
September	0.61	0.56	0.61	0.93	1.00	1.08	1.14	1.30	1.27	1.31	1.47	1.65	1.65	1.59	1.53	1.42	1.44		
October	0.58	0.56	0.67	0.89	1.04	1.09	1.16	1.25	1.36	1.27	1.52	1.61	1.70	1.58	1.52	1.44	1.42		
November	0.62	0.56	0.71	0.92	1.03	1.06	1.24	1.23	1.30	1.34	1.54	1.69	1.77	1.42	1.51	1.51	1.41		
December	0.66	0.63	0.70	0.94	1.01	1.07	1.21	1.25	1.31	1.32	1.44	1.76	1.89	1.40	1.50	1.47	1.48		
Average	0.58	0.60	0.62	0.83	0.99	1.05	1.18	1.21	1.29	1.33	1.39	1.48	1.73	1.68	1.52	1.47	1.43	1.43 YTD	
y/y Δ	(35.0%)	3.7%	3.4%	32.8%	19.3%	6.5%	11.9%	2.4%	6.7%	3.2%	4.4%	6.7%	17.2%	(2.8%)	(10.0%)	(2.8%)	(3.0%)	(1.8%) YTD	

Source: U.S. Census Bureau, Bloomberg Finance L.P.

US Multi-Family and Total Housing Starts

(In million units, annualized)

U.S. Multi-Family Housing Starts

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	0.13	0.10	0.20	0.21	0.28	0.32	0.38	0.34	0.41	0.43	0.31	0.61	0.49	0.52	0.53	0.36	0.35	0.49	39.9%
February	0.22	0.08	0.13	0.24	0.31	0.35	0.30	0.37	0.41	0.38	0.34	0.52	0.36	0.54	0.58	0.42	0.40	0.42	6.5%
March	0.15	0.09	0.17	0.22	0.39	0.32	0.33	0.36	0.35	0.44	0.35	0.39	0.48	0.53	0.54	0.27	0.41	0.51	24.1%
April	0.09	0.12	0.14	0.25	0.25	0.40	0.45	0.39	0.32	0.37	0.40	0.25	0.43	0.61	0.47	0.33	0.45	0.49	10.3%
May	0.13	0.13	0.15	0.19	0.32	0.36	0.37	0.40	0.33	0.39	0.45	0.30	0.51	0.49	0.61	0.33	0.34	0.30	(14.2%)
June	0.10	0.09	0.17	0.23	0.23	0.31	0.51	0.43	0.38	0.33	0.36	0.36	0.49	0.54	0.48	0.34	0.45	Release	7/17/26
July	0.09	0.12	0.19	0.22	0.29	0.43	0.39	0.47	0.37	0.34	0.35	0.52	0.46	0.47	0.46	0.39	0.47		
August	0.10	0.18	0.16	0.22	0.29	0.34	0.40	0.44	0.29	0.39	0.47	0.36	0.49	0.61	0.36	0.38	0.42		
September	0.08	0.15	0.23	0.25	0.27	0.36	0.48	0.29	0.34	0.38	0.38	0.36	0.48	0.59	0.39	0.31	0.48		
October	0.06	0.11	0.17	0.30	0.33	0.38	0.35	0.45	0.38	0.35	0.43	0.35	0.48	0.57	0.39	0.41	0.38		
November	0.09	0.09	0.25	0.26	0.40	0.34	0.40	0.32	0.35	0.40	0.44	0.38	0.49	0.64	0.40	0.29	0.39		
December	0.10	0.11	0.17	0.36	0.36	0.35	0.38	0.45	0.35	0.32	0.54	0.34	0.54	0.45	0.48	0.44	0.43		
Average	0.11	0.11	0.18	0.25	0.31	0.35	0.39	0.39	0.36	0.38	0.40	0.39	0.47	0.55	0.47	0.35	0.41	0.44 YTD	
y/y Δ	(60.7%)	2.2%	55.7%	38.9%	25.2%	14.6%	11.4%	(0.6%)	(9.2%)	5.4%	7.3%	(2.2%)	20.4%	15.3%	(13.4%)	(25.1%)	16.9%	13.4% YTD	

U.S. Total Housing Starts

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	0.49	0.61	0.63	0.72	0.89	0.89	1.09	1.09	1.19	1.30	1.23	1.58	1.62	1.69	1.36	1.38	1.35	1.39	2.4%
February	0.58	0.60	0.52	0.70	0.96	0.94	0.89	1.23	1.27	1.28	1.12	1.55	1.40	1.73	1.40	1.56	1.49	1.35	(9.7%)
March	0.51	0.64	0.60	0.70	1.01	0.97	0.96	1.11	1.19	1.32	1.19	1.27	1.69	1.70	1.37	1.30	1.35	1.52	13.1%
April	0.48	0.69	0.55	0.75	0.84	1.04	1.19	1.16	1.15	1.28	1.27	0.94	1.49	1.81	1.34	1.38	1.40	1.39	(0.6%)
May	0.54	0.58	0.56	0.71	0.93	1.01	1.08	1.15	1.16	1.36	1.29	1.04	1.60	1.55	1.61	1.33	1.29	1.18	(8.7%)
June	0.59	0.54	0.61	0.76	0.84	0.91	1.21	1.20	1.25	1.19	1.24	1.27	1.66	1.56	1.42	1.32	1.38	Release	7/17/26
July	0.59	0.55	0.62	0.74	0.88	1.09	1.15	1.24	1.21	1.21	1.24	1.53	1.60	1.38	1.46	1.27	1.43		
August	0.59	0.60	0.59	0.75	0.92	0.98	1.13	1.17	1.16	1.29	1.38	1.40	1.60	1.53	1.32	1.39	1.29		
September	0.59	0.59	0.65	0.85	0.85	1.02	1.22	1.07	1.18	1.25	1.31	1.47	1.58	1.48	1.37	1.36	1.32		
October	0.53	0.54	0.61	0.92	0.93	1.07	1.06	1.31	1.26	1.22	1.33	1.54	1.59	1.45	1.38	1.37	1.27		
November	0.59	0.55	0.71	0.83	1.10	1.00	1.17	1.14	1.27	1.18	1.35	1.54	1.71	1.43	1.52	1.30	1.32		
December	0.58	0.54	0.69	0.98	1.00	1.07	1.15	1.25	1.18	1.10	1.55	1.61	1.71	1.30	1.50	1.49	1.38		
Average	0.55	0.59	0.61	0.78	0.93	1.00	1.11	1.18	1.20	1.25	1.29	1.39	1.60	1.55	1.42	1.37	1.36	1.36 YTD	
y/y Δ	(38.4%)	5.7%	4.5%	28.1%	18.4%	7.8%	10.6%	6.4%	2.3%	3.5%	3.6%	8.0%	15.0%	(3.3%)	(8.3%)	(3.6%)	(1.0%)	(0.8%) YTD	

Source: U.S. Census Bureau, Bloomberg Finance L.P.

US Total Existing Home Sales & Forecast

(In million units, annualized)

U.S. Total Existing Home Sales

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	3.82	4.19	4.42	4.48	4.98	4.76	4.92	5.43	5.67	5.41	4.97	5.35	6.60	6.43	4.09	4.02	4.09	4.02	(1.7%)
February	3.97	4.27	4.16	4.58	5.06	4.78	5.08	5.26	5.45	5.48	5.31	5.63	6.13	5.80	4.38	4.19	4.15	4.13	(0.5%)
March	3.86	4.49	4.26	4.52	5.07	4.75	5.25	5.34	5.61	5.53	5.25	5.34	5.98	5.57	4.26	4.13	4.02	4.01	(0.2%)
April	3.90	4.82	4.17	4.59	5.09	4.79	5.17	5.47	5.56	5.47	5.31	4.45	5.94	5.50	4.20	4.08	4.02	4.04	0.5%
May	4.00	4.88	4.14	4.60	5.14	4.90	5.25	5.44	5.59	5.43	5.46	4.09	5.88	5.33	4.18	4.06	4.04	4.19	3.7%
June	4.10	4.45	4.23	4.47	5.11	4.94	5.36	5.47	5.49	5.45	5.39	4.87	5.99	5.11	4.16	3.96	3.98	4.09	2.8%
July	4.37	3.45	4.15	4.53	5.27	4.98	5.44	5.37	5.46	5.38	5.43	5.96	6.07	4.88	4.08	4.00	4.03	Release	8/11/26
August	4.45	3.68	4.38	4.76	5.26	4.97	5.36	5.40	5.39	5.31	5.47	6.04	5.99	4.79	4.09	3.94	4.03		
September	4.62	3.84	4.33	4.71	5.14	5.00	5.41	5.47	5.45	5.20	5.31	6.35	6.17	4.74	4.03	3.91	4.08		
October	5.02	3.83	4.34	4.79	5.04	5.12	5.28	5.56	5.46	5.16	5.29	6.56	6.21	4.50	3.90	4.02	4.11		
November	5.44	4.02	4.39	4.96	4.92	4.99	4.78	5.51	5.62	5.18	5.26	6.47	6.43	4.20	3.90	4.14	4.09		
December	4.40	4.27	4.35	4.89	4.86	5.09	5.44	5.52	5.57	5.01	5.45	6.51	6.21	4.14	3.86	4.19	4.27		
Average	4.33	4.18	4.28	4.66	5.08	4.92	5.23	5.44	5.53	5.33	5.33	5.64	6.13	5.08	4.09	4.05	4.08	4.08 YTD	
y/y Δ	5.4%	(3.4%)	2.3%	8.9%	9.1%	(3.1%)	6.2%	4.0%	1.7%	(3.5%)	(0.2%)	5.8%	8.8%	(17.1%)	(19.4%)	(1.0%)	0.6%	0.7% YTD	

U.S. Total Existing Home Sales Forecast (Fannie Mae)

Month	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	y/y Δ
March	4.50	4.93	4.68	5.05	5.36	5.60	5.51	5.20	5.49	6.29	5.98	4.32	4.14	4.09	4.05	4.33	6.7%
June	4.51	5.10	4.89	5.28	5.48	5.55	5.41	5.29	4.38	5.95	5.37	4.19	4.02	4.01	4.13	4.39	6.4%
September	4.74	5.32	5.06	5.40	5.38	5.40	5.31	5.41	6.10	6.07	4.78	4.02	3.94	4.05	4.17	4.46	7.1%
December	4.90	4.94	5.06	5.20	5.55	5.59	5.14	5.42	6.66	6.20	4.20	3.88	4.16	4.16	4.23	4.55	7.4%
Average	4.66	5.07	4.92	5.23	5.44	5.54	5.34	5.33	5.66	6.13	5.08	4.10	4.07	4.08	4.15	4.43	
y/y Δ		8.8%	(3.0%)	6.4%	4.0%	1.8%	(3.5%)	(0.2%)	6.1%	8.3%	(17.1%)	(19.3%)	(0.8%)	0.2%	1.7%	6.9%	

Source: National Association of Realtors, Fannie Mae, and Bloomberg Finance L.P.

US New Home Sales (Single Family) & Forecast

(In million units, annualized)

U.S. New Home Sales (Single Family)

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	0.34	0.35	0.31	0.34	0.45	0.44	0.52	0.51	0.57	0.59	0.60	0.69	0.90	0.81	0.65	0.68	0.67	0.58	(13.4%)
February	0.37	0.34	0.27	0.37	0.45	0.42	0.54	0.52	0.59	0.62	0.65	0.71	0.77	0.78	0.63	0.65	0.64	0.63	(1.7%)
March	0.34	0.38	0.30	0.35	0.44	0.41	0.48	0.53	0.63	0.68	0.72	0.63	0.85	0.71	0.64	0.68	0.66	0.66	1.2%
April	0.34	0.42	0.31	0.35	0.44	0.40	0.50	0.58	0.60	0.64	0.70	0.58	0.82	0.61	0.66	0.71	0.70	0.63	(10.7%)
May	0.38	0.28	0.31	0.37	0.43	0.45	0.50	0.57	0.64	0.66	0.62	0.71	0.72	0.64	0.74	0.67	0.62	0.58	(6.8%)
June	0.39	0.31	0.30	0.36	0.47	0.42	0.48	0.56	0.62	0.64	0.73	0.90	0.71	0.56	0.69	0.68	0.67	Release	7/24/26
July	0.41	0.28	0.30	0.37	0.38	0.40	0.51	0.63	0.57	0.63	0.65	0.98	0.75	0.54	0.73	0.73	0.65		
August	0.42	0.28	0.30	0.38	0.38	0.46	0.52	0.58	0.56	0.59	0.71	1.02	0.69	0.63	0.65	0.69	0.70		
September	0.39	0.32	0.30	0.39	0.40	0.47	0.46	0.56	0.64	0.60	0.71	0.98	0.73	0.55	0.68	0.72	0.71		
October	0.40	0.29	0.32	0.36	0.44	0.48	0.48	0.58	0.63	0.55	0.72	1.03	0.69	0.59	0.69	0.63	0.65		
November	0.38	0.29	0.33	0.39	0.45	0.44	0.50	0.57	0.71	0.63	0.71	0.88	0.78	0.59	0.61	0.67	0.76		
December	0.35	0.33	0.34	0.40	0.43	0.50	0.55	0.56	0.63	0.55	0.70	0.87	0.82	0.62	0.63	0.70	0.72		
Average	0.37	0.32	0.31	0.37	0.43	0.44	0.50	0.56	0.61	0.61	0.68	0.83	0.77	0.64	0.67	0.68	0.68	0.62 YTD	
y/y Δ	(22.4%)	(14.2%)	(4.6%)	20.1%	16.8%	2.4%	14.2%	11.5%	9.7%	(0.1%)	11.5%	21.3%	(7.4%)	(17.3%)	4.7%	2.8%	(0.9%)	(6.4%) YTD	

U.S. New Home Sales Forecast (Fannie Mae)

Month	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	y/y Δ
March	0.35	0.45	0.42	0.52	0.53	0.62	0.64	0.67	0.70	0.85	0.76	0.64	0.68	0.66	0.72	0.71	0.4%
June	0.36	0.45	0.43	0.49	0.56	0.60	0.63	0.66	0.71	0.75	0.60	0.70	0.69	0.67	0.72	0.71	1.4%
September	0.38	0.38	0.44	0.49	0.59	0.58	0.61	0.70	0.97	0.72	0.58	0.68	0.71	0.70	0.71	0.71	0.7%
December	0.39	0.45	0.47	0.51	0.57	0.66	0.58	0.71	0.93	0.76	0.60	0.65	0.67	0.72	0.71	0.71	0.0%
Average	0.37	0.43	0.44	0.50	0.56	0.62	0.62	0.68	0.83	0.77	0.64	0.67	0.69	0.68	0.72	0.71	
y/y Δ		16.9%	2.2%	14.1%	11.7%	9.7%	(0.1%)	11.3%	20.8%	(7.0%)	(17.2%)	4.5%	2.9%	(0.3%)	4.7%	(1.2%)	

Source: U.S. Census Bureau, Fannie Mae, and Bloomberg Finance L.P.

US Total Construction Spending

(In \$ billions, annualized)

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	975.1	822.1	760.0	830.6	862.7	986.6	1,061.2	1,161.0	1,256.0	1,338.5	1,295.5	1,491.2	1,598.7	1,807.8	1,957.6	2,191.6	1,061.2	2,195.6	(2.9%)
February	971.7	814.9	758.4	824.4	873.6	987.5	1,072.9	1,172.9	1,277.3	1,354.8	1,310.1	1,498.4	1,574.7	1,842.1	1,971.2	2,220.9	1,072.9	2,189.7	(3.4%)
March	963.8	821.8	769.2	829.3	864.5	991.2	1,086.7	1,188.8	1,274.5	1,345.7	1,319.5	1,501.7	1,609.8	1,877.4	2,000.4	2,214.1	1,086.7	2,199.4	(2.5%)
April	932.5	829.7	769.4	841.7	879.7	1,006.9	1,115.6	1,185.9	1,268.2	1,353.5	1,352.9	1,459.0	1,614.4	1,913.1	2,028.9	2,220.7	1,115.6	2,207.1	(2.5%)
May	920.0	823.4	775.3	850.9	888.8	1,007.1	1,135.4	1,190.7	1,282.1	1,360.7	1,364.6	1,448.8	1,626.6	1,924.0	2,053.9	2,232.8	1,135.4	2,210.2	(1.5%)
June	912.5	821.9	794.8	858.0	899.3	1,006.4	1,158.2	1,217.6	1,277.4	1,342.1	1,381.4	1,461.1	1,640.3	1,920.3	2,074.2	2,238.6	1,158.2	Release	8/3/26
July	911.4	799.7	792.4	858.9	917.8	1,014.9	1,154.6	1,214.5	1,275.6	1,335.0	1,413.0	1,474.6	1,661.5	1,923.9	2,083.6	2,240.4	1,154.6		
August	900.1	802.0	812.0	862.7	929.1	1,015.1	1,164.4	1,221.3	1,274.0	1,339.1	1,426.0	1,490.0	1,665.4	1,918.1	2,108.5	2,246.0	1,164.4		
September	887.2	807.7	813.7	866.6	938.1	1,021.9	1,163.9	1,232.4	1,280.0	1,323.2	1,438.2	1,512.5	1,673.5	1,917.9	2,117.2	2,263.1	1,163.9		
October	871.4	801.4	808.4	875.4	951.2	1,046.4	1,151.5	1,241.1	1,283.9	1,315.9	1,445.1	1,538.3	1,688.7	1,914.4	2,139.6	2,263.8	1,151.5		
November	858.5	804.2	812.1	872.1	964.1	1,044.9	1,145.5	1,264.9	1,309.3	1,302.7	1,464.9	1,552.5	1,731.9	1,919.5	2,170.3	2,262.6	1,145.5		
December	842.6	789.4	821.4	871.8	979.2	1,056.4	1,156.4	1,270.4	1,314.4	1,292.0	1,469.3	1,577.5	1,762.4	1,926.9	2,174.8	2,258.8	1,156.4		
Average	912.2	811.5	790.6	853.5	912.3	1,015.4	1,130.5	1,213.5	1,281.1	1,333.6	1,390.0	1,500.5	1,654.0	1,900.4	2,073.4	2,237.8	1,130.5	2,200.4 YTD	
y/y Δ	(15.2%)	(11.0%)	(2.6%)	8.0%	6.9%	11.3%	11.3%	7.3%	5.6%	4.1%	4.2%	7.9%	10.2%	14.9%	9.1%	7.9%	11.3%	(2.6%) YTD	

Source: U.S. Census Bureau, Bloomberg Finance L.P.

30-Year Fixed Mortgage Rates

(End of month %)

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	5.34	5.07	4.78	3.88	3.60	4.26	3.79	3.76	4.04	4.18	4.37	3.63	2.88	3.78	6.38	6.96	7.05	6.21	(11.9%)
February	5.25	5.08	4.85	3.88	3.61	4.31	3.92	3.84	3.99	4.34	4.40	3.62	3.25	4.30	7.03	7.30	6.89	6.12	(11.2%)
March	4.89	5.12	4.84	3.97	3.67	4.38	3.79	3.65	3.99	4.27	4.08	3.86	3.27	4.90	6.81	7.25	6.83	6.48	(5.1%)
April	4.92	5.13	4.70	3.81	3.43	4.26	3.86	3.63	3.89	4.42	4.09	3.52	3.11	5.42	6.85	7.55	6.90	6.35	(8.0%)
May	5.23	4.92	4.55	3.74	4.10	4.15	3.90	3.65	3.82	4.36	4.03	3.52	3.10	5.35	7.08	7.29	7.01	6.63	(5.4%)
June	5.39	4.68	4.57	3.69	4.39	4.15	4.17	3.53	3.87	4.40	3.80	3.27	3.13	5.83	7.15	7.26	6.80	6.55	(3.7%)
July	5.33	4.56	4.47	3.57	4.39	4.20	3.90	3.36	3.85	4.43	3.87	3.09	2.98	5.28	7.26	7.04	6.78		
August	5.27	4.33	4.26	3.55	4.46	4.07	3.90	3.40	3.77	4.41	3.69	3.10	3.03	5.95	7.53	6.80	6.62		
September	5.16	4.47	4.03	3.43	4.28	4.12	3.84	3.34	3.80	4.57	3.72	3.08	3.18	7.06	7.74	6.68	6.36		
October	5.16	4.35	4.13	3.45	4.15	3.97	4.01	3.51	3.83	4.75	3.75	3.06	3.14	7.22	8.06	7.28	6.26		
November	4.93	4.57	4.02	3.41	4.38	3.92	3.82	3.98	3.82	4.68	3.73	2.94	3.23	6.67	7.57	7.12	6.36		
December	5.33	4.99	3.94	3.40	4.54	3.99	3.90	4.06	3.85	4.51	3.86	2.87	3.27	6.66	6.99	7.28	6.32		
Average	5.18	4.77	4.43	3.65	4.08	4.15	3.90	3.64	3.88	4.44	3.95	3.30	3.13	5.70	7.20	7.15	6.68	6.39	YTD
y/y Δ	(12.1%)	(7.9%)	(7.2%)	(17.6%)	11.9%	1.6%	(6.0%)	(6.6%)	6.4%	14.6%	(11.1%)	(16.5%)	(5.0%)	82.1%	26.4%	(0.7%)	(6.6%)	(7.6%)	YTD

Source: Bankrate.com, Bloomberg Finance L.P.

US Producer Price Index (Lumber & Plywood)

(Index)

Month	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	154.3	159.1	170.4	167.0	192.9	207.5	209.6	194.4	200.6	224.5	210.8	206.9	319.3	392.2	269.3	255.1	265.0	266.9	0.7%
February	152.8	167.0	170.6	169.8	197.6	210.9	207.1	193.5	206.6	235.5	215.6	211.7	334.6	412.3	271.1	256.1	270.1	272.6	0.9%
March	149.6	170.3	171.4	173.3	204.6	214.2	204.8	195.9	209.3	239.9	213.7	218.8	358.3	435.8	268.2	258.6	278.3	272.5	(2.1%)
April	149.0	178.9	171.0	174.7	208.8	211.8	203.4	199.9	213.3	238.5	213.1	204.4	386.9	382.2	265.8	261.8	274.6	281.6	2.5%
May	148.2	184.0	166.8	178.4	202.5	214.0	199.4	200.9	215.3	247.8	210.7	208.7	453.4	387.2	265.4	256.9	271.0	283.1	4.5%
June	148.6	174.2	165.9	178.4	194.5	213.2	198.8	201.9	213.1	252.6	205.6	222.5	450.8	325.0	262.0	255.7	267.6	Release	7/15/26
July	154.0	171.0	165.9	175.2	191.6	213.8	200.0	203.5	213.3	246.7	208.1	238.9	355.0	320.8	273.2	252.4	266.0		
August	154.7	168.1	167.4	179.5	195.5	216.7	196.6	205.0	218.2	231.9	206.3	267.4	293.1	310.9	260.8	255.2	267.8		
September	156.2	166.5	166.5	180.1	196.9	217.5	192.8	203.5	218.0	228.9	207.7	313.4	272.9	298.5	262.2	258.8	262.3		
October	154.6	165.0	167.2	175.8	200.3	213.9	193.8	202.3	223.0	217.7	206.0	295.2	285.6	289.4	255.3	261.0	261.3		
November	156.1	165.2	165.2	180.7	203.2	211.8	195.8	200.4	224.2	211.3	205.8	263.7	296.4	281.7	249.7	265.3	262.0		
December	158.6	167.6	166.5	184.9	203.4	210.1	193.5	200.4	222.9	211.9	207.2	284.3	335.8	271.0	253.8	266.8	261.8		
Average	153.06	169.74	167.90	176.48	199.32	212.95	199.6	200.1	214.8	232.3	209.2	244.7	345.2	342.3	263.1	258.6	267.3	275.3 YTD	
y/y Δ	(8.0%)	10.9%	(1.1%)	5.1%	12.9%	6.8%	(6.3%)	0.3%	7.3%	8.1%	(9.9%)	16.9%	41.1%	(0.8%)	(23.1%)	(1.7%)	3.4%	1.3% YTD	

Source: Bureau of Labor Statistics, Bloomberg Finance L.P.

Global Light Vehicle Production

(In million units; auto production data is rounded.)

Region	2019	2020	2021	2022	2023	2024	2025	2026E	2027E
EMEA	23	18	18	18	20	20	19	19	19
y/y growth		(21%)	(2%)	1%	12%	(4%)	(0%)	(4%)	2%
China	25	24	25	26	29	30	33	32	32
y/y growth		(4%)	5%	6%	10%	4%	10%	(4%)	1%
Asia Pacific, ex China	22	17	19	21	23	22	22	22	22
y/y growth		(19%)	8%	11%	9%	(4%)	3%	(0%)	(0%)
North America	16	13	13	14	16	15	15	15	15
y/y growth		(20%)	0%	10%	10%	(1%)	(1%)	(2%)	0%
South America	3	2	3	3	3	3	3	3	3
y/y growth		(32%)	16%	8%	3%	2%	3%	2%	8%
Total	89.0	74.6	77.2	82.3	90.5	89.6	93.1	90.7	91.6
y/y growth		(16.1%)	3.5%	6.7%	9.9%	(1.0%)	3.9%	(2.6%)	1.0%

Source: J.P. Morgan estimates.

Light Vehicle Sales: United States

(In million units, annualized)

US	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	16.66	17.64	17.34	17.07	16.60	16.84	16.63	15.04	15.74	15.00	15.60	14.85	(4.8%)
February	16.40	17.51	17.32	16.96	16.56	16.83	15.67	14.07	14.89	15.81	16.00	15.75	(1.6%)
March	17.50	16.77	16.72	17.40	17.50	11.37	17.75	13.33	14.82	15.49	17.77	16.34	(8.0%)
April	16.93	17.49	16.97	17.07	16.40	8.58	18.51	14.29	15.91	15.74	17.27	15.92	(7.8%)
May	17.69	17.22	16.70	16.81	17.30	12.21	16.99	12.68	15.05	15.90	15.65	16.08	2.7%
June	17.29	16.99	16.59	17.38	17.30	13.05	15.36	13.00	15.68	15.29	15.34	16.52	7.7%
July	17.50	17.75	16.69	16.68	16.82	14.52	14.75	13.35	15.74	15.82	16.41	Release	8/3/26
August	17.76	17.13	16.03	16.60	16.97	15.19	13.06	13.18	15.04	15.13	16.07		
September	17.97	17.65	18.47	17.40	17.19	16.34	12.18	13.49	15.67	15.77	16.39		
October	17.91	17.80	17.98	17.50	16.55	16.21	12.99	14.90	15.50	16.04	15.32		
November	17.97	17.56	17.40	17.40	17.09	15.55	12.86	14.14	15.32	16.50	15.60		
December	17.19	18.05	17.76	17.50	16.70	16.27	12.44	13.31	15.83	16.80	16.02		
Average	17.40	17.46	17.16	17.15	16.92	14.41	14.93	13.73	15.43	15.77	16.12	15.91	YTD
y/y Δ	5.7%	0.4%	(1.7%)	(0.1%)	(1.4%)	(14.8%)	3.6%	(8.0%)	12.4%	2.2%	2.2%	(2.2%)	YTD

Source: WARD's Automotive Group, Bloomberg Finance L.P.

Light Vehicle Sales: China

(In million units, monthly)

China	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	2.32	2.50	2.52	2.81	2.37	1.93	2.50	2.53	1.65	2.44	2.42	2.35	(3.2%)
February	1.59	1.58	1.94	1.72	1.48	0.31	1.45	1.74	1.98	1.58	2.13	1.81	(15.2%)
March	2.24	2.44	2.54	2.66	2.52	1.43	2.53	2.23	2.45	2.69	2.92	2.90	(0.6%)
April	1.99	2.12	2.08	2.32	1.98	2.07	2.25	1.18	2.16	2.36	2.59	2.53	(2.5%)
May	1.90	2.09	2.10	2.29	1.91	2.19	2.13	1.86	2.38	2.42	2.69	2.63	(2.1%)
June	1.80	2.07	2.17	2.27	2.06	2.30	2.02	2.50	2.62	2.55	2.90		
July	1.50	1.85	1.97	1.89	1.81	2.11	1.86	2.42	2.39	2.26	2.59		
August	1.66	2.07	2.19	2.10	1.96	2.19	1.80	2.38	2.58	2.45	2.86		
September	2.02	2.56	2.71	2.39	2.27	2.57	2.07	2.61	2.86	2.81	3.23		
October	2.22	2.65	2.70	2.38	2.28	2.57	2.33	2.50	2.85	3.05	3.32		
November	2.51	2.94	2.96	2.55	2.46	2.77	2.52	2.33	2.97	3.32	3.43		
December	2.79	3.06	3.06	2.66	2.66	2.83	2.79	2.56	3.16	3.49	3.27		
Total	24.56	27.94	28.94	28.04	25.75	25.27	26.25	26.85	30.05	31.43	34.35	12.21	YTD
y/y Δ	4.6%	13.7%	3.6%	(3.1%)	(8.1%)	(1.9%)	3.9%	2.3%	11.9%	4.6%	9.3%	(4.2%)	YTD

Source: China Automotive Information Net, Bloomberg Finance L.P.

Light Vehicle Sales: Europe

(In million units, monthly)

Europe	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	1.17	1.20	1.31	1.43	1.34	1.26	0.97	0.94	0.99	1.16	1.14	1.10	(3.9%)
February	1.13	1.24	1.26	1.34	1.29	1.22	1.01	0.94	1.02	1.16	1.12	1.13	0.6%
March	1.85	1.94	2.13	2.05	1.97	1.05	1.61	1.26	1.57	1.62	1.60	1.76	10.3%
April	1.41	1.51	1.42	1.56	1.51	0.35	1.24	0.90	1.12	1.28	1.27	1.35	6.4%
May	1.34	1.51	1.62	1.65	1.60	0.71	1.27	1.03	1.28	1.30	1.29	1.33	2.9%
June	1.62	1.70	1.75	1.82	1.67	1.31	1.50	1.17	1.44	1.53	1.43		
July	1.38	1.32	1.41	1.50	1.48	1.49	1.14	0.96	1.21	1.24	1.29		
August	0.98	1.02	1.11	1.35	1.23	1.07	0.88	0.83	1.08	0.98	1.00		
September	1.58	1.67	1.69	1.30	1.47	1.53	1.13	1.15	1.36	1.34	1.45		
October	1.32	1.33	1.45	1.30	1.40	1.36	0.96	1.00	1.24	1.30	1.35		
November	1.32	1.42	1.52	1.37	1.41	1.27	1.03	1.12	1.29	1.28	1.32		
December	1.42	1.45	1.45	1.28	1.50	1.46	1.12	1.23	1.30	1.36	1.46		
Total	16.53	17.31	18.11	17.93	17.87	14.08	13.88	12.53	14.91	15.56	15.72	6.66	YTD
y/y Δ	2.7%	4.8%	4.6%	(0.9%)	(0.4%)	(21.2%)	(1.4%)	(9.7%)	19.0%	4.3%	1.0%	3.8%	YTD

Source: Bloomberg Finance L.P.

Electric Vehicle Sales: United States, EU-5 and China

(In units)

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	y/y Δ
January	26,544	59,479	127,461	120,435	282,368	571,817	521,772	898,340	1,011,207	856,000	(15%)
February	36,567	58,423	87,907	83,603	222,927	489,543	655,909	628,796	953,131	734,000	(23%)
March	60,652	103,322	159,274	113,856	415,683	702,638	854,243	1,032,204	1,384,711	1,295,000	(6%)
April	53,230	101,525	128,121	86,870	326,051	464,234	751,583	932,910	1,213,557	1,189,000	(2%)
May	64,836	117,765	138,860	114,693	370,492	559,059	847,452	1,066,091	1,336,804	1,309,000	(2%)
June	71,826	111,498	185,968	169,935	442,018	757,780	981,240	1,170,665	1,416,402		
July	69,745	113,689	111,195	189,722	395,844	671,314	917,267	1,142,422	1,344,000		
August	78,474	131,079	123,987	189,819	396,931	706,166	1,018,436	1,272,901	1,395,000		
September	94,172	150,492	130,761	266,706	523,944	859,566	1,051,352	1,437,301	1,742,000		
October	94,001	163,401	118,675	276,751	501,444	776,529	1,039,342	1,478,178	1,703,000		
November	117,450	188,688	134,529	330,298	609,246	878,245	1,133,549	1,590,148	1,672,000		
December	135,519	221,744	191,546	439,589	744,733	1,018,968	1,275,159	1,658,368	1,745,000		
Total	903,016	1,521,108	1,638,284	2,382,277	5,231,681	8,455,859	11,047,304	14,308,324	16,916,812	5,383,000	
y/y Δ		68%	8%	45%	120%	62%	31%	30%	18%	(9%)	
Vehicle change		618,092	117,176	743,993	2,849,404	3,224,178	2,591,445	3,261,020	2,608,488	(516,410)	

Note: Includes EV + PHEV

Source: Company Data, J.P. Morgan estimates

Electric Vehicle Sales: United States

(In units)

US	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	y/y Δ
January	12,226	11,720	26,095	32,793	34,974	56,610	95,348	111,979	126,238	76,000	(40%)
February	12,614	15,374	23,931	31,145	38,736	64,553	106,785	120,966	125,367	82,000	(35%)
March	15,238	19,442	25,440	18,832	48,673	75,143	111,098	131,469	150,119	97,000	(35%)
April	14,222	16,171	19,557	9,579	40,926	79,228	110,517	128,446	134,517	93,000	(31%)
May	15,923	20,255	23,798	13,930	58,151	78,814	118,719	141,509	133,533	105,000	(21%)
June	15,460	21,381	26,330	17,905	48,406	82,784	125,152	131,558	121,747		
July	15,006	26,229	23,785	23,578	51,067	80,854	130,518	134,917	154,000		
August	15,261	27,412	26,222	28,490	42,685	77,161	133,842	150,459	179,000		
September	15,825	27,772	26,730	33,339	44,039	74,449	129,505	124,654	172,000		
October	14,534	27,908	24,931	32,039	49,795	80,359	117,213	132,653	84,000		
November	16,399	31,974	29,004	32,204	52,450	82,639	117,328	147,070	82,000		
December	18,979	37,985	35,210	42,154	69,762	94,639	138,490	167,467	101,000		
Total	181,687	283,623	311,033	315,988	579,664	927,233	1,434,515	1,623,147	1,563,521	453,000	YTD
y/y Δ		56%	10%	2%	83%	60%	55%	13%	(4%)	(32%)	YTD
Vehicle change		101,936	27,410	4,955	263,676	347,569	507,282	188,632	(59,626)	(216,774)	

Note: Includes EV + PHEV

Source: Company Data, J.P. Morgan estimates

Electric Vehicle Sales: China

(In units)

China	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	y/y Δ
January	6,122	35,488	84,803	41,610	175,270	419,106	331,542	671,000	744,000	596,000	(20%)
February	16,693	32,254	48,898	11,500	108,429	321,176	439,068	388,000	686,000	464,000	(32%)
March	28,568	61,393	106,639	46,759	211,862	460,792	545,739	709,000	991,000	852,000	(14%)
April	30,478	71,876	89,621	63,249	178,681	283,000	524,730	674,000	905,000	850,000	(6%)
May	38,960	83,581	95,630	73,895	191,820	360,000	580,000	790,000	1,021,000	950,000	(7%)
June	43,002	72,230	137,291	92,510	238,018	532,000	666,000	856,000	1,070,000		
July	44,461	73,589	66,677	90,074	232,757	486,000	641,000	878,000	1,003,000		
August	54,740	90,013	79,740	102,740	260,823	529,000	716,000	1,027,000	1,079,000		
September	61,176	106,704	72,549	128,890	327,227	611,000	746,000	1,123,000	1,307,000		
October	67,929	120,611	66,118	148,508	326,968	555,400	767,000	1,196,000	1,400,000		
November	87,576	139,625	76,479	187,179	407,546	601,000	841,000	1,277,000	1,354,000		
December	101,326	166,203	125,420	223,089	498,616	642,000	945,000	1,302,000	1,390,000		
Total	581,031	1,053,567	1,049,865	1,210,003	3,158,017	5,800,474	7,743,079	10,891,000	12,950,000	3,712,000	YTD
y/y Δ		81%	0%	15%	161%	84%	33%	41%	19%	(15%)	YTD
Vehicle change		472,536	(3,702)	160,138	1,948,014	2,642,457	1,942,605	3,147,921	2,059,000	(635,000)	

Note: Includes EV + PHEV

Source: Company Data, J.P. Morgan estimates

Electric Vehicle Sales: Europe-5

(In units)

Europe	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	y/y Δ
January	8,196	12,271	16,563	46,032	72,124	96,101	94,882	115,361	140,969	184,000	31%
February	7,260	10,795	15,078	40,958	75,762	103,814	110,056	119,830	141,764	188,000	33%
March	16,846	22,487	27,195	48,265	155,148	166,703	197,406	191,735	243,592	346,000	42%
April	8,530	13,478	18,943	14,042	106,444	102,006	116,336	130,464	174,040	246,000	41%
May	9,953	13,929	19,432	26,868	120,521	120,245	148,733	134,582	182,271	254,000	39%
June	13,364	17,887	22,347	59,520	155,594	142,996	190,088	183,107	224,655		
July	10,278	13,871	20,733	76,070	112,020	104,460	145,749	129,505	187,000		
August	8,473	13,654	18,025	58,589	93,423	100,005	168,594	95,442	137,000		
September	17,171	16,016	31,482	104,477	152,678	174,117	175,847	189,647	263,000		
October	11,538	14,882	27,626	96,204	124,681	140,770	155,129	149,525	219,000		
November	13,475	17,089	29,046	110,915	149,250	194,606	175,221	166,078	236,000		
December	15,214	17,556	30,916	174,346	176,355	282,329	191,669	188,901	254,000		
Total	140,298	183,918	277,386	856,286	1,494,000	1,728,152	1,869,710	1,794,177	2,403,291	1,218,000	YTD
y/y Δ		31%	51%	209%	74%	16%	8%	(4%)	34%	38%	YTD
Vehicle change		43,620	93,468	578,900	637,714	234,152	141,558	(75,533)	609,114	335,364	

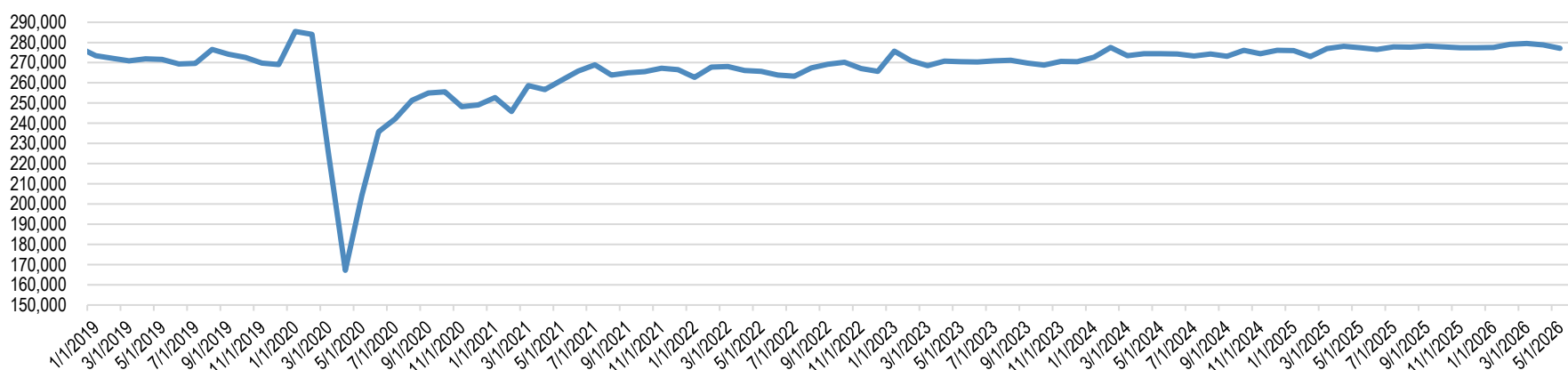
Note: Represents EU-5 (France, Germany, UK, Italy and Spain); includes EV + PHEV

Source: Company Data, J.P. Morgan estimates

U.S. Vehicle Miles Traveled

Millions of miles

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
Jan	242,519	246,965	249,790	250,411	246,531	254,967	262,611	266,509	268,361	273,421	285,452	252,702	262,682	275,775	272,717	276,030	277,559	0.6%
Feb	241,803	245,208	250,746	249,845	249,499	254,414	261,752	265,234	267,857	272,141	284,040	245,840	267,848	270,918	277,559	272,963	279,053	2.2%
Mar	248,076	246,519	249,737	248,349	251,120	255,735	262,392	265,123	268,556	270,920	224,870	258,686	268,103	268,528	273,437	276,982	279,570	0.9%
Apr	249,112	245,553	246,547	247,844	251,959	257,796	264,735	270,670	273,320	271,928	167,174	256,647	266,100	270,814	274,438	278,063	278,865	0.3%
May	247,042	243,131	248,096	249,071	252,289	257,531	264,565	266,153	267,556	271,676	204,304	261,366	265,748	270,548	274,506	277,390	277,166	(0.1%)
Jun	247,723	245,757	247,577	249,233	252,054	258,177	264,364	266,376	269,325	269,353	235,771	265,907	263,849	270,301	274,262	276,500		
Jul	249,293	245,196	245,308	247,384	252,111	258,012	265,645	269,548	270,452	269,606	242,159	268,899	263,304	270,863	273,378	277,886		
Aug	249,208	244,641	246,538	250,869	252,472	259,020	264,145	267,211	267,900	276,508	251,358	263,810	267,354	271,249	274,317	277,741		
Sep	249,084	245,393	245,805	249,145	253,485	258,671	263,573	266,506	269,877	274,195	254,995	265,052	269,221	269,771	273,150	278,291		
Oct	249,386	245,952	245,176	249,407	254,117	257,672	266,664	268,718	270,155	272,669	255,505	265,589	270,175	268,783	276,125	277,821		
Nov	247,614	246,579	247,779	249,394	253,099	260,006	265,314	268,212	269,052	269,800	248,198	267,303	267,155	270,657	274,417	277,443		
Dec	245,054	249,896	245,651	245,957	255,291	261,788	267,135	272,319	277,525	269,023	249,061	266,539	265,697	270,471	276,121	277,441		
Total	2,965,914	2,950,790	2,968,750	2,986,909	3,024,027	3,093,789	3,172,895	3,212,579	3,239,936	3,261,240	2,902,887	3,138,340	3,197,236	3,248,678	3,294,427	3,324,551	1,392,213 YTD	
Y/Y Δ	0.3%	(0.5%)	0.6%	0.6%	1.2%	2.3%	2.6%	1.3%	0.9%	0.7%	(11.0%)	8.1%	1.9%	1.6%	1.4%	0.9%	0.8%	YTD



Source: U.S. Department of Transportation, Bloomberg Finance L.P.

Gasoline Demand

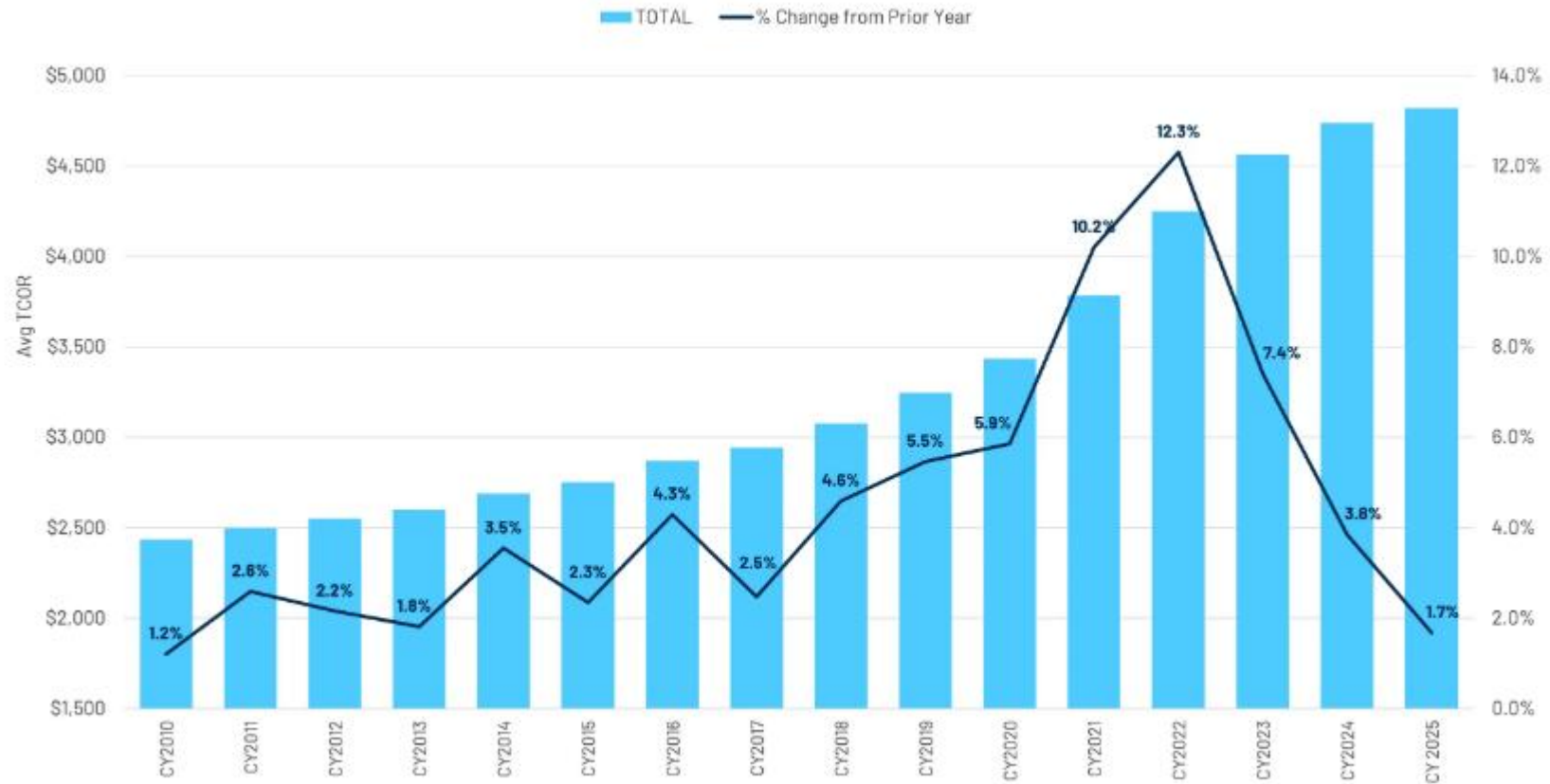
(Billions of gallons)

Month	2019	2020	2021	2022	2023	2024	2025	2026 YTD	y/y Δ
January	12.5	12.2	11.1	11.3	11.7	11.8	12.0	11.9	(0.3%)
February	11.5	11.8	10.0	10.9	12.3	12.0	12.3	12.3	0.2%
March	12.9	12.2	12.0	12.4	12.7	12.8	12.5	12.7	1.9%
April	12.7	7.7	12.1	12.2	12.7	12.3	12.5	12.8	2.2%
May	13.1	9.7	12.9	12.8	13.1	12.9	12.6	12.7	0.7%
June	13.0	10.9	12.5	12.4	13.3	13.2	13.0	12.9	(0.6%)
July	13.3	12.0	13.4	12.5	12.8	13.2	12.8		
August	13.5	12.5	13.2	12.9	12.9	12.9	12.9		
September	12.7	11.5	12.4	12.8	12.3	12.5	12.6		
October	13.4	11.6	13.1	12.4	12.7	12.9	12.7		
November	12.7	11.3	12.5	12.5	12.7	12.7	12.7		
December	13.0	11.1	12.7	12.3	12.6	12.7	12.7		
Total	154.4	134.5	147.9	147.4	151.9	152.1	151.2	75.3 YTD	
Δ Y/Y		(12.9%)	9.9%	(0.4%)	3.0%	0.1%	(0.6%)		
YTD January-June	75.8	64.5	70.5	72.1	75.8	75.1	74.8	75.3 YTD	
Δ YTD		(14.9%)	9.3%	2.2%	5.1%	(0.9%)	(0.3%)	0.7% YTD	

Source: U.S. Dept of Energy, Bloomberg Finance L.P.

Average Total Cost of Repair (TCOR)

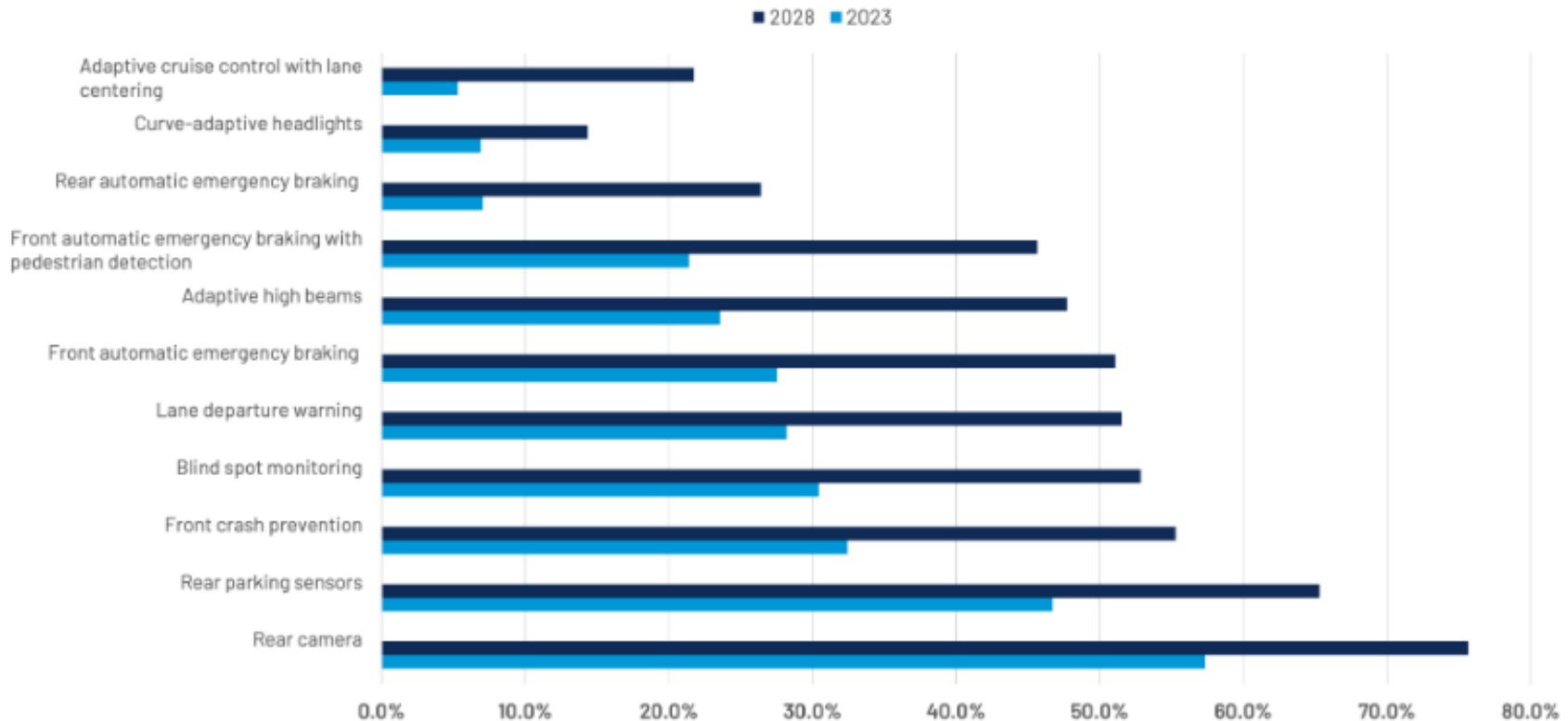
(The average total cost of repair (TCOR) has risen from >\$2,500 in 2010 to ~\$4,800 in 2025)



Source: CCC Intelligent Solutions (CCCIS)

Advanced Driver Assistance System (ADAS)

(Rear cameras, automatic emergency braking, blind-spot monitoring, and lane departure systems are standard features on most new vehicles)

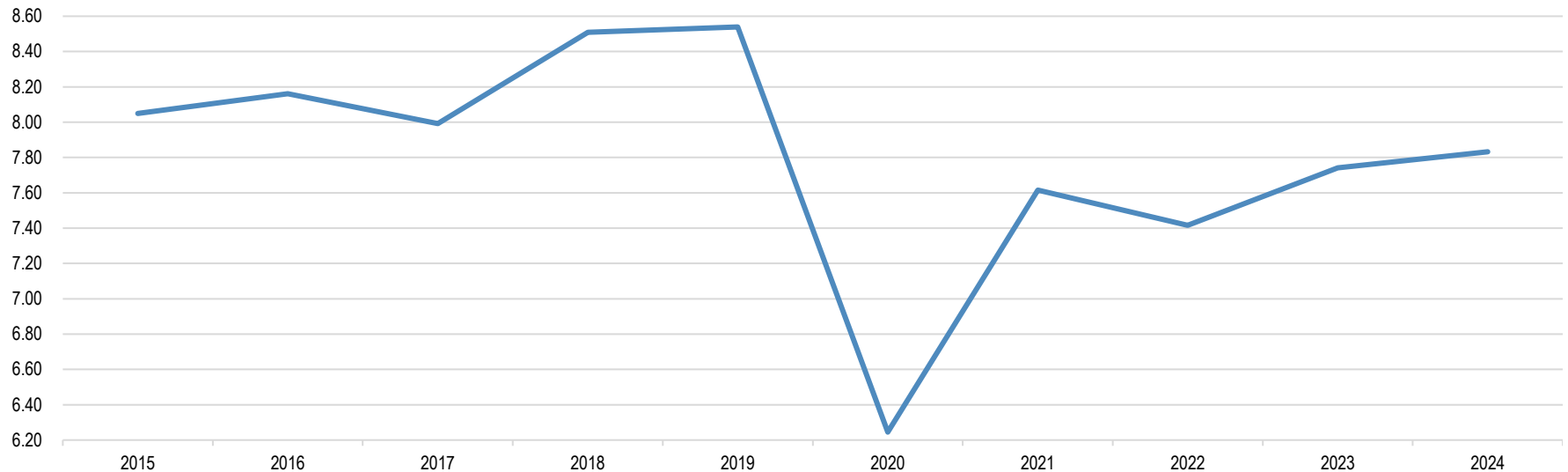


Source: Highway Loss Data Institute (HLDI), CCC Intelligent Solutions (CCCIS)

Vehicles Involved in Property-Damage-Only (2015-2024)

(The number of vehicles involved in PDO crashes has decreased over the past 10 years)

	January	February	March	April	May	June	July	August	September	October	November	December	Total
2015	722,838	666,992	647,452	604,414	648,813	608,670	624,668	623,333	654,272	748,088	741,686	758,197	8,049,420
2016	784,784	749,789	718,751	662,380	679,695	663,083	618,895	731,695	655,313	655,678	609,426	631,325	8,160,813
2017	641,177	582,502	687,166	632,517	710,605	633,256	582,097	668,197	655,991	744,842	721,836	732,077	7,992,263
2018	740,593	661,753	712,514	663,542	734,293	669,739	639,987	698,960	679,590	787,929	776,018	744,401	8,509,320
2019	735,378	678,723	681,913	689,548	720,109	669,085	670,518	693,935	668,860	806,693	763,441	761,135	8,539,339
2020	698,432	653,038	487,429	279,861	412,804	462,633	495,786	532,855	527,921	626,093	541,862	526,275	6,244,989
2021	554,013	545,557	618,068	623,964	645,270	660,400	616,641	651,956	654,162	744,063	653,944	648,127	7,616,165
2022	588,885	585,059	621,776	589,702	604,688	576,761	565,986	608,027	627,745	684,650	665,237	697,546	7,416,061
2023	638,399	606,871	695,725	634,237	649,825	603,694	552,592	662,910	658,033	711,122	688,287	640,377	7,742,073
2024	695,503	608,776	649,340	641,853	681,712	590,954	585,110	627,833	634,630	727,007	709,512	680,659	7,832,889
Total	9,883,648	9,232,310	9,345,020	8,701,454	9,350,578	8,827,104	8,594,662	9,291,419	9,285,958	10,431,921	10,101,856	10,147,638	113,193,568



Source: National Highway Traffic Safety Administration (NHTSA)

North & Central America Beverage Can Supply / Demand

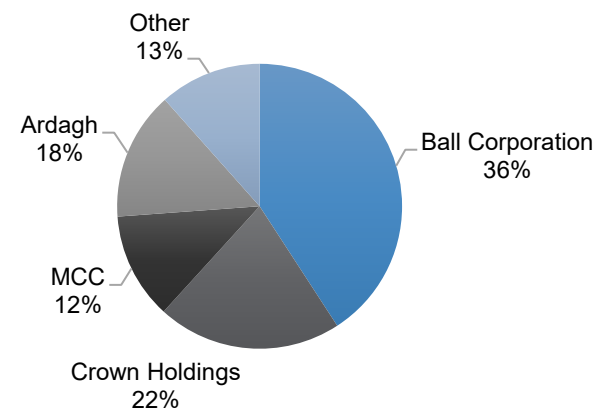
(in billions of cans.)

	2019	2020	2021	2022E	2023E	2024E	2025E	2026E	2027E
1b can expansion at Fort Worth, Texas - start up 2H:20 (July 2020)		0.5	0.5						
1b can expansion at Rome, Georgia - start-up 2H:20 (mid-August 2020)		0.3	0.7						
4b new can facility in Pittston, PA - Start-up 2Q:21 (June 2021) Expandable			0.6	1.0	2.4	0.0			
4b new can facility in Glendale, AZ - start up 1Q:21 (March 2021) Expandable			1.8	1.0	1.2				
Capacity closure in Phoenix, AZ (Rexam plant) 4Q:22				(2.0)					
Capacity closure in St. Paul, MN (Rexam plant) 1Q:23					(2.0)				
Capacity closure in Walkill, NY - 3 lines, mostly specialty (est mid-2023)					(2.4)				
~2b North Las Vegas, NV - startup mid-2023 (indefinitely delayed)									
Capacity closure in Kent, WA - 2 lines (est 1Q:24)						(0.75)	(0.75)		
~2b Concord, NC - startup early 2024 (on hold pending customer decision)									
1b acquired Florida Can Manufacturing in Winter Haven, FL (January 2025)							1.0		
~2b new can facility in Millersburg, OR (1 line) est. ramp mid 2026 - expandable to two lines								0.50	1.50
Other and debottlenecking		1.7	0.0	(1.7)	(3.0)	(0.2)	2.05	0.0	0.0
Ball Corporation	48.0	50.5	54.1	52.4	48.6	47.7	50.0	50.5	52.0
~1b can expansion, third line, Weston/Toronto Ontario (4Q:19)		0.8	0.2						
~1b can expansion, third line, Nichols, NY (2Q:20)		0.5	0.5						
~2.4b new can facility in Bowling Green, KY (3Q:21)			0.7	0.2	0.9	0.6			
~1b can expansion, third line, at Olympia, WA (4Q:21)			0.4	0.7					
est. ~2.5b new can facility in Martinsville, VA (line 1 Nov'22, line 2 Feb'23)				0.1	0.7	1.2	0.00	0.35	0.15
1.2b can expansion, second line at Monterrey, Mexico (end 2022)				0.0	0.7	0.5			
~2.5b cans in Mesquite, NV (line 1 July'23, line 2 October'23)					0.3	1.7	0.00	0.35	0.15
Capacity closure in Batesville, MS - 2 lines (2b cans)						(2.0)			
Other and debottlenecking		0.1	0.5	(1.1)	(1.5)	0.0	0.0	0.00	0.20
Crown Holdings	24.0	25.4	27.7	27.6	28.7	30.7	30.7	31.4	31.9
MCC	16.0	16.0	16.0	16.0	16.0	16.0	16.0	16.0	16.0
Olive Branch, MS - 2 new lines (2H:21 ramp up), est 3b cans			1.0	0.8	1.2				
Winston Salem, NC - line 1 end of 2021, line 20 1Q:22, est 3b cans			0.3	0.7	1.5	0.5			
Huron, OH - 3 brownfield lines (1Q:22) est. 4.5b cans			0.0	0.5	1.35	1.1	1.55	0.0	
NA capacity shut down in 2023 (1-2b cans)					(1.5)				
Whitehouse, OH plant closure						(1.0)	(1.0)		
Arizona - 2 new lines, line 1 beginning 1H:24, 3.5b cans - POSTPONED						0.0	0.0	0.0	
Other and debottlenecking				(1.0)			0.8	0.0	0.5
Ardagh	18.0	18.0	19.3	20.3	22.9	23.5	24.8	24.8	25.3
~4.5b can facility in Olyphant, PA (3 lines +1 line added; start-up end 2022)	0.0	0.0	1.0	2.0	0.0				
~3b can facility in Muncie, IN (3 lines, start-up end 2022)				0.0	1.0	1.5	0.15	0.15	0.20
CANPACK Group	0.0	0.0	1.0	3.0	4.0	5.5	5.7	5.8	6.0
Other	9.5	10.5	14.3	20.6	15.7	14.6	11.8	12.0	12.1
N&C America (US, Canada, Mexico) Capacity	115.5	120.4	132.5	140.0	136.0	138.0	139.0	140.5	143.3
<u>less Mexico</u>	<u>(15.0)</u>	<u>(15.7)</u>	<u>(17.7)</u>	<u>(17.7)</u>	<u>(16.9)</u>	<u>(17.4)</u>	<u>(17.4)</u>	<u>(17.4)</u>	<u>(17.4)</u>
North America Capacity	100.5	104.7	114.8	122.3	119.1	120.6	121.6	123.1	125.9
% y/y		4.2%	9.6%	6.5%	-2.7%	1.3%	0.8%	1.2%	2.3%
North America Demand	107.2	111.2	121.8	121.8	118.8	120.4	123.8	125.0	127.5
% y/y		3.7%	9.5%	0.0%	-2.4%	1.3%	2.9%	1.0%	1.9%
NA Utilization Rate		106%	106%	100%	100%	100%	102%	102%	101%
Canada capacity	(3.0)	(3.0)	(4.0)	(4.0)	(3.5)	(3.5)	(3.5)	(3.5)	(3.5)
US Capacity	97.5	101.7	110.8	118.3	115.6	117.1	118.1	119.6	122.4
US Imports Needed/(US Oversupply)	9.7	9.5	11.0	3.5	3.2	3.3	5.7	5.5	5.1

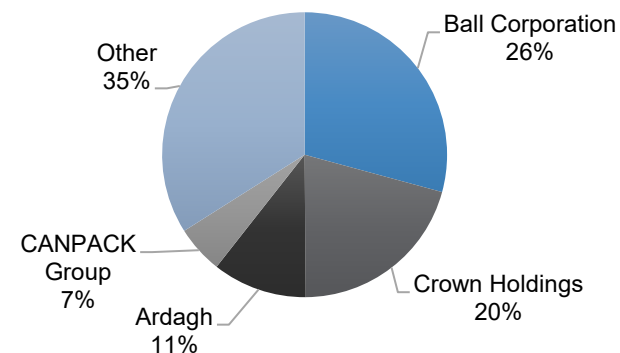
Beverage Can Market Share

(in billions of cans.)

US, Canada, and Mexico Beverage Can Market Share (2025E)		
	billion cans	share
Ball Corporation	50.0	36%
Crown Holdings	30.7	22%
Ardagh	24.8	18%
MCC	16.0	11.5%
Other	17.5	12.5%
Total	139.0	100.0%



Global Beverage Can Market Share (2025E)		
	billion cans	share
Ball Corporation	110.4	26%
Crown Holdings	84.4	20%
Ardagh	48.3	11.5%
CANPACK Group	31.4	7.5%
Other	150.0	35%
Total	424.6	100.0%



Source: Company reports and J.P. Morgan estimates (Note: Sum of company market share may not match total due to rounding)

Global Beverage Can Market (2025E)

(Shipments and Capacity in billions of units.)

Region	Global Can Industry Shipments 2025E	Industry Split by Region 2025E	Ball Can Capacity 2025E	Ball Market Share 2025E	Crown Can Capacity 2025E	Crown Market Share 2025E
North & Central America (US, Canada, Mexico)	139.0	33%	50	36%	31	22%
Europe/Russia	97.0	23%	38	39%	16	17%
Asia/Pacific (China, Southeast Asia)	93.8	22%	NM	NM	16	17%
South America	43.4	10%	20	46%	13	29%
AMEI (North Africa, Middle East, India)	24.9	6%	2	8%	9	36%
Sub-Sahara Africa	8.0	2%	0	0%	0	0%
Other	18.4	4%	0	NM	NM	NM
Total	424.6	100%	110	26%	84	20%

Source: Company reports and J.P. Morgan estimates (Note: Sum of regions may not match total due to rounding)

Global Beverage Can Supply / Demand

(in billions of cans.)

	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Ball Corporation	100.7	105.4	112.5	108.4	104.1	105.1	110.4	114.6	117.9
% y/y		4.7%	6.7%	-3.6%	-4.0%	1.0%	5.0%	3.8%	2.9%
Crown Holdings	70.0	73.0	79.6	82.8	81.8	84.1	84.4	86.7	91.1
% y/y		4.3%	9.0%	4.0%	-1.2%	2.9%	0.4%	2.7%	5.0%
Ardagh	38.9	38.9	41.3	43.4	45.8	46.8	48.3	49.1	50.7
% y/y		0.0%	6.0%	5.1%	5.7%	2.1%	3.4%	1.7%	3.1%
CANPACK	19.9	22.7	26.2	27.8	28.8	33.5	31.44	33.1	34.8
% y/y		14.1%	15.4%	6.1%	3.6%	16.3%	-6.2%	5.2%	5.1%
Other, incl. MCC	120.5	129.9	125.5	135.6	139.5	143.3	150.0	157.2	164.4
% y/y		1.0%	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%
Total Capacity	350.0	369.9	385.0	397.9	400.0	412.8	424.6	440.7	458.9
% y/y		5.7%	4.1%	3.4%	0.5%	3.2%	2.8%	3.8%	4.1%
Demand	350.0	369.9	397.7	409.4	408.3	420.7	435.3	449.5	465.3
% y/y		5.7%	7.5%	2.9%	-0.3%	3.0%	3.5%	3.3%	3.5%
Utilization rates		100%	103%	103%	102%	102%	103%	102%	101%
Over/(Under) Supply	0.0	0.0	(12.7)	(11.5)	(8.4)	(7.9)	(10.8)	(8.8)	(6.4)

	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
<i>Regional split</i>									
NA	33%	33%	34%	35%	34%	33%	33%	32%	31%
SA	10%	11%	11%	10%	10%	10%	10%	10%	10%
EMEA	26%	26%	29%	28%	28%	28%	29%	29%	30%
AP	20%	21%	20%	21%	22%	22%	22%	22%	23%
Other	11%	10%	6%	6%	6%	6%	6%	6%	6%
	100%	100%	100%	100%	100%	100%	100%	100%	100%
<i>Regional growth</i>									
NA			9.5%	0.0%	-2.4%	1.3%	2.9%	1.0%	1.9%
SA			4.0%	-5.0%	2.0%	3.0%	0.8%	3.0%	3.8%
EMEA			8.0%	4.0%	-5.3%	3.8%	4.2%	4.7%	4.2%
AP			6.0%	10.0%	8.0%	5.0%	5.0%	5.0%	5.0%
Other			4.5%	4.5%	2.0%	2.0%	2.0%	2.0%	2.0%

Source: Company reports and J.P. Morgan estimates

U.S. Corn Supply and Demand

(Supply and use in millions of bushels.)

Marketing Year	Supply							Disappearance						Ending Stocks	Stocks- to-Use Ratio
								Domestic Use				Exports	Total Use + Exports		
	Beginning Stocks	Yield Bu/acre	Planted Acres	Harvested Acres	Production	Imports	Supply	Food, + Seed	Ethanol	Feed + Residual	Total Domestic Use				
2005/06	2,114	148	81.8	75.1	11,112	9	13,235	1,416	1,603	6,115	9,134	2,134	11,268	1,967	17.5%
2006/07	1,967	149.1	78.3	70.6	10,531	12	12,510	1,422	2,119	5,540	9,081	2,125	11,206	1,304	11.6%
2007/08	1,304	150.7	93.5	86.5	13,038	20	14,362	1,393	3,049	5,858	10,300	2,437	12,737	1,624	12.8%
2008/09	1,624	153.9	86.0	78.6	12,092	14	13,729	1,316	3,709	5,182	10,207	1,849	12,056	1,673	13.9%
2009/10	1,673	164.7	86.4	79.5	13,092	8	14,774	1,370	4,591	5,125	11,086	1,980	13,066	1,708	13.1%
2010/11	1,708	152.8	88.2	81.4	12,447	28	14,182	1,407	5,019	4,795	11,221	1,834	13,055	1,128	8.6%
2011/12	1,128	147.2	91.9	84.0	12,360	29	13,517	1,428	5,000	4,557	10,985	1,543	12,528	989	7.9%
2012/13	989	123.1	97.3	87.4	10,755	160	11,904	1,397	4,641	4,315	10,353	730	11,083	821	7.4%
2013/14	821	158.1	95.4	87.5	13,829	36	14,686	1,369	5,124	5,040	11,534	1,920	13,454	1,232	9.2%
2014/15	1,232	171.0	90.6	83.1	14,216	32	15,479	1,401	5,200	5,280	11,881	1,867	13,748	1,731	12.6%
2015/16	1,731	168.4	88.0	80.8	13,602	68	15,401	1,424	5,224	5,114	11,763	1,901	13,664	1,737	12.7%
2016/17	1,737	174.6	94.0	86.7	15,148	57	16,942	1,453	5,432	5,470	12,355	2,294	14,649	2,293	15.7%
2017/18	2,293	176.6	90.2	82.7	14,609	36	16,939	1,452	5,605	5,304	12,361	2,438	14,798	2,140	14.5%
2018/19	2,140	176.4	88.9	81.3	14,340	28	16,509	1,415	5,378	5,429	12,222	2,066	14,288	2,221	15.5%
2019/20	2,221	167.5	89.7	81.3	13,620	42	15,883	1,429	4,857	5,900	12,186	1,777	13,963	1,919	13.7%
2020/21	1,919	171.4	90.7	82.3	14,111	24	16,055	1,439	5,028	5,607	12,074	2,747	14,821	1,235	8.3%
2021/22	1,235	176.7	92.9	85.0	15,018	24	16,277	1,437	5,320	5,671	12,427	2,472	14,900	1,377	9.2%
2022/23	1,377	173.4	88.2	78.7	13,651	39	15,066	1,382	5,176	5,486	12,044	1,662	13,706	1,360	9.9%
2023/24	1,360	177.3	94.6	86.5	15,341	28	16,729	1,391	5,489	5,831	12,711	2,255	14,966	1,763	11.8%
2024/25E	1,763	179.3	90.9	83.0	14,892	22	16,677	1,379	5,436	5,438	12,253	2,873	15,126	1,551	10.3%
2025/26E (USDA)	1,551	186.5	98.8	91.3	17,021	28	18,600	1,355	5,550	6,350	13,255	3,325	16,580	2,020	12.2%
2025/26E (mid stock est)	1,551	186.5	98.8	91.3	17,021	28	18,600	1,380	5,525	6,325	13,230	3,275	16,505	2,095	12.7%
2025/26E (high stock est)	1,551	186.5	98.8	91.3	17,021	28	18,600	1,405	5,500	6,300	13,205	3,225	16,430	2,170	13.2%
2026/27E (low stock est)	2,020	182.0	95.0	87.1	15,856	25	17,900	1,330	5,625	6,125	13,080	3,225	16,305	1,595	9.8%
2026/27E (USDA)	2,020	183.0	95.3	87.4	16,000	25	18,045	1,355	5,600	6,100	13,055	3,200	16,255	1,790	11.0%
2026/27E (high stock est)	2,020	184.0	95.6	87.7	16,131	25	18,176	1,380	5,575	6,075	13,030	3,175	16,205	1,971	12.2%

Source: USDA & J.P. Morgan estimates

Note: The 2026/27 USDA estimate is based on the July 2026 WASDE report.

U.S. Corn Supply/Demand Scenarios (2026/27)

(Supply and use in millions of bushels.)

Marketing Year	Supply							Disappearance						Ending Stocks	Stocks- to-Use Ratio	
								Domestic Use								
	Beg. Stocks	Yield Bu/acre	Planted Acres	Harv. Acres	Prod.	Imports	Supply	Food, + Seed	Ethanol	Feed + Residual	Total Domestic Use	Exports	Total Use + Exports			
2026 US Corn Planted Acre Changes																
2026/27E (Low Acres)	2,020	183.0	95.0	87.1	15,943	25	17,988	1,355	5,600	6,100	13,055	3,175	16,230	1,758	10.8%	
2026/27E (Mid Acres)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,600	6,100	13,055	3,175	16,230	1,816	11.2%	
2026/27E (High Acres)	2,020	183.0	95.6	87.7	16,044	25	18,089	1,355	5,600	6,100	13,055	3,175	16,230	1,859	11.5%	
2026 US Corn Yield Changes																
2026/27E (Low Yields)	2,020	181.0	95.3	87.4	15,826	25	17,871	1,355	5,600	6,100	13,055	3,175	16,230	1,641	10.1%	
2026/27E (Mid Yields)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,600	6,100	13,055	3,175	16,230	1,816	11.2%	
2026/27E (High Yields)	2,020	185.0	95.3	87.4	16,175	25	18,220	1,355	5,600	6,100	13,055	3,175	16,230	1,990	12.3%	
2026 US Corn Export Changes																
2026/27E (High Exports)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,600	6,100	13,055	3,400	16,455	1,591	9.7%	
2026/27E (Mid Exports)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,600	6,100	13,055	3,200	16,255	1,791	11.0%	
2026/27E (Low Exports)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,600	6,100	13,055	3,000	16,055	1,991	12.4%	
2026 US Corn Ethanol Demand																
2026/27E (High Ethanol Demand)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,700	6,100	13,155	3,175	16,330	1,716	10.5%	
2026/27E (Mid Ethanol Demand)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,600	6,100	13,055	3,175	16,230	1,816	11.2%	
2026/27E (Low Ethanol Demand)	2,020	183.0	95.3	87.4	16,000	25	18,046	1,355	5,500	6,100	12,955	3,175	16,130	1,916	11.9%	

Source: USDA & J.P. Morgan estimates

World Corn Supply and Demand

Corn-World	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25E	2025/26E	2026/27E
Beginning Stocks (M/mt)	131	115	123	146	216	280	313	348	341	328	315	297	313	304	314	296	299
Area Harvested (M/ha)	166	176	184	189	189	187	195	195	193	195	201	208	203	208	207	215	214
Yield (M mt/ha)	5.11	5.18	4.91	5.47	5.62	5.45	5.78	5.59	5.87	5.80	5.65	5.88	5.75	5.92	5.97	6.17	6.07
Production (M/mt)	849	910	903	1,034	1,061	1,021	1,129	1,087	1,133	1,128	1,134	1,221	1,166	1,231	1,234	1,328	1,297
Total Supply (M/mt)	981	1,025	1,026	1,179	1,277	1,301	1,442	1,434	1,474	1,456	1,449	1,518	1,479	1,535	1,548	1,624	1,596
Imports (M/mt)	94	101	99	125	125	140	138	153	166	167	185	185	173	198	186	197	204
Exports (M/mt)	92	117	95	131	143	121	162	150	183	173	183	207	180	193	187	221	210
Net Exports (M/mt)	(2)	16	(4)	6	18	(19)	23	(3)	16	5	(2)	22	7	(5)	1	24	6
Feed& Residual (M/mt)	517	525	530	580	594	618	666	675	703	716	727	741	730	769	784	825	833
Food, Seed, Ind. (M/mt)	351	361	354	377	385	390	405	422	426	419	427	441	438	457	466	476	482
Net Exports (M/mt)	(2)	16	(4)	6	18	(19)	23	(3)	16	5	(2)	22	7	(5)	1	24	6
Total Use (M/mt)	866	902	881	963	997	988	1,095	1,094	1,146	1,141	1,152	1,204	1,175	1,221	1,252	1,325	1,320
Ending Stocks (M/mt)	115	123	146	216	280	313	348	341	328	315	297	313	304	314	296	299	275
Stocks-to-Use Ratio	13.3%	13.6%	16.5%	22.4%	28.1%	31.7%	31.8%	31.2%	28.6%	27.6%	25.8%	26.0%	25.8%	25.7%	23.7%	22.5%	20.8%

Source: USDA & J.P. Morgan estimates

Key Global Corn Producers, Exporters, and Importers (2026/27E)

World Corn	2026/27E	2026/27E	
Producers	Production	Production	
Regions	M tons	M bushels	% share
U.S.	406.42	16,000	31%
China	307.00	12,086	24%
Brazil	139.00	5,472	11%
Argentina	55.00	2,165	4%
Europe	53.78	2,117	4%
S.E. Asia	31.20	1,228	2%
Ukraine	30.00	1,181	2%
Mexico	24.60	968	2%
South Africa	16.50	650	1%
Canada	16.30	642	1%
Russia	15.80	622	1%
Other	201.49	7,932	16%
World Corn Producers	1,297.09	51,064	100%

World Corn	2026/27E	2026/27E	
Exporters	Exports	Exports	
Regions	M tons	M bushels	% share
U.S.	81.28	3,200	39%
Brazil	44.00	1,732	21%
Argentina	38.00	1,496	18%
Ukraine	23.00	905	11%
Russia	4.00	157	2%
South Africa	2.20	87	1%
Canada	3.00	118	1%
Europe	1.90	75	1%
S.E. Asia	0.57	22	0%
Mexico	0.02	1	0%
China	0.02	1	0%
Other	11.89	468	6%
World Corn Exporters	209.88	8,263	100%

World Corn	2026/27E	2026/27E	
Importers	Imports	Imports	
Regions	M tons	M bushels	% share
Mexico	27.70	1,090	14%
S.E. Asia	24.55	966	12%
Europe	22.50	886	11%
Japan	15.50	610	8%
Egypt	13.20	520	6%
South Korea	11.50	453	6%
China	6.00	236	3%
Canada	2.00	79	1%
U.S.	0.64	25	0%
Russia	0.05	2	0%
Ukraine	0.01	0	0%
Other	80.35	3,163	39%
World Corn Importers	204.00	8,031	100%

Total US Corn Exports (Monthly)

(Metric Tons 000)

Calendar Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026YTD
January	2,519	3,365	3,195	665	2,732	4,761	4,913	2,614	3,881	6,537	7,204
February	2,918	5,080	3,954	7,031	3,273	6,153	6,204	2,419	5,086	5,549	6,803
March	5,197	7,417	6,500	3,772	3,926	7,806	8,044	5,403	5,663	7,008	7,108
April	4,609	5,085	6,681	4,822	5,795	9,586	6,224	4,991	6,195	6,958	9,009
May	4,194	4,651	8,161	5,508	4,980	7,761	6,285	5,170	6,165	8,067	6,447
June	6,726	5,547	6,163	2,519	4,872	6,135	6,198	4,418	4,802	6,374	7,048
July	5,002	3,907	5,596	3,101	4,806	6,123	3,914	1,914	4,228	6,666	
August	5,121	3,381	6,515	2,933	3,933	3,179	1,328	2,597	5,602	5,452	
September	5,432	3,112	4,572	1,607	2,743	2,497	2,210	2,671	3,406	4,988	
October	3,497	2,212	4,674	2,332	4,111	3,397	1,899	2,319	4,266	7,236	
November	2,834	2,847	5,523	2,405	3,519	3,753	1,631	3,718	3,712	7,435	
December	4,584	2,571	3,833	2,031	4,884	5,006	4,224	3,683	4,364	6,378	
Total	52,633	49,174	65,367	38,726	49,575	66,156	53,074	41,918	57,369	78,648	43,619YTD
y/y Δ	23.3%	(6.6%)	32.9%	(40.8%)	28.0%	33.4%	(19.8%)	(21.0%)	36.9%	37.1%	7.7%YTD
Marketing Year	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26YTD
Sept 1 – Aug 31	45,750	54,780	57,506	48,952	42,694	66,760	57,763	39,490	54,012	68,359	69,656YTD
y/y Δ	0.8%	19.7%	5.0%	(14.9%)	(12.8%)	56.4%	(13.5%)	(31.6%)	36.8%	26.6%	23.9%YTD

(Million Bushels)

Calendar Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026YTD
January	99	132	126	26	108	187	193	103	153	257	284
February	115	200	156	277	129	242	244	95	200	218	268
March	205	292	256	148	155	307	317	213	223	276	280
April	181	200	263	190	228	377	245	196	244	274	355
May	165	183	321	217	196	306	247	204	243	318	254
June	265	218	243	99	192	242	244	174	189	251	277
July	197	154	220	122	189	241	154	75	166	262	
August	202	133	256	115	155	125	52	102	221	215	
September	214	123	180	63	108	98	87	105	134	196	
October	138	87	184	92	162	134	75	91	168	285	
November	112	112	217	95	139	148	64	146	146	293	
December	180	101	151	80	192	197	166	145	172	251	
Total	2,072	1,936	2,573	1,525	1,952	2,604	2,089	1,650	2,258	3,096	1,717YTD
y/y Δ	23.3%	(6.6%)	32.9%	(40.8%)	28.0%	33.4%	(19.8%)	(21.0%)	36.9%	37.1%	7.7%YTD
Marketing Year	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26YTD
Sept 1 – Aug 31	1,801	2,157	2,264	1,927	1,681	2,628	2,274	1,555	2,126	2,691	2,742YTD
y/y Δ	0.8%	19.7%	5.0%	(14.9%)	(12.8%)	56.4%	(13.5%)	(31.6%)	36.8%	26.6%	29.9%YTD

Source: USDA, Bloomberg Finance L.P.

Total Global Corn Imports to China

(Metric Tons 000)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026YTD
January	580	10	160	390	400	660	3,010	2,750	2,240	3,590	100	370
February	603	60	140	100	170	270	1,780	1,930	3,090	2,600	80	170
March	51	580	5	60	420	320	1,930	2,410	2,190	1,710	80	220
April	141	1,160	-	380	660	890	1,850	2,210	1,000	1,180	180	50
May	404	1,040	40	760	750	640	3,160	2,080	1,660	1,050	190	60
June	873	70	380	520	720	880	3,570	2,210	1,850	920	160	
July	1,108	29	910	330	390	910	2,860	1,540	1,680	1,090	60	
August	608	27	380	330	230	1,020	3,230	1,800	1,200	430	40	
September	170	20	250	40	140	1,080	3,530	1,530	1,650	310	60	
October	43	10	70	80	90	1,140	1,300	550	2,040	250	360	
November	19	30	20	120	100	1,230	790	740	3,590	300	560	
December	133	140	450	420	730	2,250	1,330	870	4,950	340	800	
Total	4,731	3,176	2,805	3,530	4,800	11,290	28,340	20,620	27,140	13,770	2,670	870YTD
y/y Δ	81.9%	(32.9%)	(11.7%)	25.8%	36.0%	135.2%	151.0%	(27.2%)	31.6%	(49.3%)	(80.6%)	38.1%YTD
Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sep 1 – Aug 31	5,368	3,341	2,215	3,660	4,400	6,650	27,090	23,880	18,600	24,800	2,090	2,650YTD
y/y Δ	64.7%	(37.8%)	(33.7%)	65.2%	20.2%	51.1%	57.7%	(11.8%)	(22.1%)	33.3%	(91.6%)	44.8%YTD

(Million Bushels)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026YTD
January	23	0	6	15	16	26	118	108	88	141	4	15
February	24	2	6	4	7	11	70	76	122	102	3	7
March	2	23	0	2	17	13	76	95	86	67	3	9
April	6	46	-	15	26	35	73	87	39	46	7	2
May	16	41	2	30	30	25	124	82	65	41	7	2
June	34	3	15	20	28	35	141	87	73	36	6	
July	44	1	36	13	15	36	113	61	66	43	2	
August	24	1	15	13	9	40	127	71	47	17	2	
September	7	1	10	2	6	43	139	60	65	12	2	
October	2	0	3	3	4	45	51	22	80	10	14	
November	1	1	1	5	4	48	31	29	141	12	22	
December	5	6	18	17	29	89	52	34	195	13	31	
Total	186	125	110	139	189	444	1,116	812	1,068	542	105	34YTD
y/y Δ	81.9%	(32.9%)	(11.7%)	25.8%	36.0%	135.2%	151.0%	(27.2%)	31.6%	(49.3%)	(80.6%)	38.1%YTD
Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sep 1 – Aug 31	211	132	87	144	173	262	1,066	940	732	976	82	104YTD
y/y Δ	64.7%	(37.8%)	(33.7%)	65.2%	20.2%	51.1%	307.4%	(11.8%)	(22.1%)	33.3%	(91.6%)	44.8%YTD

Corn Economics for the U.S. Farmer

	Planting 2014	Planting 2015	Planting 2016	Planting 2017	Planting 2018	Planting 2019	Planting 2020	Planting 2021	Planting 2022	Planting 2023	Planting 2024	Planting 2025E	Planting 2026E
Yield/acre*	231	200	228	227	237	208	214	221	235	232	244	242	241
Price/bu*	\$3.76	\$3.72	\$3.49	\$3.40	\$3.60	\$3.54	\$4.34	\$5.79	\$6.29	\$4.50	\$4.24	\$4.09	\$4.58
Crop Revenue	\$869	\$744	\$796	\$772	\$853	\$736	\$929	\$1,280	\$1,478	\$1,044	\$1,035	\$990	\$1,104
ARC/PLC or ACRE	4	45	12	1	0	10	0	0	0	0	3	55	25
Other Gov't payments	0	0	0	0	1	110	68	0	0	0	43	44	0
Crop insurance proceeds	10	31	2	6	2	14	9	14	2	22	10	4	0
Gross revenue	\$883	\$820	\$810	\$779	\$856	\$870	\$1,006	\$1,294	\$1,480	\$1,066	\$1,090	\$1,093	\$1,129
Fertilizers	171	166	154	135	131	154	143	154	240	289	218	210	229
Pesticides	67	66	64	73	76	76	78	90	128	124	118	116	116
Seed	120	118	116	115	115	114	112	114	117	129	127	130	129
Drying	28	15	13	16	14	22	22	20	26	24	19	14	15
Storage	12	14	11	15	15	15	10	6	8	11	16	14	14
Crop Insurance	24	24	22	24	22	22	21	31	39	24	26	30	31
Total direct costs	\$422	\$403	\$380	\$378	\$373	\$403	\$386	\$415	\$558	\$601	\$524	\$514	\$534
Total power costs	\$132	\$124	\$119	\$122	\$128	\$128	\$131	\$143	\$167	169	179	178	181
Total overhead costs	\$63	\$65	\$64	\$69	\$73	\$75	\$77	\$76	\$83	101	106	114	118
Total non-land costs	\$617	\$592	\$563	\$569	\$574	\$606	\$594	\$634	\$808	\$871	\$809	\$806	\$833
Operator and land return	\$266	\$228	\$247	\$210	\$282	\$264	\$412	\$660	\$672	\$195	\$281	\$287	\$296
Land costs (e.g. cash rent)	293	278	273	267	274	275	279	311	358	359	337	326	321
Farmer return	(\$27)	(\$50)	(\$26)	(\$57)	\$8	(\$11)	\$133	\$349	\$314	(\$164)	(\$56)	(\$39)	(\$25)

Source: Farmdoc Daily (May 2026) and J.P. Morgan estimates

Note: The amount of pre-harvest hedging will have a meaningful effect on crop prices and revenues. The crop price for 2026E represents the December 2026 harvest price.

Soybean Economics for the U.S. Farmer

	Planting 2014	Planting 2015	Planting 2016	Planting 2017	Planting 2018	Planting 2019	Planting 2020	Planting 2021	Planting 2022	Planting 2023	Planting 2024	Planting 2025E	Planting 2026E
Yield/acre*	64	66	69	68	74	64	66	72	71	75	75	75	76
Price/bu*	\$10.67	\$9.39	\$9.99	\$9.70	\$8.80	\$8.87	\$10.76	\$13.40	\$14.56	\$11.30	\$10.41	\$10.24	\$11.91
Crop Revenue	\$683	\$620	\$689	\$660	\$651	\$568	\$710	\$965	\$1,034	\$848	\$781	\$768	\$905
ARC/PLC or ACRE	4	45	12	1	0	10	0	0	0	0	2	55	25
Other Gov't payments	0	0	0	0	122	94	45	0	0	0	30	31	0
Crop insurance proceeds	5	7	1	3	4	9	4	5	4	5	9	2	0
Gross revenue	\$692	\$672	\$702	\$664	\$777	\$681	\$759	\$970	\$1,038	\$853	\$821	\$856	\$930
Fertilizers	58	56	49	44	42	45	42	45	71	87	65	63	66
Pesticides	41	40	40	44	46	46	47	54	77	75	71	70	70
Seed	77	76	74	73	73	73	71	72	74	83	81	84	83
Drying	1	1	1	1	1	0	0	0	0	0	0	0	0
Storage	7	8	8	8	8	8	5	5	4	6	9	8	8
Crop Insurance	18	16	14	16	14	15	14	16	26	10	10	10	11
Total direct costs	\$202	\$197	\$186	\$186	\$184	\$187	\$179	\$192	\$252	\$261	\$236	\$235	\$238
Total power costs	\$117	\$107	\$106	\$108	\$102	\$113	\$113	\$124	\$149	158	155	158	160
Total overhead costs	\$59	\$59	\$61	\$64	\$67	\$72	\$72	\$69	\$77	96	108	109	113
Total non-land costs	\$378	\$363	\$353	\$358	\$353	\$372	\$364	\$385	\$478	\$515	\$499	\$502	\$511
Operator and land return	\$314	\$309	\$349	\$306	\$424	\$309	\$395	\$585	\$560	\$338	\$322	\$354	\$419
Land costs (e.g. cash rent)	293	278	273	267	274	275	279	311	358	359	337	326	321
Farmer return	\$21	\$31	\$76	\$39	\$150	\$34	\$116	\$274	\$202	(\$22)	(\$15)	\$28	\$98

Source: Farmdoc Daily (May 2026) and J.P. Morgan estimates

Note: The amount of pre-harvest hedging will have a meaningful effect on crop prices and revenues. The crop price for 2026E represents the November 2026 harvest price.

U.S. Soybean Supply and Demand

(Supply and use in millions of bushels.)

Marketing Year	Supply							Disappearance				Ending Stocks	Stocks-to-Use Ratio
	Beginning Stocks	Yield Bu/acre	Planted Acres	Harvested Acres	Production	Imports	Supply	Crushings	Exports	Seeds + Residual	Total Use + Exports		
2005/06	256	43.1	72.0	71.3	3068	3	3,327	1,739	940	199	2,878	449	15.6%
2006/07	449	42.9	75.5	74.6	3197	9	3,655	1,808	1,116	157	3,081	574	18.6%
2007/08	574	41.7	64.7	64.1	2677	10	3,261	1,803	1,159	94	3,056	205	6.7%
2008/09	205	39.7	75.7	74.7	2967	13	3,185	1,662	1,279	106	3,047	138	4.5%
2009/10	138	44.0	77.5	76.4	3359	15	3,512	1,752	1,499	110	3,361	151	4.5%
2010/11	151	43.5	77.4	76.6	3329	14	3,495	1,648	1,505	127	3,280	215	6.6%
2011/12	215	41.9	75.0	73.8	3094	16	3,325	1,703	1,365	87	3,155	169	5.4%
2012/13	169	40.0	77.2	76.1	3042	41	3,252	1,689	1,317	105	3,111	141	4.5%
2013/14	141	44.0	76.8	76.3	3358	72	3,570	1,734	1,638	107	3,478	92	2.6%
2014/15	92	47.5	83.3	82.6	3,927	33	4,052	1,873	1,842	146	3,862	191	4.9%
2015/16	191	48.0	82.7	81.7	3,926	24	4,140	1,886	1,942	115	3,944	197	5.0%
2016/17	197	51.9	83.5	82.7	4,296	22	4,516	1,901	2,166	147	4,214	302	7.2%
2017/18	302	49.3	90.2	89.5	4,412	22	4,735	2,055	2,134	109	4,297	438	10.2%
2018/19	438	50.6	89.2	87.6	4,428	14	4,880	2,092	1,752	127	3,971	909	22.9%
2019/20	909	47.4	76.1	74.9	3,552	15	4,476	2,165	1,679	108	3,952	525	13.3%
2020/21	525	51.0	83.4	82.6	4,216	20	4,761	2,141	2,266	97	4,504	257	5.7%
2021/22	257	51.7	87.2	86.3	4,464	16	4,737	2,204	2,152	107	4,463	274	6.1%
2022/23	274	49.6	87.5	86.2	4,269.6	25	4,569	2,212	1,980	113	4,305	264	6.1%
2023/24	264	50.6	83.6	82.3	4,162	21	4,447	2,285	1,700	119	4,105	342	8.3%
2024/25E	342	50.7	87.3	86.2	4,374	30	4,746	2,445	1,892	84	4,421	325	7.3%
2025/26E (low stock est)	325	53.0	81.2	80.4	4,262	25	4,612	2,670	1,540	112	4,322	290	6.7%
2025/26E (USDA)	325	53.0	81.2	80.4	4,262	25	4,612	2,650	1,520	112	4,282	330	7.7%
2025/26E (high stock est)	325	53.0	81.2	80.4	4,262	25	4,612	2,630	1,500	112	4,242	370	8.7%
2026/27E (low stock est)	330	52.8	85.1	84.1	4,441	25	4,795	2,770	1,680	110	4,560	235	5.2%
2026/27E (USDA)	330	53.0	85.4	84.4	4,475	25	4,830	2,750	1,660	110	4,520	310	6.9%
2026/27E (high stock est)	330	53.2	85.7	84.7	4,506	25	4,860	2,730	1,640	110	4,480	380	8.5%

Source: USDA & J.P. Morgan estimates

Note: The 2026/27 USDA estimate is based on the July 2026 WASDE report.

U.S. Soybean Supply/Demand Scenarios (2026/27)

(Supply and use in millions of bushels.)

Marketing Year	Supply							Disappearance				Ending Stocks	Stocks-to-Use Ratio
	Beginning	Yield	Planted	Harvested	Production	Imports	Supply	Crushings	Exports	Seeds	Total Use		
	Stocks	Bu/acre	Acres	Acres						+ Residual	+ Exports		
2026 US Soy Planted Acre Changes													
2026/27E (Low Acres)	330	53.0	85.1	84.1	4,458	25	4,812	2,750	1,660	110	4,520	292	6.5%
2026/27E (Mid Acres)	330	53.0	85.4	84.4	4,473	25	4,828	2,750	1,660	110	4,520	308	6.8%
2026/27E (High Acres)	330	53.0	85.7	84.7	4,489	25	4,843	2,750	1,660	110	4,520	323	7.2%
2026 US Soy Yield Changes													
2026/27E (Low Yields)	330	52.5	85.4	84.4	4,431	25	4,786	2,750	1,660	110	4,520	266	5.9%
2026/27E (Mid Yields)	330	53.0	85.4	84.4	4,473	25	4,828	2,750	1,660	110	4,520	308	6.8%
2026/27E (High Yields)	330	53.5	85.4	84.4	4,515	25	4,870	2,750	1,660	110	4,520	350	7.7%
2026 US Soy Export Changes													
2026/27E (High Exports)	330	53.0	85.4	84.4	4,473	25	4,828	2,750	1,710	110	4,570	258	5.6%
2026/27E (Mid Exports)	330	53.0	85.4	84.4	4,473	25	4,828	2,750	1,660	110	4,520	308	6.8%
2026/27E (Low Exports)	330	53.0	85.4	84.4	4,473	25	4,828	2,750	1,610	110	4,470	358	8.0%
2026 US Soy Crushings													
2026/27E (High Crushings)	330	53.0	85.4	84.4	4,473	25	4,828	2,800	1,660	110	4,570	258	5.6%
2026/27E (Mid Crushings)	330	53.0	85.4	84.4	4,473	25	4,828	2,750	1,660	110	4,520	308	6.8%
2026/27E (Low Crushings)	330	53.0	85.4	84.4	4,473	25	4,828	2,700	1,660	110	4,470	358	8.0%

Source: USDA & J.P. Morgan estimates

World Soybean Supply and Demand

Soybean-World	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25E	2025/26E	2026/27E
Beginning Stocks (M/mt)	63	73	58	58	64	79	80	96	100	114	95	99	94	102	115	126	125
Area Harvested (M/ha)	104	103	110	113	119	121	121	125	126	124	130	132	137	141	146	143	146
Yield (M mt/ha)	2.55	2.34	2.44	2.50	2.70	2.61	2.91	2.74	2.88	2.75	2.84	2.74	2.76	2.82	2.93	3.01	3.02
Production (M/mt)	265	241	269	283	321	316	351	344	363	342	370	361	378	396	428	429	442
Total Supply (M/mt)	328	315	327	342	385	395	431	439	463	456	465	459	472	498	543	555	567
Imports (M/mt)	90	95	97	113	124	134	145	154	146	165	166	155	169	178	179	186	189
Exports (M/mt)	91	92	100	113	126	133	148	153	149	166	165	154	172	178	184	187	190
Net Exports (M/mt)	2	(3)	3	(0)	2	(1)	3	(1)	3	1	(1)	(0)	3	(1)	5	1	1
Crush (M/mt)	222	229	232	243	265	276	288	296	299	313	318	316	316	331	359	372	384
Food (M/mt)	15	15	16	16	17	18	18	19	20	21	22	22	23	24	25	26	26
Feed & Waste (M/mt)	15	15	18	19	22	24	25	25	26	27	27	28	28	29	28	31	32
Total Use (M/mt)	253	259	265	278	304	317	332	340	346	360	367	366	367	384	412	429	442
Ending Stocks (M/mt)	73	58	58	64	79	80	96	100	114	95	99	94	102	115	126	125	124
Stocks-to-Use Ratio	29.1%	22.4%	22.0%	22.9%	26.1%	25.1%	28.8%	29.3%	33.1%	26.4%	26.9%	25.6%	27.8%	30.0%	30.6%	29.2%	28.1%

Key Global Soybean Producers, Exporters, and Importers (2026/27E)

World Soy	2026/27E	2026/27E	
Producers	Production	Production	
Regions	M tons	M bushels	% share
Brazil	186.00	6,834	42%
U.S.	121.79	4,475	28%
Argentina	50.00	1,837	11%
China	21.00	772	5%
Paraguay	11.10	408	3%
Europe	2.98	109	1%
S.E. Asia	0.38	14	0%
Other	48.45	1,780	11%
World Soy Producers	441.70	16,230	100%

World Soy	2026/27E	2026/27E	
Exporters	Exports	Exports	
Regions	M tons	M bushels	% share
Brazil	118.00	4,336	62%
U.S.	45.18	1,660	24%
Paraguay	7.35	270	4%
Argentina	6.20	228	3%
Europe	0.25	9	0%
China	0.10	4	0%
S.E. Asia	0.02	1	0%
Other	13.31	489	7%
World Soy Exporters	190.41	6,996	100%

World Soy	2026/27E	2026/27E	
Importers	Imports	Imports	
Regions	M tons	M bushels	% share
China	115.00	4,225	61%
Europe	13.20	485	7%
S.E. Asia	11.56	425	6%
Mexico	6.75	248	4%
Argentina	6.50	239	3%
Brazil	0.80	29	0%
U.S.	0.68	25	0%
Other	34.78	1,278	18%
World Soy Importers	189.27	6,954	100%

Total US Soybean Exports (Monthly)

(Metric Tons 000)

Calendar Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026YTD
January	5,238	5,798	5,188	908	5,925	8,520	5,682	7,479	4,600	5,699	6,550
February	5,687	4,822	4,708	8,630	2,863	5,285	4,481	6,033	6,613	3,478	4,218
March	3,257	4,114	3,809	3,459	2,127	2,202	3,545	3,582	2,929	3,221	3,928
April	1,333	2,659	1,959	2,259	2,426	1,589	2,970	2,003	1,671	2,377	3,364
May	777	1,372	3,194	2,714	1,672	1,125	2,309	1,061	1,603	1,457	2,341
June	1,068	1,728	3,028	2,867	1,372	819	2,702	1,177	1,079	1,255	1,569
July	2,210	2,053	3,019	3,160	2,884	1,043	1,864	1,087	1,220	2,038	
August	3,928	2,744	3,738	5,249	3,824	972	2,868	1,813	2,084	1,864	
September	3,440	4,068	3,021	2,854	4,566	1,845	1,788	2,061	1,866	2,067	
October	9,972	8,251	4,364	6,480	12,013	8,737	8,033	7,694	10,537	5,501	
November	11,017	10,923	5,669	6,654	10,481	10,489	9,024	8,455	9,155	3,871	
December	9,043	5,359	3,958	4,857	11,889	9,543	9,128	4,486	6,823	3,747	
Total	56,970	53,888	45,655	50,090	62,041	52,167	54,394	46,930	50,181	36,574	21,968YTD
y/y Δ	17.0%	(5.4%)	(15.3%)	9.7%	23.9%	(15.9%)	4.3%	(13.7%)	6.9%	(27.1%)	25.6%YTD
Marketing Year	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	50,536	58,760	57,243	46,257	43,937	60,503	57,034	52,208	44,495	49,770	37,154YTD
y/y Δ	1.9%	16.3%	(2.6%)	(19.2%)	(5.0%)	37.7%	(5.7%)	(8.5%)	(14.8%)	11.9%	(19.0%)YTD

(Million Bushels)

Calendar Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026YTD
January	192	213	191	33	218	313	209	275	169	209	241
February	209	177	173	317	105	194	165	222	243	128	155
March	120	151	140	127	78	81	130	132	108	118	144
April	49	98	72	83	89	58	109	74	61	87	124
May	29	50	117	100	61	41	85	39	59	54	86
June	39	63	111	105	50	30	99	43	40	46	58
July	81	75	111	116	106	38	68	40	45	75	
August	144	101	137	193	141	36	105	67	77	68	
September	126	149	111	105	168	68	66	76	69	76	
October	366	303	160	238	441	321	295	283	387	202	
November	405	401	208	244	385	385	332	311	336	142	
December	332	197	145	178	437	351	335	165	251	138	
Total	2,093	1,980	1,678	1,840	2,280	1,917	1,999	1,724	1,844	1,344	807YTD
y/y Δ	17.0%	(5.4%)	(15.3%)	9.7%	23.9%	(15.9%)	4.3%	(13.7%)	6.9%	(27.1%)	25.6%YTD
Marketing Year	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	1,857	2,159	2,103	1,700	1,614	2,223	2,096	1,918	1,635	1,829	1,365YTD
y/y Δ	1.9%	16.3%	(2.6%)	(19.2%)	(5.0%)	37.7%	(5.7%)	(8.5%)	(14.8%)	11.9%	(19.0%)YTD

Source: USDA, Bloomberg Finance L.P.

China Soybean Imports from the U.S.

(Metric Tons 000)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
January	6,643,600	4,634,265	6,777,647	5,814,257	135,772	3,393,933	6,667,200	6,333,079	5,358,868	3,325,940	4,918,848	38,002
February	4,008,724	3,888,122	4,426,103	3,344,521	907,754	2,707,350	5,285,630	3,706,156	6,227,082	1,631,314	4,213,515	1,450,735
March	3,800,983	4,572,560	4,222,970	3,100,553	1,511,080	1,711,663	7,178,737	3,372,742	4,830,298	2,180,305	2,436,146	1,846,646
April	1,856,419	2,024,574	1,777,058	4,947	1,754,852	665,591	2,152,835	1,637,769	1,824,128	2,445,784	1,377,889	3,332,744
May	623,342	552,221	1,470,025	489,540	977,024	491,697	244,431	1,732,550	944,850	1,265,939	1,627,316	1,660,975
June	2,771	101,582	387,093	630,253	614,806	267,553	54,806	773,134	527,823	1,309,225	1,579,247	
July	57,385	3,962	499,922	308,127	911,888	38,331	42,277	382,018	142,150	475,392	420,874	
August	6,370	364,003	179,490	265,378	1,675,034	166,370	17,576	287,864	120,073	202,383	227,205	
September	201	1,355,702	937,262	132,248	1,730,206	1,168,591	169,439	1,153,318	133,692	1,711,508	-	
October	512,139	2,430,768	1,325,207	66,955	1,147,339	3,399,545	775,331	773,167	228,264	541,434	-	
November	4,254,200	5,617,534	4,661,950	-	2,555,402	6,040,965	3,634,612	3,379,041	2,295,004	2,791,259	-	
December	6,646,722	8,112,419	6,189,320	69,298	3,093,488	5,836,215	6,088,817	6,015,683	3,848,991	4,254,887	-	
Total	28,412,856	33,657,712	32,854,047	14,226,077	17,014,645	25,887,804	32,311,691	29,546,521	26,481,223	22,135,370	16,801,040	8,329,102YTD
y/y Δ	(5.4%)	18.5%	(2.4%)	(56.7%)	19.6%	52.2%	24.8%	(8.6%)	(10.4%)	(16.4%)	(24.1%)	(42.8%) YTD

Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	29,697,233	27,554,551	37,256,731	27,071,315	8,756,711	17,968,923	38,088,808	28,893,511	31,296,481	19,342,233	26,100,128	8,329,102YTD
y/y Δ	9.8%	(7.2%)	35.2%	(27.3%)	(67.7%)	105.2%	112.0%	(24.1%)	8.3%	(38.2%)	34.9%	(65.1%) YTD

(Million Bushels)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
January	244	170	249	214	5	125	245	233	197	122	181	1
February	147	143	163	123	33	99	194	136	229	60	155	53
March	140	168	155	114	56	63	264	124	177	80	90	68
April	68	74	65	0	64	24	79	60	67	90	51	122
May	23	20	54	18	36	18	9	64	35	47	60	61
June	0	4	14	23	23	10	2	28	19	48	58	
July	2	0	18	11	34	1	2	14	5	17	15	
August	0	13	7	10	62	6	1	11	4	7	8	
September	0	50	34	5	64	43	6	42	5	63	-	
October	19	89	49	2	42	125	28	28	8	20	-	
November	156	206	171	-	94	222	134	124	84	103	-	
December	244	298	227	3	114	214	224	221	141	156	-	
Total	1,044	1,237	1,207	523	625	951	1,187	1,086	973	813	617	306YTD
y/y Δ	(5.4%)	18.5%	(2.4%)	(56.7%)	19.6%	52.2%	24.8%	(8.6%)	(10.4%)	(16.4%)	(24.1%)	(42.8%) YTD

Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	1,091	1,012	1,369	995	322	660	1,400	1,062	1,150	711	959	306YTD
y/y Δ	9.8%	(7.2%)	35.2%	(27.3%)	(67.7%)	105.2%	112.0%	(24.1%)	8.3%	(38.2%)	34.9%	(65.1%) YTD

Source: China Customs General Administration, Bloomberg Finance L.P.

China Soybean Imports from Brazil

(Metric Tons 000)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
January	28,000	620,743	253,028	2,073,687	4,933,786	2,934,102	998,241	2,215,417	1,831,905	4,074,606	2,220,226	4,258,110
February	-	267,146	687,350	1,746,069	1,986,344	2,205,215	33,450	1,291,542	408,681	2,889,270	1,369,191	2,303,675
March	507,343	1,259,725	1,749,322	2,332,587	2,792,529	2,098,600	315,334	2,865,065	1,670,363	3,022,848	951,006	1,404,128
April	3,433,274	4,815,048	6,150,254	4,400,989	5,785,671	5,939,011	5,076,886	6,302,379	5,299,792	5,915,219	4,599,604	4,753,386
May	5,426,828	7,016,647	7,939,055	9,123,507	6,302,462	8,862,098	9,234,898	7,794,408	10,937,566	8,813,079	12,110,220	9,956,308
June	6,659,720	6,874,940	6,417,862	7,838,563	5,498,144	10,512,014	10,476,377	7,237,928	9,533,481	9,720,823	10,615,869	
July	6,369,442	5,977,850	7,635,939	7,044,374	6,420,960	8,177,767	7,875,167	6,971,234	9,228,483	9,122,663	10,393,104	
August	5,533,289	4,956,337	6,079,284	7,945,943	6,684,762	8,149,091	9,037,448	6,246,981	9,086,367	10,237,604	10,485,379	
September	5,130,244	3,756,358	5,938,073	7,591,955	4,788,870	7,245,096	5,936,015	5,577,015	6,880,459	8,445,203	10,960,517	
October	3,884,696	1,452,048	3,376,114	6,530,470	3,793,215	4,233,132	3,296,974	2,795,188	4,814,108	5,553,752	7,115,742	
November	2,138,217	725,002	2,760,485	5,069,352	3,857,470	2,743,092	3,753,112	2,536,543	5,289,371	3,939,924	5,852,857	
December	1,015,785	314,709	1,941,927	4,386,214	4,832,272	1,177,055	2,116,701	2,564,458	4,979,799	2,935,355	5,655,026	
Total	40,126,838	38,036,553	50,928,693	66,083,710	57,676,485	64,276,273	58,150,603	54,398,158	69,960,375	74,650,346	83,328,741	22,675,607YTD
y/y Δ	25.4%	(5.2%)	33.9%	29.8%	(12.7%)	11.4%	(9.5%)	(6.5%)	28.6%	6.7%	10.3%	6.7%YTD

Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	34,674,041	43,957,378	43,160,211	56,522,318	63,982,649	66,149,725	58,446,176	56,027,756	61,469,842	75,759,849	73,598,833	52,259,749YTD
y/y Δ	4.4%	26.8%	(1.8%)	31.0%	13.2%	3.4%	(11.6%)	(4.1%)	9.7%	23.2%	(2.9%)	24.1%YTD

(Million Bushels)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
January	1	23	9	76	181	108	37	81	67	150	82	156
February	-	10	25	64	73	81	1	47	15	106	50	85
March	19	46	64	86	103	77	12	105	61	111	35	52
April	126	177	226	162	213	218	187	232	195	217	169	175
May	199	258	292	335	232	326	339	286	402	324	445	366
June	245	253	236	288	202	386	385	266	350	357	390	
July	234	220	281	259	236	300	289	256	339	335	382	
August	203	182	223	292	246	299	332	230	334	376	385	
September	189	138	218	279	176	266	218	205	253	310	403	
October	143	53	124	240	139	156	121	103	177	203	261	
November	79	27	101	186	142	101	138	93	194	145	215	
December	37	12	71	161	178	43	78	94	183	108	208	
Total	1,474	1,398	1,871	2,428	2,119	2,362	2,137	1,999	2,571	2,743	3,025	833YTD
y/y Δ	25.4%	(5.2%)	33.9%	29.8%	(12.7%)	11.4%	(9.5%)	(6.5%)	28.6%	6.7%	10.3%	6.7%YTD

Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	1,274	1,615	1,586	2,077	2,351	2,431	2,148	2,059	2,259	2,784	2,704	1,920YTD
y/y Δ	4.4%	26.8%	(1.8%)	31.0%	13.2%	3.4%	(11.6%)	(4.1%)	9.7%	23.2%	(2.9%)	24.1%YTD

Source: China Customs General Administration, Bloomberg Finance L.P.

China Soybean Imports from Argentina

(Metric Tons 000)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
January	-	187,794	68,750	89,150	985,247	1,370,371	55,890	122,511	1,296,597	160,005	111,603	1,669,459
February	-	36,300	-	-	841,394	432,177	45,230	-	59,800	52,967	-	1,597,585
March	-	30,480	10,000	-	323,101	336,088	-	-	72,600	-	-	401,405
April	-	66,643	56,000	-	-	56,165	-	-	-	-	-	-
May	68,390	64,000	66,000	-	-	-	-	-	-	-	-	-
June	1,171,882	486,612	712,845	120,320	255,391	305,825	82,800	-	-	-	-	-
July	2,278,154	1,621,361	1,474,260	297,103	1,073,228	1,532,449	660,839	139,573	118,092	116,150	561,027	-
August	1,778,893	1,908,583	1,379,207	547,942	654,555	977,486	136,906	197,771	361	1,291,114	1,050,883	-
September	1,619,825	1,452,544	738,472	153,510	976,486	1,003,721	524,460	416,395	-	612,746	1,173,355	-
October	919,018	1,099,123	772,756	33,200	959,936	812,982	923,776	160,988	1,077	1,362,782	1,567,095	-
November	735,446	1,060,493	665,030	36,119	1,417,112	497,027	979,257	1,153,405	54,300	242,227	1,776,926	-
December	865,838	-	639,970	186,640	1,305,119	131,362	338,112	1,459,387	389,776	264,306	1,651,084	-
Total	9,437,446	8,013,933	6,583,290	1,463,984	8,791,569	7,455,653	3,747,270	3,650,030	1,992,603	4,102,297	7,891,973	3,668,449YTD
y/y Δ	57.2%	(15.1%)	(17.9%)	(77.8%)	500.5%	(15.2%)	(49.7%)	(2.6%)	(45.4%)	105.9%	92.4%	NMYTD

Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	7,676,511	8,541,900	7,379,222	3,870,743	4,542,385	9,669,214	3,426,757	3,225,460	4,737,625	2,065,389	4,205,574	9,836,909YTD
y/y Δ	13.9%	11.3%	(13.6%)	(47.5%)	17.4%	112.9%	(64.6%)	(5.9%)	46.9%	(56.4%)	103.6%	279.3%YTD

(Million Bushels)

Calendar Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
January	-	7	3	3	36	50	2	5	48	6	4	61
February	-	1	-	-	31	16	2	-	2	2	-	59
March	-	1	0	-	12	12	-	-	3	-	-	15
April	-	2	2	-	-	2	-	-	-	-	-	-
May	3	2	2	-	-	-	-	-	-	-	-	-
June	43	18	26	4	9	11	3	-	-	-	-	-
July	84	60	54	11	39	56	24	5	4	4	21	-
August	65	70	51	20	24	36	5	7	0	47	39	-
September	60	53	27	6	36	37	19	15	-	23	43	-
October	34	40	28	1	35	30	34	6	0	50	58	-
November	27	39	24	1	52	18	36	42	2	9	65	-
December	32	-	24	7	48	5	12	54	14	10	61	-
Total	347	294	242	54	323	274	138	134	73	151	290	135YTD
y/y Δ	57.2%	(15.1%)	(17.9%)	(77.8%)	500.5%	(15.2%)	(49.7%)	(2.6%)	(45.4%)	105.9%	92.4%	NMYTD

Marketing Year	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26
Sept 1 – Aug 31	282	314	271	142	167	355	126	119	174	76	155	361YTD
y/y Δ	13.9%	11.3%	(13.6%)	(47.5%)	17.4%	112.9%	(64.6%)	(5.9%)	46.9%	(56.4%)	103.6%	279.3%YTD

Source: China Customs General Administration, Bloomberg Finance L.P.

Key Global Wheat Producers, Exporters, and Importers (2026/27E)

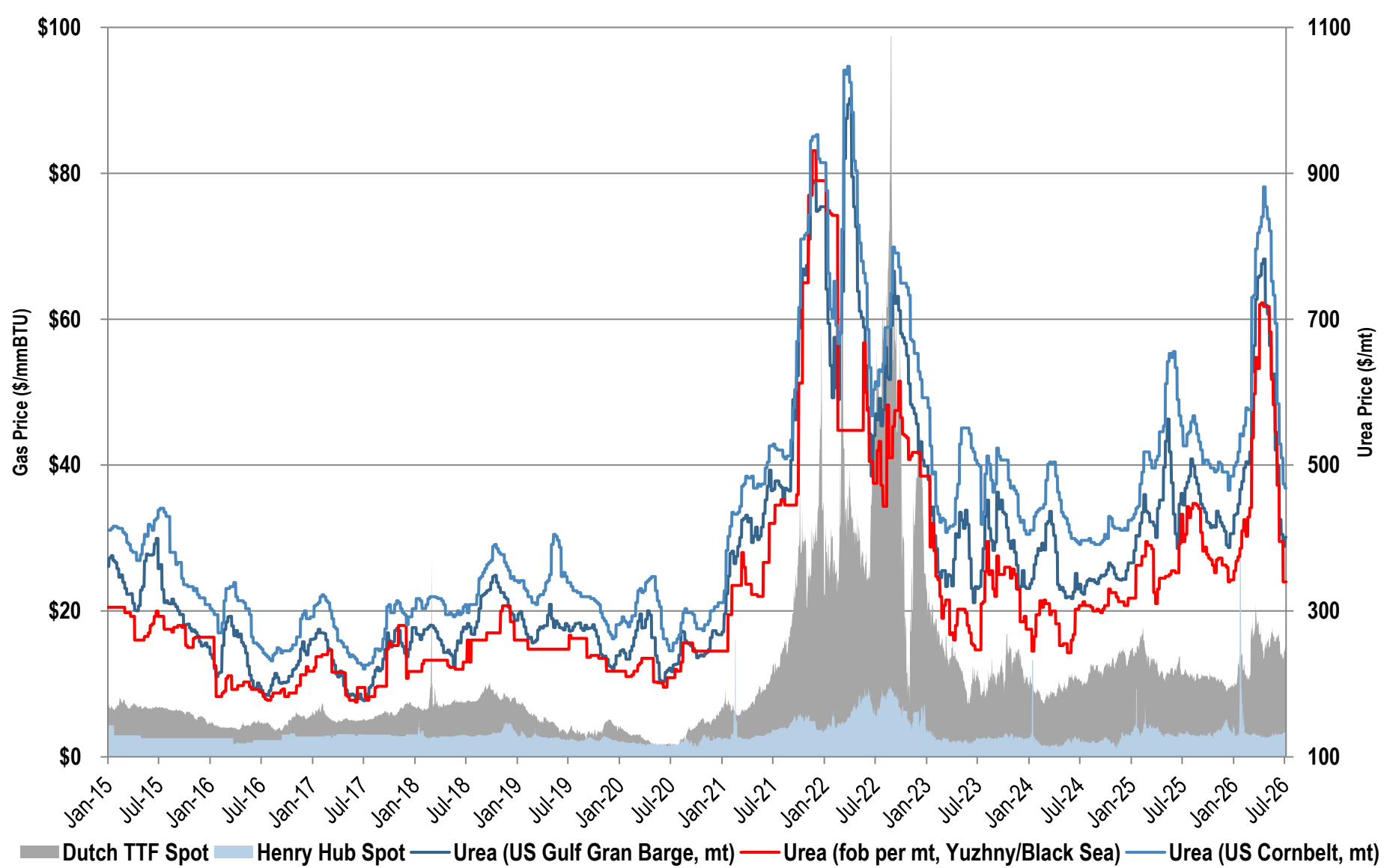
World Wheat	2026/27E	2026/27E	
Producers	Production	Production	
Regions	M tons	M bushels	% share
Europe	136.00	4,997	17%
China	141.00	5,181	17%
India	121.00	4,446	15%
Russia	88.50	3,252	11%
U.S.	41.81	1,536	5%
Canada	34.00	1,249	4%
Australia	28.00	1,029	3%
Ukraine	24.00	882	3%
Middle East	23.24	854	3%
North Africa	23.10	849	3%
Argentina	21.00	772	3%
Kazakhstan	15.00	551	2%
UK	13.50	496	2%
Brazil	6.70	246	1%
Other	103.12	3,789	13%
World Wheat Producers	819.97	30,128	100%

World Wheat	2026/27E	2026/27E	
Exporters	Exports	Exports	
Regions	M tons	M bushels	% share
Russia	47.50	1,745	22%
Europe	31.00	1,139	15%
Canada	27.50	1,010	13%
Australia	22.00	808	10%
U.S.	21.09	775	10%
Argentina	15.00	551	7%
Ukraine	14.50	533	7%
Kazakhstan	9.00	331	4%
North Africa	2.45	90	1%
Brazil	2.00	73	1%
India	2.00	73	1%
China	1.00	37	0%
Middle East	0.68	25	0%
UK	0.60	22	0%
Other	16.73	615	8%
World Wheat Exporters	213.05	7,828	100%

World Wheat	2026/27E	2026/27E	
Importers	Imports	Imports	
Regions	M tons	M bushels	% share
S.E. Asia	31.70	1,165	15%
North Africa	29.60	1,088	14%
Middle East	19.94	733	10%
Bangladesh	7.70	283	4%
Brazil	7.20	265	3%
Europe	6.50	239	3%
Nigeria	6.20	228	3%
China	6.00	220	3%
Japan	5.55	204	3%
U.S.	3.81	140	2%
UK	2.50	92	1%
Kazakhstan	0.70	26	0%
Canada	0.65	24	0%
Russia	0.30	11	0%
Other	79.82	2,933	38%
World Wheat Importers	208.17	7,649	100%

Nitrogen Fertilizer Pricing: Global Urea Pricing vs Natural Gas

(prices as of July 9th, 2026)



Domestic urea (nitrogen) fertilizer prices are generally a function of the delivered costs for offshore producers.

Source: Bloomberg Finance L.P. & J.P. Morgan Estimates

~15M Tons Urea Exports Affected by Closure of SOH

(Export shipments in 1,000 metric tons.)

Urea Export Regions	2023	2024	2025	Main Destinations
Russia	8,548	9,565	9,716	U.S., Brazil, India
China	4,255	262	5,567	India, Mexico, Vietnam
Iran	4,848	5,759	5,408	Brazil, Turkey, South Africa
Qatar	5,308	5,589	5,230	U.S., Brazil, India
Egypt	4,830	4,296	4,345	Turkey, Italy, France
Saudi Arabia	4,123	4,489	4,260	Thailand, India, Bangladesh
Algeria	3,248	2,805	3,202	Brazil, France, U.S.
Nigeria	2,548	2,826	3,110	Brazil, India, U.S.
Oman	2,988	3,269	3,103	India, Argentina, Thailand
Malaysia	1,899	2,010	1,917	Thailand, Australia, India
Other	11,774	12,736	12,289	
Global Urea Exports	54,369	53,606	58,147	
% of shipments through SOH	26%	30%	26%	

Ammonia Export Regions	2023	2024	2025	Main Destinations
Trinidad	3,478	3,231	3,619	U.S., Morocco, Brazil
Saudi Arabia	2,615	2,359	2,112	India, S. Korea, Morocco
Indonesia	1,789	1,709	1,748	S. Korea, China, India
U.S.	1,264	1,053	1,369	Morocco, Norway, Belgium
Canada	992	962	1,063	U.S.
Algeria	1,025	1,077	976	Turkey, U.K., Portugal
Russia	545	669	524	Belgium, Lithuania, Turkey
Egypt	671	603	516	Spain, Netherlands, Jordan
Oman	537	571	542	India, Turkey, Morocco
Iran	631	803	307	India, Turkey, China
Other	3,071	3,354	4,201	
Global Ammonia Exports	16,618	16,391	16,977	
% of shipments through SOH	20%	19%	14%	

Note: Morocco does not ship through SOH, but some of its sulphur imports come from the Persian region affected by the closure of the Strait of Hormuz (SOH).

Source: Nutrien Fact Book (CRU), J.P. Morgan estimates

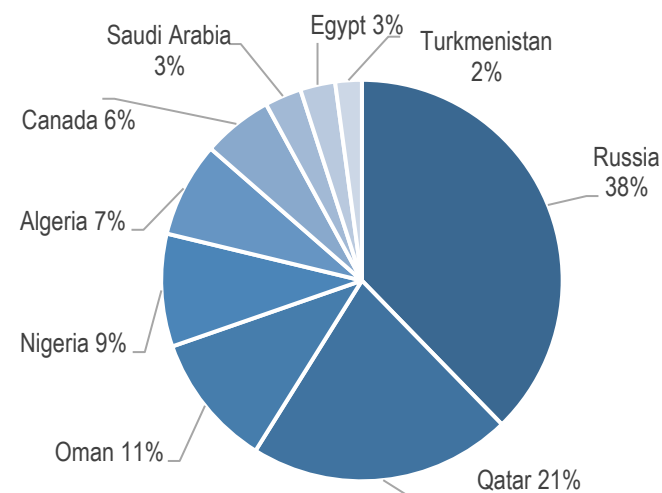
U.S. Nitrogen Imports

(Annual, import shipments in -000- metric tons)

	Ammonia	% change	Urea	% change	UAN	% change
2005	7,954	10.5%	5,674	15.0%	2,822	40.3%
2006	7,221	(9.2%)	5,029	(11.4%)	1,978	(29.9%)
2007	7,970	10.4%	6,546	30.2%	3,202	61.9%
2008	7,320	(8.2%)	5,460	(16.6%)	6,950	117.1%
2009	5,506	(24.8%)	4,728	(13.4%)	1,165	(83.2%)
2010	6,742	22.4%	6,632	40.3%	2,207	89.4%
2011	6,812	1.0%	5,860	(11.6%)	3,493	58.3%
2012	6,296	(7.6%)	6,915	18.0%	3,010	(13.8%)
2013	6,033	(4.2%)	6,467	(6.5%)	3,160	5.0%
2014	5,035	(16.5%)	7,942	22.8%	3,125	(1.1%)
2015	5,250	4.3%	8,092	1.9%	3,112	(0.4%)
2016	4,668	(11.1%)	7,142	(11.7%)	2,811	(9.7%)
2017	3,763	(19.4%)	6,091	(14.7%)	2,621	(6.8%)
2018	3,079	(18.2%)	5,540	(9.0%)	2,396	(8.6%)
2019	2,459	(20.1%)	4,905	(11.5%)	2,867	19.6%
2020	2,416	(1.8%)	4,564	60.9%	2,388	(16.7%)
2021	2,531	4.8%	6,268	37.3%	2,499	4.6%
2022	2,350	(7.1%)	4,897	(21.9%)	1,585	(36.6%)
2023	2,090	(11.1%)	4,812	(1.7%)	2,440	54.0%
2024	2,073	(0.8%)	5,098	5.9%	2,140	(12.3%)
2025	2,029	(2.1%)	5,232	2.6%	2,167	1.3%
2026 YTD (Apr)	725	5.6%	3,461	16.0%	968	7.7%

Source: US ITC, Bloomberg Finance L.P., Chemical Market Analytics by OPIS, a Dow Jones Company, & J.P. Morgan Estimates

Urea Imports by Country-of-Origin 2026 YTD (Apr)



Urea Import Source	2025	% of total	2024	% of total
Russia	1,885	36%	1,286	25%
Qatar	1,125	21%	1,035	20%
Canada	506	10%	559	11%
Algeria	411	8%	456	9%
Nigeria	401	8%	336	7%
Saudi Arabia	288	6%	484	9%
Trinidad and Tobago	155	3%	89	2%
Netherlands	109	2%	133	3%
Bahrain	100	2%	0	0%
Other	252	5%	719	14%

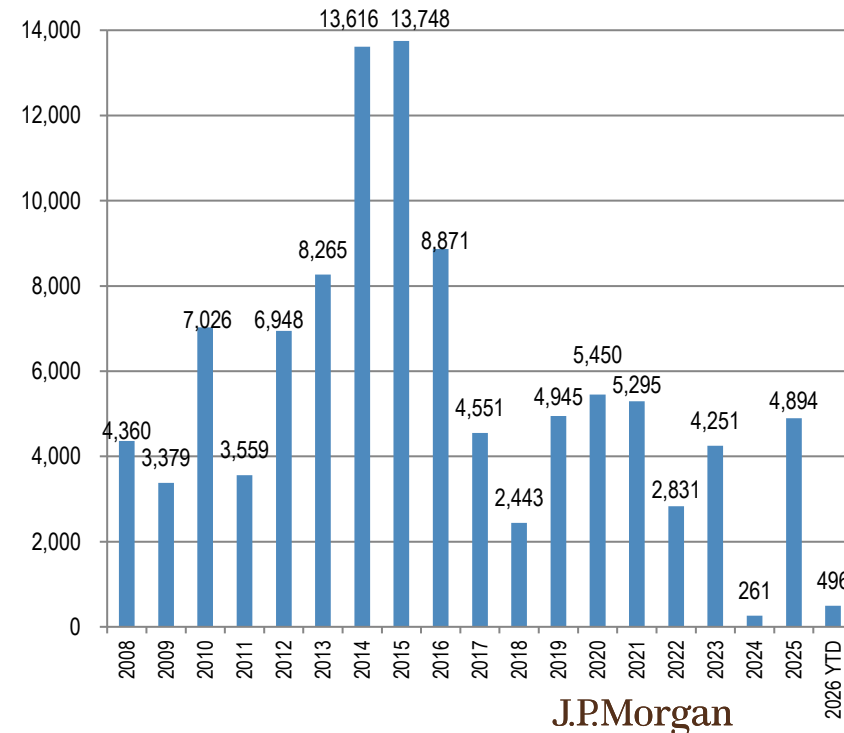
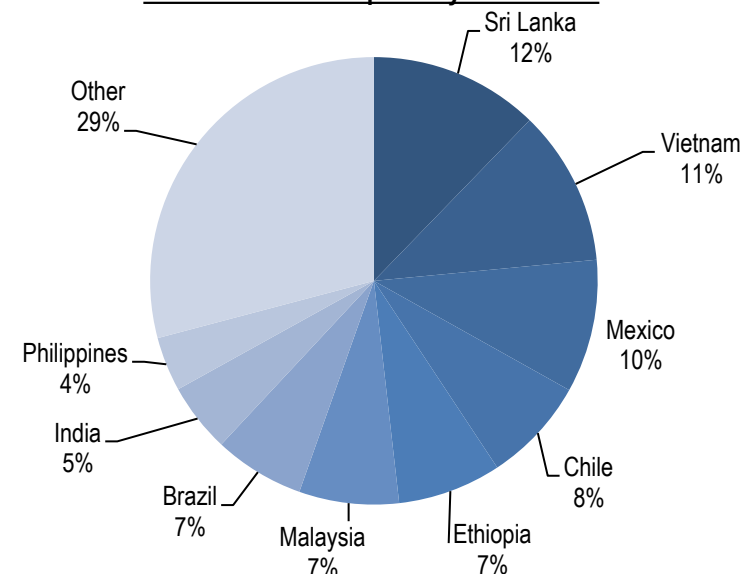
China Urea Exports

(Export shipments by month in -000- metric tons)

Urea	2018	2019	2020	2021	2022	2023	2024	2025	2026	% chg
Jan	142.8	589.5	366.1	291.8	85.1	240.8	12.6	2.6	307.9	N.M.
Feb	51.5	393.9	148.8	142.8	152.0	165.4	7.5	1.4	111.5	N.M.
Mar	86.6	325.2	271.3	367.4	65.6	119.7	4.2	2.3	58.7	N.M.
Apr	117.1	97.8	111.7	539.2	149.7	75.8	5.2	2.3	15.0	N.M.
May	162.5	215.5	599.8	601.0	85.6	182.8	34.6	2.4	3.4	39.9%
Jun	143.2	144.7	222.6	482.4	186.1	223.9	74.2	66.2		
Jul	104.4	269.0	181.2	243.0	150.7	323.4	79.4	567.2		
Aug	124.9	657.8	200.5	261.3	351.2	314.0	25.5	796.7		
Sep	190.8	545.8	827.9	1,091.5	347.4	1,184.4	9.3	1,371.2		
Oct	234.1	651.9	1,092.0	739.8	350.9	561.6	3.5	1,202.2		
Nov	444.3	457.0	846.0	499.8	372.6	515.5	2.2	601.8		
Dec	640.9	596.6	582.0	35.4	534.3	343.6	2.3	278.3		
Total	2,443.0	4,944.6	5,450.1	5,295.3	2,831.3	4,251.0	260.6	4,893.5	496.5	
% Δ	(46.3%)	102.4%	10.2%	(2.8%)	(46.5%)	50.1%	(93.9%)	1778.1%	N.M.	

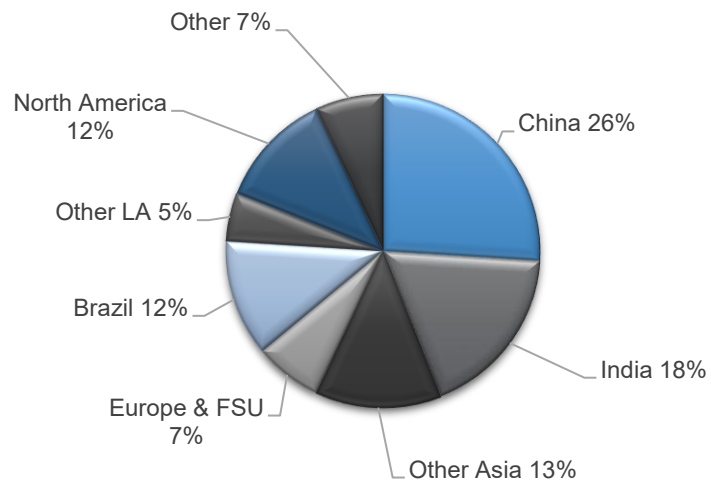
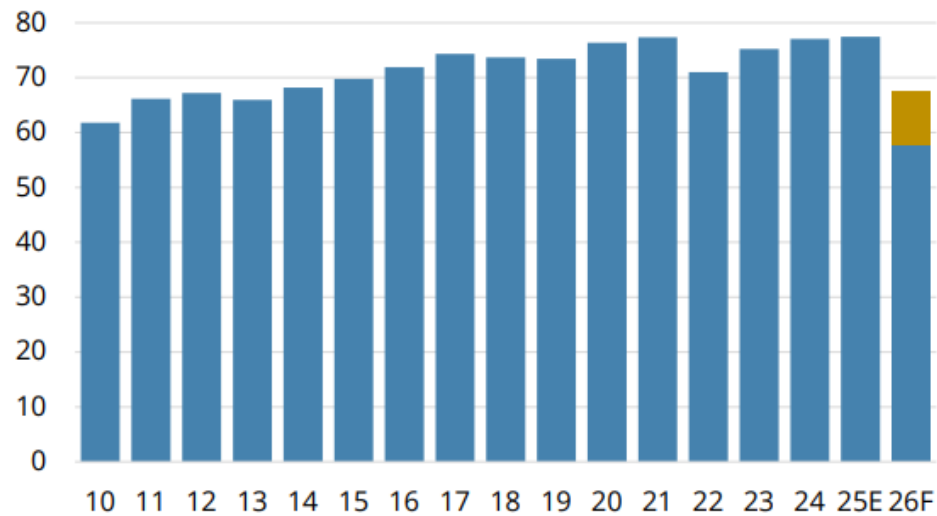
Source: Bloomberg Finance L.P., Chemical Market Analytics by OPIS, a Dow Jones Company & J.P. Morgan Estimates

2025 China Urea Exports by Destination



Global DAP/MAP/TSP/NPS Consumption

million metric tons



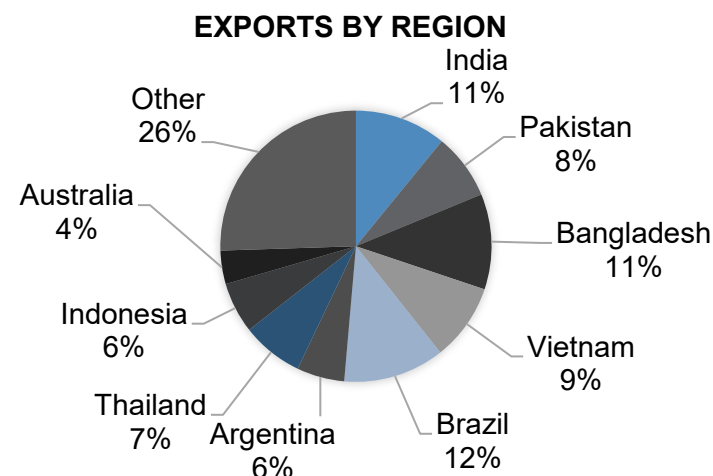
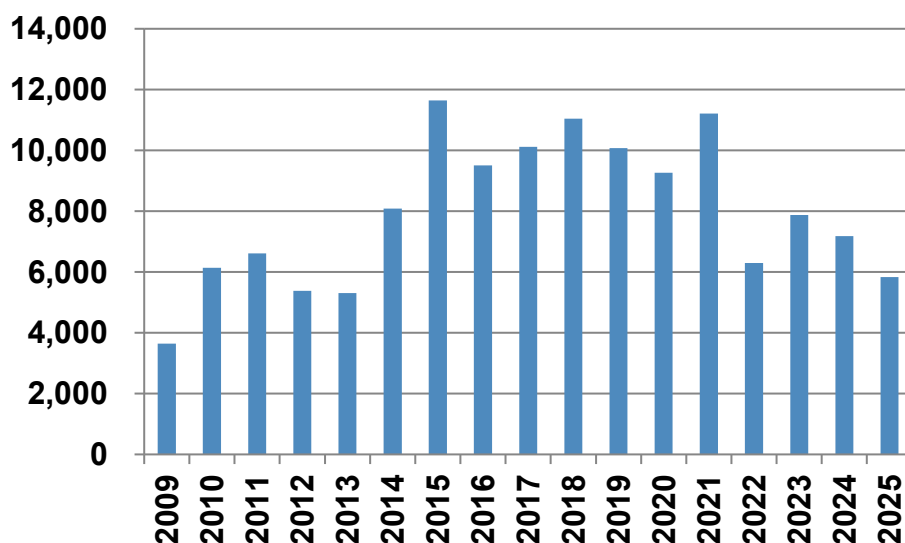
Source: Mosaic (Argus, IFA, CRU).

Chinese DAP/MAP/TSP

(Export shipments by month in 1,000 metric tons.)

DAP/MAP/TSP	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD	% Δ
Jan	125	141	187	354	338	264	412	627	353	390	360	550	202	112	88	(21.8%)
Feb	108	65	374	566	193	587	314	450	228	553	167	292	81	118	90	(23.8%)
Mar	61	85	489	664	492	562	486	678	1,008	799	560	500	95	17	69	299.7%
Apr	57	144	360	636	458	542	467	969	701	905	493	552	664	89	101	13.1%
May	170	114	339	1,400	775	857	945	1,082	616	1,458	438	866	800	89	70	(21.4%)
Jun	258	394	614	1,058	956	1,225	1,358	890	894	1,622	538	998	878	731	Release	7/20/26
Jul	573	671	637	1,111	869	1,181	1,139	1,431	1,014	1,777	541	788	816	1,412		
Aug	1,387	1,015	1,198	1,048	1,168	1,486	1,003	806	1,228	1,338	766	819	943	1,007		
Sep	991	586	863	1,600	725	855	1,253	972	817	1,006	755	488	823	594		
Oct	619	794	856	961	1,211	1,083	1,873	645	891	1,048	803	625	624	630		
Nov	604	627	957	1,098	1,214	703	1,230	990	788	250	438	864	884	773		
Dec	429	667	1,214	1,144	1,107	772	561	532	722	63	441	532	375	265		
Total	5,382	5,304	8,087	11,638	9,507	10,116	11,042	10,071	9,261	11,209	6,300	7,873	7,184	5,837	417	YTD
% Change	(18.6%)	(1.4%)	52.5%	43.9%	(18.3%)	6.4%	9.2%	(8.8%)	(8.0%)	21.0%	(43.8%)	25.0%	(8.7%)	(18.4%)	(2.0%)	YTD

Note: China's lithium-iron-phosphate production for LFP batteries has grown from 1.1m tons in 2022 to 3.8m tons in 2025. LFP production could approach 5m tons in 2026E. The finished fertilizer equivalent (DAP) to LFP is about 1:1, according to MOS.



Source: Bloomberg Finance L.P., CNCIC, and J.P. Morgan estimates

U.S. DAP/MAP Import By Major Regions

(Import shipments in 1,000 metric tons.)

Year	Morocco	Russia	SaudiArabia	Jordan	Other	Total	Δ %
2018	1,453	917	92	0	313	2,775	
2019	1,747	738	262	0	180	2,926	5.5%
2020	828	370	290	32	464	1,983	(32.2%)
2021	3	109	845	529	989	2,476	24.9%
2022	5	211	549	0	404	1,169	(52.8%)
2023	0.4	208	1,037	163	750	2,158	84.7%
2024	2	9	1,052	269	787	2,118	(1.9%)
2025	4	11	542	0	562	1,120	(47.1%)
2025 YTD (Jan-Apr)	0.4	4	475	0	335	815	
2026 YTD (Jan-Apr)	1.5	3	414	34	226	679	(16.7%)

Note: Other notable regions include Mexico and Egypt.

Source: U.S. ITC.

3M Tons of P205 (Equivalent ~6M Tons of DAP) Exports Affected by Closure of SOH

(Export shipments in 1,000 metric tons.)

Export Regions	2021	2022	2023	Main Destinations
Morocco	4,226	4,772	4,984	India, Brazil, Australia
China	3,667	3,278	2,661	India, Bangladesh, Brazil
Saudi Arabia	2,398	2,527	2,947	India, U.S., Brazil
Russia	2,296	2,123	2,301	Brazil, Mexico, India
U.S.	1,199	1,059	846	Canada, Colombia, Australia
Jordan	307	359	365	U.S., India, Iraq
Egypt	188	265	290	U.S., India, Turkey
Tunisia	320	244	268	U.S., Italy, Turkey
Australia	156	166	198	India, U.S., Pakistan
Mexico	135	182	193	U.S., Canada, Dom. Republic
Other	794	867	1,079	
Global Exports (P205)	15,686	15,842	16,132	
% of shipments through SOH	15%	16%	18%	

Note:

It takes ~0.5 tons of P205 to produce a ton of DAP/MAP.

Morocco does not ship through SOH, but some of its sulphur imports come from the Persian region affected by the closure of the Strait of Hormuz (SOH).

Source: Nutrien Fact Book (CRU), J.P. Morgan estimates

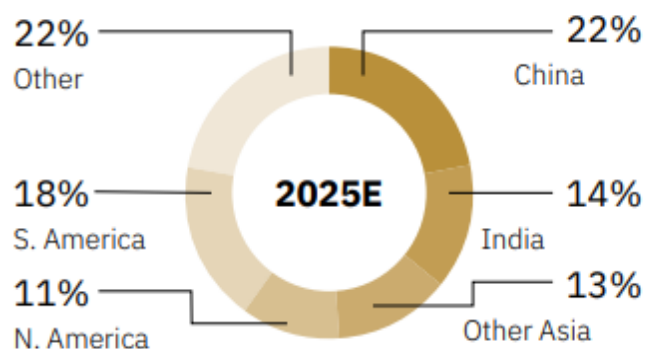
Phosacid (P2O5) by Region

(Capacity and Demand in 1,000 metric tons.)

Region	Companies	Nameplate Capacity	Demand
China	Yuntianhua, Guizhou Kailin, and Wengfu in China	20,500	10,400
Morocco	OCP	8,100	
U.S.	Mosaic, Nutrien, J.R. Simplot, Itafos	7,500	3,900
Russia	PhosAgro, Eurochem	4,200	1,400
Saudi Arabia	Ma'aden	3,100	
India	IFFCO	2,600	7,300
Tunisia	GCT	2,100	
Brazil	Mosaic/Vale	1,800	6,900
Jordan		1,100	
Mexico		900	
Other		9,700	18,700
Total Fertilizer			48,600
Total Industrial			3,400
Total		61,600	52,000

Phosphate demand

(percent)

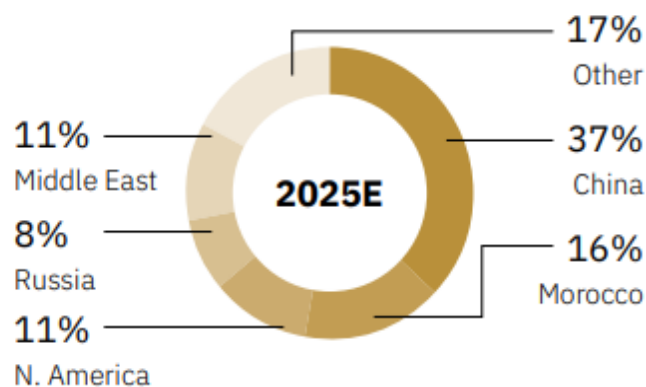


Note:

It takes ~0.5 tons of P2O5 to produce a ton of DAP/MAP.

Phosphate production

(percent)



Source: Nutrien, CRU, TFI

Major Global Phosphate: Imports and Exports

(Imports and Exports in 1,000 metric tons)

Imports

U.S. DAP/MAP	2018	2019	2020	2021	2022E	2023E	2024E	2025E	2026E YTD
1Q	952	1,367	669	839	335	573	875	698	655
2Q	441	345	501	471	188	444	369	183	23 Apr
3Q	554	661	207	692	274	722	488	100	
4Q	867	675	609	480	388	420	386	140	
Total	2,814	3,047	1,986	2,481	1,185	2,158	2,118	1,120	679 YTD
% Change	52.8%	8.3%	(34.8%)	24.9%	(52.3%)	82.2%	(1.9%)	(47.1%)	(11.7%) YTD

Brazil DAP/MAP	2018	2019	2020	2021	2022E	2023E	2024E	2025E	2026E YTD
1Q	478	620	861	1,041	621	599	449	939	720
2Q	612	966	1,614	1,284	1,725	1,054	661	1,073	1,060
3Q	1,603	1,493	1,543	1,666	1,364	1,273	2,023	907	
4Q	988	1,105	1,149	1,574	625	1,352	1,314	748	
Total	3,681	4,184	5,168	5,565	4,335	4,278	4,447	3,668	1,780 YTD
% Change	(2.5%)	13.7%	23.5%	7.7%	(22.1%)	(1.3%)	4.0%	(17.5%)	(11.5%) YTD

India DAP	2018	2019	2020	2021	2022E	2023E	2024E	2025E	2026E YTD
1Q	418	1,032	589	220	1,450	1,371	462	474	312
2Q	2,063	1,877	1,227	1,339	772	2,288	1,150	1,219	1,087
3Q	2,001	1,370	2,413	981	2,130	1,357	876	3,370	
4Q	1,506	1,034	1,022	1,745	2,697	1,721	2,491	2,465	
Total	5,988	5,313	5,251	4,285	7,049	6,736	4,979	7,527	1,399 YTD
% Change	49.7%	(11.3%)	(1.2%)	(18.4%)	64.5%	(4.4%)	(26.1%)	51.2%	(17.4%) YTD

Exports

U.S. DAP/MAP	2018	2019	2020	2021	2022E	2023E	2024E	2025E	2026E YTD
1Q	879	873	873	685	837	697	605	578	832
2Q	778	1,025	979	673	777	689	515	452	295 Apr
3Q	753	975	739	615	681	469	421	477	
4Q	871	881	620	584	658	653	687	485	
Total	3,280	3,754	3,210	2,557	2,953	2,508	2,229	1,992	1,127 YTD
% Change	(13.8%)	14.5%	(14.5%)	(20.3%)	15.5%	(15.1%)	(11.1%)	(11.3%)	38.2% YTD

China DAP/MAP	2018	2019	2020	2021	2022E	2023E	2024E	2025E	2026E YTD
1Q	987	1,550	1,395	1,505	945	1,168	241	111	130
2Q	2,519	2,580	1,873	3,608	1,228	2,170	2,204	745	1 Apr/May
3Q	3,117	2,747	2,817	3,638	1,905	1,811	2,416	2,907	
4Q	3,335	1,990	2,177	1,291	1,467	1,923	1,706	1,593	
Total	9,958	8,868	8,263	10,042	5,545	7,072	6,567	5,356	131 YTD
% Change	11.9%	(10.9%)	(6.8%)	21.5%	(44.8%)	27.5%	(7.1%)	(18.4%)	(29.0%) YTD

Morocco DAP/MAP	2018	2019	2020	2021	2022E	2023E	2024E	2025E	2026E YTD
1H	2,600	2,300	3,700	3,100	2,500	2,700	3,200	3,600	2,400 est 1H26
3Q	1,500	2,100	2,300	1,900	2,100	1,900	1,800	2,000	
4Q	1,300	1,600	1,000	1,900	1,500	2,000	2,000	1,200	
Total	5,400	6,000	7,000	6,900	6,100	6,600	7,000	7,400	2,400 est YTD
% Change	17.4%	11.1%	16.7%	(1.4%)	(11.6%)	8.2%	6.1%	5.7%	(33%) est YTD

Source: Nutrien Crop Input Fundamentals, OCP Company reports, US ITC, Argus, and J.P. Morgan estimates

U.S. Duties Imposed on Phosphate Fertilizer Imports from Morocco and Russia

Time Line

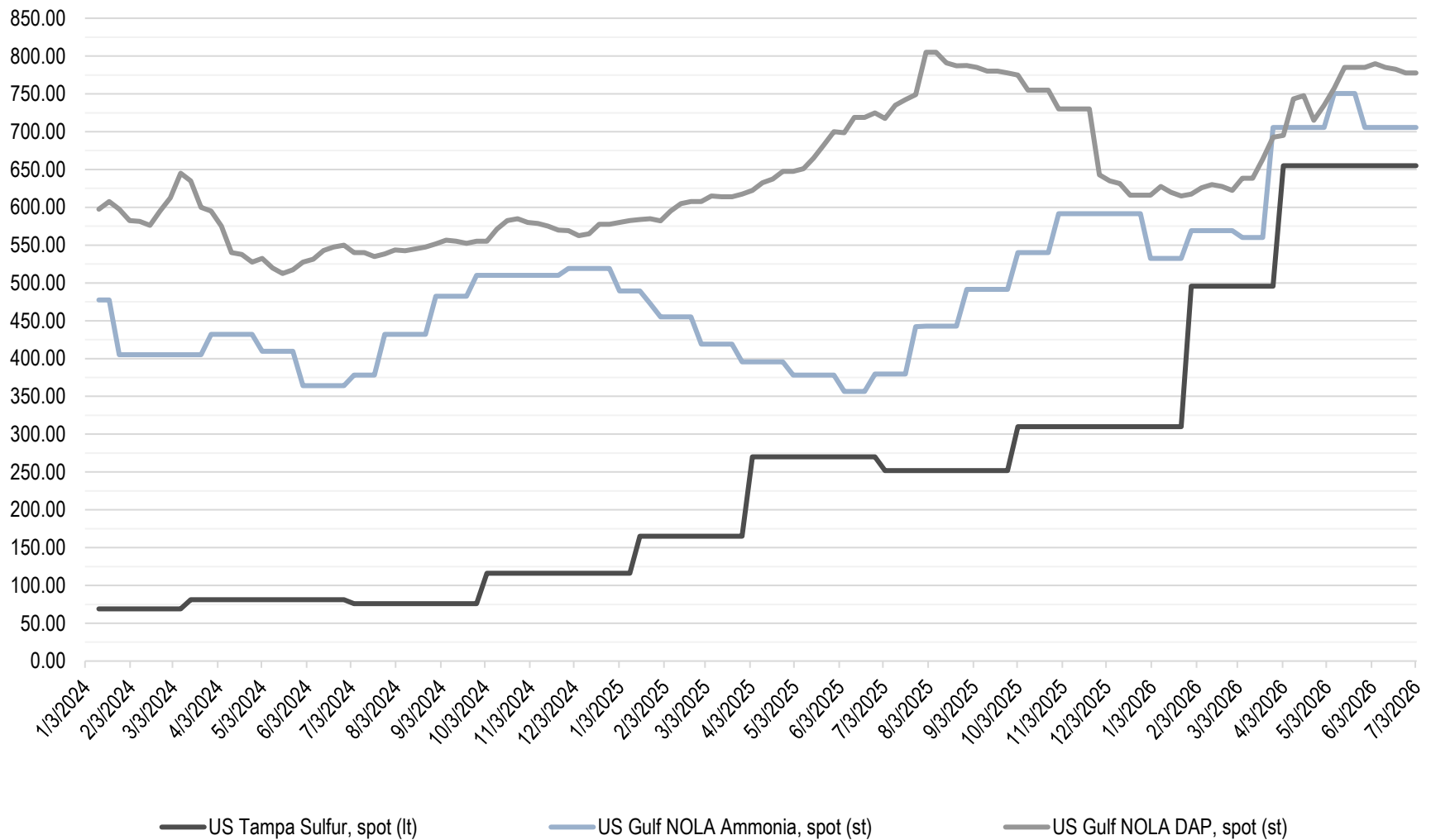
6/26/2020	Mosaic filed a petition with the U.S. Department of Commerce (DOC) and the International Trade Commission (ITC)
11/24/2020	The DOC set preliminary duties
2/9/2021	Final decision by the DOC
3/11/2020	Final decision by the ITC
April 2021	Issuance of final order
November 2023	DOC announced final results of an administrative review of CVD orders originally issued in April 2021
April 2024	Second administrative review
March 2026	5-year sunset review began; Final ruling expected by April of 2027
June 2026	Proclamation to suspend Moroccan CVDs for 8 months until March 2027; while sunset review continues

Countervailing Duties Issued by the U.S. Department of Commerce:

16.6%	CVD on Moroccan phosphate (OCP) after the second review; applies retroactively from 2022-2026
=>	Plus 10% tariff for imports from Morocco
18.2%	CVD imposed on Russian phosphate (Phosagro)

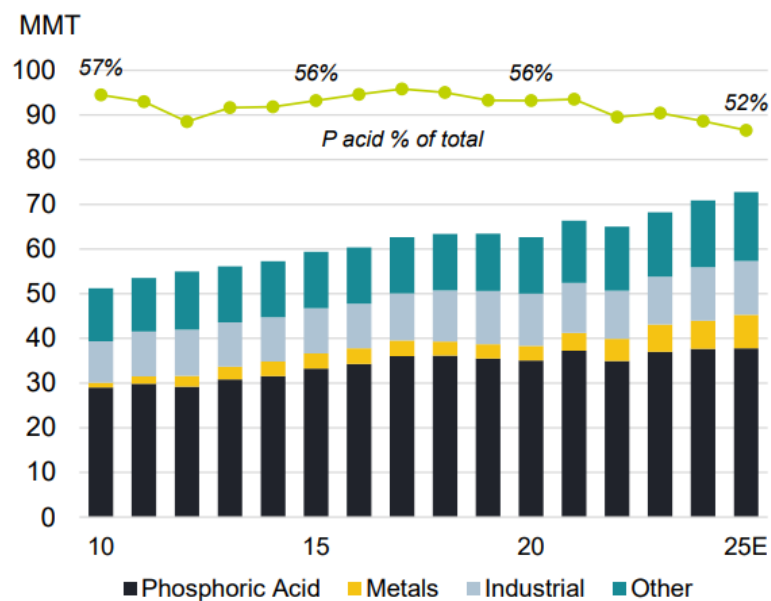
NOLA DAP versus NOLA Ammonia versus Tampa Sulfur

(Prices as of July 3, 2026)

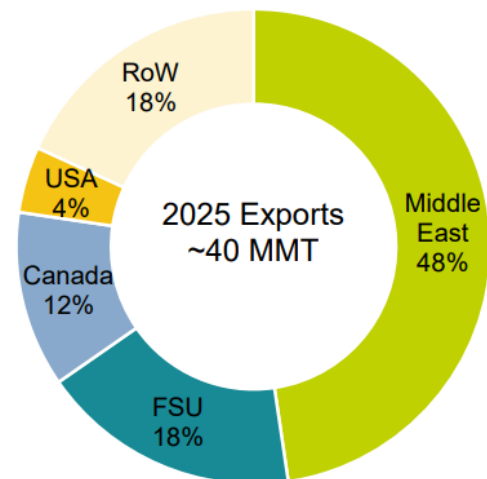


Global Sulphur Demand and Sulphur Exports

Global Sulphur Demand by Use



Global Sulphur Exports



Source: Mosaic, CRU

FSU Sulphur Shipments

(Export shipments in 1,000 metric tons.)

Sulphur Exports from Ust-Luga										
Destination	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD (May)
Morocco	605	1,031	1,021	688	1,003	1,621	2,129	3,357	2,845	589
Brazil	610	1,137	1,267	1,259	1,058	725	367	339	483	103
India	0	0	0	0	0	0	361	0	63	0
Israel	278	314	340	294	186	290	313	60	0	0
China		0	127	0	0	138	59	0	77	0
U.S.	253	670	191	477	608	57	0	0	0	0
Other	90	685	901	828	1,011	896	1,921	914	597	64
Total Sulphur Exports from Ust-Luga	1,835	3,838	3,846	3,545	3,867	3,728	5,149	4,669	4,065	755

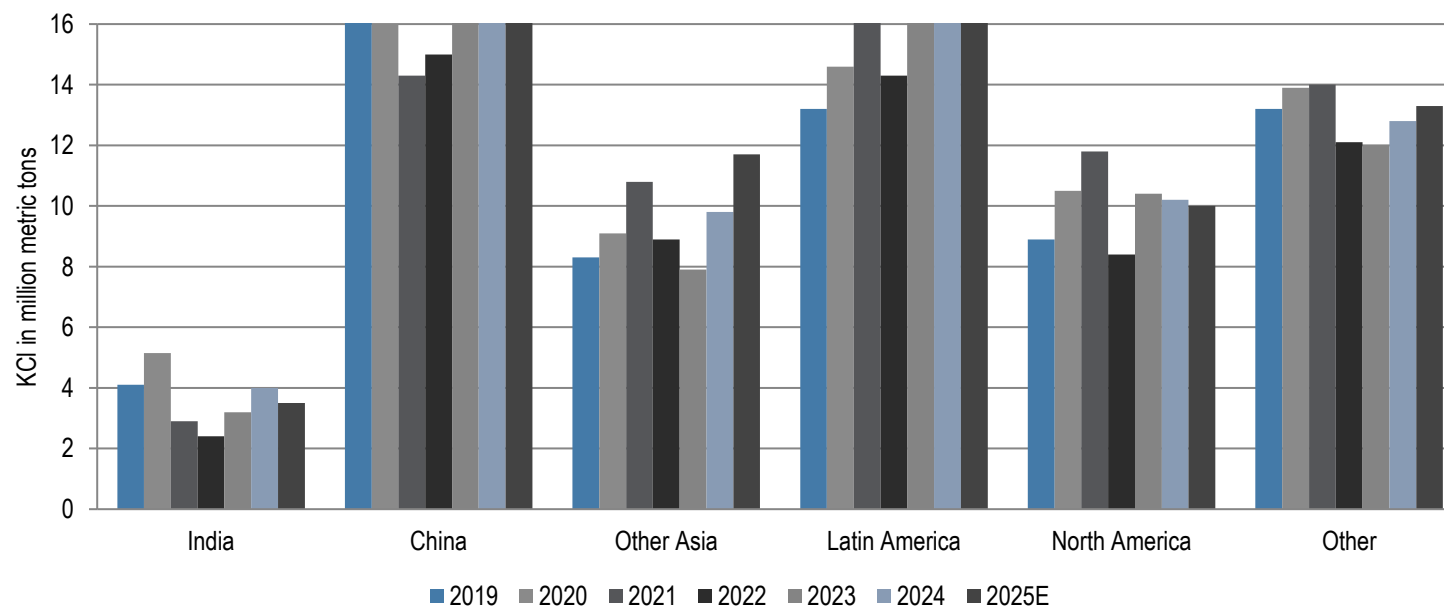
Sulphur Rail Shipments to Port of Ust-Luga										
Regions	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD (January)
Russia (Gazprom)	440	1,300	1,368	930	473	500	786	986	341	0
Kazakhstan (TCO, Kashagan)	1,476	2,350	2,281	2,363	2,908	2,683	3,061	3,251	3,212	161
Total Sulphur Shipments to Port of Ust-Luga	1,915	3,650	3,648	3,293	3,381	3,183	3,847	4,238	3,554	161

Source: Argus, J.P. Morgan calculations

World Potash Consumption

(KCl in millions of metric tons.)

Year	India	China	Other Asia	Latin America	North America	Other	TOTAL
2019	4.1	16.1	8.3	13.2	8.9	13.2	63.8
2020	5.1	16.0	9.1	14.6	10.5	13.9	69.2
2021	2.9	14.3	10.8	16.3	11.8	14.0	70.1
2022	2.4	15.0	8.9	14.3	8.4	12.1	61.1
2023	3.2	18.0	7.9	16.5	10.4	12.0	68.0
2024	4.0	19.0	9.8	17.0	10.2	12.8	72.8
2025E	3.5	19.0	11.7	17.0	10.0	13.3	74.5
2026E - low estimate	3.5	19.0	10.0	17.0	10.0	13.0	72.5
2026E - mid estimate	4.0	19.5	10.5	17.5	10.3	13.5	75.3
2026E - high estimate	4.5	20.0	11.0	18.0	10.5	14.0	78.0



Potash Market and Share Data

(KCl in millions of metric tons.)

	Total Potash	Nutrien	Mosaic	K+S	ICL Group	Belarus	Russia	Uralkali	Eurochem-Usolskiy	Eurochem-VolgaKaliy
	Consumption	Shipments	Shipments	Shipments	Shipments	Shipments	Shipments	Shipments	Shipments	Shipments
2019	63.8	11.5	7.8	3.9	4.1	10.5	9.9	9.0	0.9	0.0
% share		18.1%	12.3%	6.1%	6.5%	16.5%	15.5%			
2020	69.2	12.8	9.4	4.7	4.7	11.8	11.2	9.2	2.0	0.0
% share		18.5%	13.6%	6.8%	6.7%	17.0%	16.1%			
Δ y/y	8.5%	11.3%	19.8%	20.9%	13.0%	12.4%	13.1%	1.9%	126.8%	NM
2021	70.1	13.6	8.2	4.7	4.4	10.0	12.3	9.9	2.2	0.2
% share		19.4%	11.7%	6.7%	6.3%	14.3%	17.5%			
Δ y/y	1.2%	6.2%	(12.9%)	0.2%	(5.0%)	(15.0%)	10.2%	8.2%	8.8%	NM
2022	61.1	12.5	8.1	4.4	4.5	3.0	7.5	4.9	2.2	0.3
% share		20.5%	13.3%	7.3%	7.4%	5.0%	12.2%			
Δ y/y	(12.8%)	(8.0%)	(1.0%)	(5.3%)	1.5%	(69.8%)	(39.4%)	(50.8%)	3.3%	59.4%
2023	68.0	13.2	8.9	4.6	4.7	9.4	10.9	7.9	2.4	0.7
% share		19.4%	13.0%	6.8%	6.9%	13.9%	16.1%			
Δ y/y	11.3%	5.4%	9.5%	4.1%	4.1%	211.1%	46.7%	62.6%	5.5%	88.0%
2024	72.8	13.9	8.7	4.4	4.6	10.8	13.8	10.4	2.4	0.9
% share		19.1%	12.0%	6.0%	6.3%	14.8%	19.0%			
Δ y/y	7.1%	5.1%	(1.4%)	(5.8%)	(2.7%)	14.2%	26.3%	31.5%	3.6%	45.5%
2025	74.5	14.3	9.0	4.4	4.3	12.0	14.5	10.7	2.7	1.1
% share		19.1%	12.0%	5.9%	5.8%	16.1%	19.5%			
Δ y/y	2.3%	2.6%	2.6%	1.8%	(5.2%)	11.3%	5.2%	2.5%	10.7%	20.7%
CAGR	2.6%	3.6%	2.3%	2.3%	0.8%	2.2%	6.7%	2.9%	20.6%	NM

Source: Company reports, Argus, Bloomberg Finance L.P. and J.P. Morgan estimates

Potash Imports by Region: China and India

China	M tons	% change
2016 full year	6.83	(28%)
2017 full year	7.54	10%
2018 full year	7.49	(1%)
2019 full year	9.12	22%
2020 full year	8.85	(3%)
2021 full year	7.67	(13%)
2022 full year	7.97	4%
2023 full year	11.57	45%
2024 full year	12.63	9%
2025 full year	12.61	(0%)
2026 YTD	5.81 YTD	25% YTD
1Q:25	3.55	(8%)
2Q:25	2.73	5%
3Q:25	2.39	(15%)
4Q:25	3.95	16%
1Q:26	4.52	29%
2Q:26 (April)	1.23 QTD	10% QTD
3Q:26		
4Q:26		

India	M tons	% change
2016 full year	3.85	(4%)
2017 full year	4.51	17%
2018 full year	4.73	5%
2019 full year	4.15	(12%)
2020 full year	5.10	23%
2021 full year	2.90	(43%)
2022 full year	2.41	(17%)
2023 full year	3.15	31%
2024 full year	4.00	27%
2025 full year	2.96	(26%)
2026 YTD	1.55 YTD	38% YTD
1Q:25	0.86	97%
2Q:25	0.26	(72%)
3Q:25	0.86	27%
4Q:25	0.98	(50.%)
1Q:26	0.90	4%
2Q:26	0.65	154%
3Q:26		
4Q:26		

Source: Nutrien, Mosaic, CRU, Fertecon, US ITC and J.P. Morgan estimates

Potash Imports by Region: Brazil and U.S.

Brazil	M tons	% change
2016 full year	8.71	11%
2017 full year	9.67	11%
2018 full year	10.52	9%
2019 full year	10.45	(1%)
2020 full year	11.24	7%
2021 full year	12.78	14%
2022 full year	11.82	(8%)
2023 full year	13.52	14%
2024 full year	14.09	4%
2025 full year	13.67	(3%)
2026 YTD	7.28	5%
1Q:25	2.63	6%
2Q:25	4.28	(4%)
3Q:25	4.31	4%
4Q:25	2.45	(18%)
1Q:26	3.16	20%
2Q:26	4.13	(4%)
3Q:26		
4Q:26		

U.S	M tons	% change
2016 full year	8.77	(14%)
2017 full year	12.16	39%
2018 full year	12.83	6%
2019 full year	11.49	(10%)
2020 full year	11.89	3%
2021 full year	13.45	13%
2022 full year	11.04	(18%)
2023 full year	11.85	7%
2024 full year	13.97	18%
2025 full year	13.25	(5%)
2026 YTD	5.25 YTD	9% YTD
1Q:25	3.32	(7%)
2Q:25	3.23	(18%)
3Q:25	3.63	29%
4Q:25	3.06	(16%)
1Q:26	3.68	11%
2Q:26 (April)	1.57 QTD	5% QTD
3Q:26		
4Q:26		

Source: Nutrien, Mosaic, CRU, Fertecon, US ITC and J.P. Morgan estimates

Russia/Belarus Potash Rail Shipments to Russian Ports & China

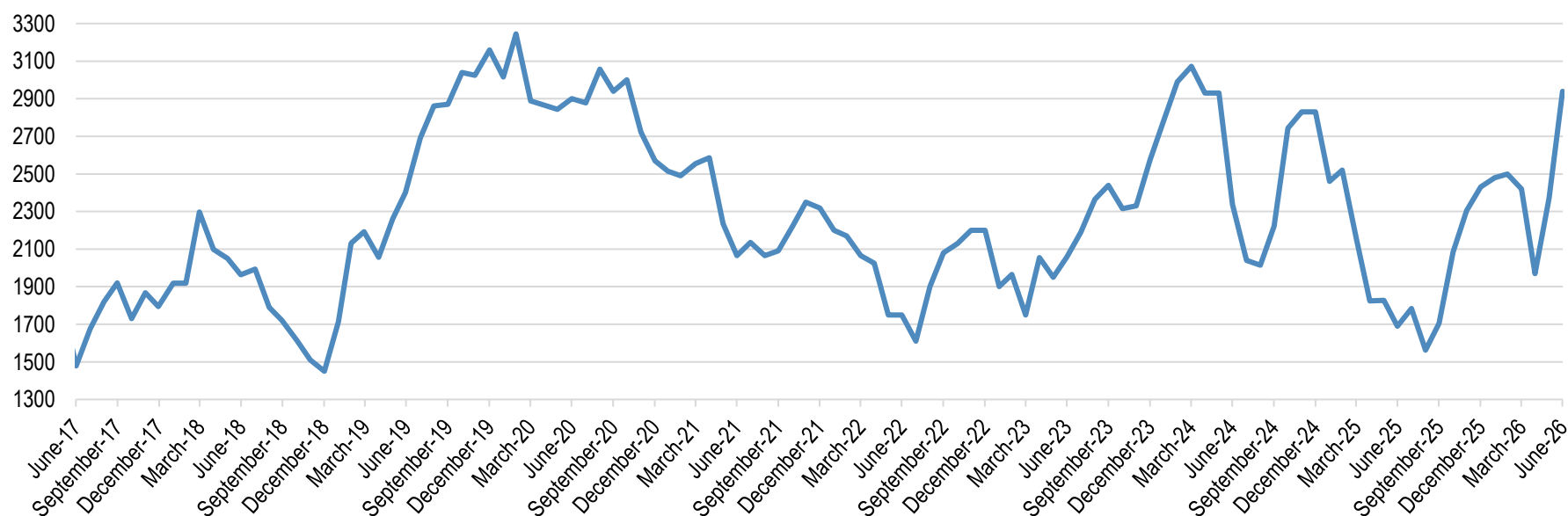
(Export shipments in 1,000 metric tons)

Year	Belarus	% Δ	Russia	% Δ		Eurochem-Usolskiy	Eurochem-Volgakaliy	Uralkali	Dest. China
2019	18		9,862			878	0	8,985	640
2020	119	NM	11,152	13.1%		1,990	5	9,156	1,983
2021	174	47.0%	12,294	10.2%		2,165	218	9,912	1,816
2022	3,030	NM	7,454	(39.4%)		2,236	347	4,872	2,316
2023	9,426	211.1%	10,933	46.7%		2,359	652	7,922	2,878
2024	10,764	14.2%	13,813	14.4%		2,445	948	10,420	3,113
2025	11,977	11.3%	14,534	5.2%		2,706	1,144	10,683	3,390
2026 YTD	1,076	5.7%	1,461	15.8%		334	152	975	372
Months	Belarus	% Δ	Russia	% Δ		Eurochem-Usolskiy	Eurochem-Volgakaliy	Uralkali	Dest. China
Jan-24	763	32.0%	1,151	89.5%		185	42	924	342
Feb-24	729	20.9%	1,082	102.3%		214	77	791	268
Mar-24	797	5.5%	1,044	77.4%		100	41	903	188
Apr-24	924	31.7%	1,139	43.5%		196	101	842	130
May-24	958	22.8%	1,229	46.4%		243	93	893	149
Jun-24	943	32.4%	1,030	5.5%		214	111	704	166
Jul-24	954	27.3%	1,137	8.7%		219	88	830	162
Aug-24	952	19.4%	1,208	2.7%		240	84	884	289
Sep-24	937	0.3%	1,087	3.2%		207	73	807	313
Oct-24	898	(3.8%)	1,145	1.9%		191	78	876	285
Nov-24	983	0.6%	1,253	15.6%		191	89	973	407
Dec-24	927	2.4%	1,309	17.9%		245	71	993	413
2024	10,764	14.2%	13,813	14.4%		2,445	948	10,420	3,113
Jan-25	1,018	33.4%	1,261	9.6%		237	98	927	301
Feb-25	877	20.4%	1,231	13.8%		236	13	882	302
Mar-25	1,061	33.2%	1,400	34.0%		282	160	958	343
Apr-25	1,123	21.5%	1,285	12.9%		295	101	890	286
May-25	1,031	7.6%	1,156	(5.9%)		247	90	820	150
Jun-25	930	(1.3%)	1,017	(1.2%)		197	77	743	125
Jul-25	1,084	13.6%	1,284	12.9%		279	88	917	208
Aug-25	1,019	7.0%	1,316	8.9%		284	141	891	312
Sep-25	975	4.1%	1,125	3.4%		174	68	883	361
Oct-25	1,003	11.7%	1,151	0.6%		141	79	931	374
Nov-25	929	(5.5%)	1,071	(14.5%)		96	60	914	259
Dec-25	927	(0.0%)	1,237	(5.5%)		238	71	928	369
2025 YTD	11,977	11.3%	14,534	5.2%		2,706	1,144	10,683	3,390
Jan-26	1,076	5.7%	1,461	15.8%		334	152	975	372
2026 YTD	1,076	5.7%	1,461	15.8%		334	152	975	372

China Potash Port Inventories

(Inventories in 1,000 metric tons.)

	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
January	1,005	1,212	1,318	2,850	2,105	1,918	1,713	3,015	2,515	2,200	1,900	2,790	2,460	2,480
February	927	1,410	1,351	2,910	2,140	1,918	2,131	3,245	2,490	2,170	1,965	2,990	2,520	2,500
March	986	1,220	1,240	2,157	2,090	2,297	2,192	2,889	2,555	2,065	1,750	3,073	2,157	2,420
April	1,159	812	1,255	2,065	2,000	2,098	2,056	2,866	2,586	2,025	2,055	2,930	1,824	1,970
May	1,090	1,009	1,157	2,000	1,800	2,050	2,262	2,844	2,234	1,750	1,950	2,930	1,827	2,370
June	1,431	982	1,060	1,907	1,479	1,963	2,400	2,900	2,065	1,750	2,060	2,338	1,689	2,940
July	1,399	1,138	1,130	1,807	1,676	1,994	2,691	2,878	2,135	1,610	2,190	2,040	1,784	
August	1,418	1,269	1,365	1,850	1,820	1,790	2,862	3,058	2,066	1,900	2,365	2,015	1,563	
September	1,482	1,145	1,336	1,955	1,920	1,721	2,870	2,940	2,090	2,080	2,440	2,222	1,705	
October	1,477	1,251	1,460	1,828	1,730	1,615	3,040	3,000	2,220	2,130	2,315	2,744	2,085	
November	1,410	1,211	1,663	1,893	1,868	1,510	3,025	2,722	2,350	2,200	2,330	2,830	2,305	
December	1,410	1,120	2,150	1,890	1,795	1,450	3,160	2,570	2,318	2,200	2,574	2,830	2,430	



Potash (KCl) Capacity Expansion

(Capacity in millions of metric tons; operating capacity)

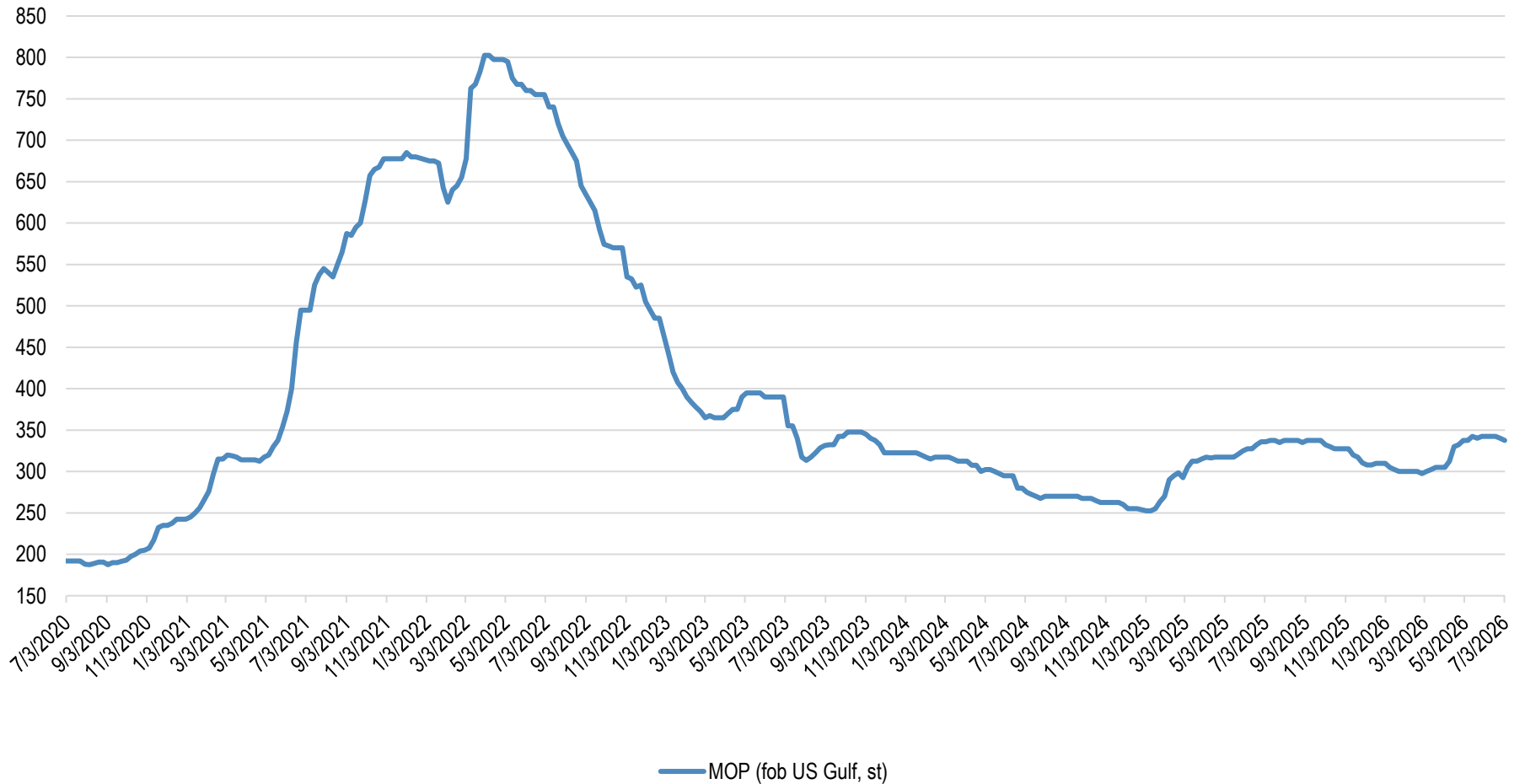
Million Tonnes KCl	2016	2017	2018	2019	2020	2021E	2022E	2023E	2024E	2025E	2026E	2027E & Beyond
Nutrien (Canada)	12.4	12.6	13.0	12.1	12.8	13.8	14.6	15.2	14.0	14.4	14.7	20.6
Lanigan SK	2.0	1.9	2.0	1.8	2.0	2.9	2.8	3.1	3.0	3.2	3.5	3.8
Rocanville SK	3.0	5.0	5.8	5.8	6.0	5.0	5.2	5.2	5.1	5.0	4.5	6.5
Allan SK	2.6	2.1	2.0	1.7	2.0	2.8	2.9	3.0	2.4	2.7	3.1	4.0
Cory SK	1.4	0.8	0.8	0.8	0.8	1.8	2.1	2.2	2.1	2.1	2.2	3.0
Patience Lake SK (solution)	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Vanscoy SK (legacy AGU)	3.1	2.5	2.1	1.7	1.7	1.1	1.3	1.4	1.1	1.1	1.1	3.0
Mosaic (Canada, US)	9.4	10.6	11.2	10.9	9.6	8.9	10.5	10.8	10.8	10.8	10.8	12.8
Belle Plaine, SK (solution)	2.5	2.7	2.9	3.1	3.1	3.0	3.0	3.0	3.0	3.0	3.0	3.9
Colonsay, SK	1.2	2.1	2.1	1.5	0.0	0.4	1.5	1.5	1.5	1.5	1.5	2.6
Esterhazy, SK	5.7	5.8	6.2	6.3	6.5	5.5	6.0	6.3	6.3	6.3	6.3	6.3
Belaruskali (Belarus)	10.1	11.5	12.1	12.3	12.3	12.3	6.2	9.5	10.8	12.0	12.0	12.6
Soligorsk 2	0.9	1.2	1.3	1.3	1.3	1.3	0.7	1.1	1.2	1.2	1.2	1.4
Soligorsk 3	2.3	2.6	2.7	2.8	2.8	2.8	1.4	2.1	2.4	2.7	2.7	2.8
Soligorsk 4	2.3	2.6	2.7	2.8	2.8	2.8	1.4	2.1	2.4	2.7	2.7	2.8
Soligorsk 5	2.3	2.6	2.7	2.8	2.8	2.8	1.4	2.1	2.4	2.7	2.7	2.8
Soligorsk 6	2.3	2.5	2.7	2.8	2.8	2.8	1.4	2.1	2.4	2.7	2.7	2.8
ICL (Israel)	4.3	4.6	4.9	4.2	4.7	4.4	4.5	4.7	4.6	4.4	4.4	6.5
Dead Sea Works, Israel	3.7	3.9	4.1	4.1	4.1	3.9	4.0	4.1	4.1	3.5	3.5	5.0
Iberpotash, Spain	0.6	0.7	0.8	0.1	0.5	0.5	0.5	0.5	0.5	0.9	0.9	1.5
Uralkali (Russia)	10.5	10.8	10.8	10.3	10.3	10.4	5.0	8.0	10.4	10.7	10.7	16.6
Solikamsk 1 (former Silvinit)	0.9	0.9	0.9	0.8	0.8	0.8	0.3	0.5	0.8	0.8	0.8	1.0
Solikamsk 2 (former Silvinit)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.5	0.7	0.7	2.3
Solikamsk 3 (former Silvinit)	2.7	2.7	2.7	2.6	2.5	2.4	1.5	1.9	2.6	2.6	2.6	2.5
Berezniki 2	1.9	1.9	1.9	1.8	1.8	1.8	1.0	1.3	0.0	0.0	0.0	0.0
Berezniki 3	2.2	2.3	2.3	2.2	2.2	2.2	1.0	1.7	3.0	3.0	3.0	2.8
Berezniki 4	2.8	3.0	3.0	2.9	2.9	2.9	1.0	2.2	3.0	3.0	3.0	3.0
Polovodovsky field	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.5
Ust-Yavinsky at Berezniki	0.0	0.0	0.0	0.0	0.1	0.3	0.2	0.3	0.5	0.6	0.6	2.5
K+S (Germany)	5.0	5.5	6.8	6.4	7.3	7.4	7.1	7.3	7.0	7.2	7.4	8.0
K+S in Germany & Europe	5.0	5.0	5.0	4.4	5.0	5.0	4.8	5.0	5.0	5.0	5.0	4.0
K+S, Bethune mine in Canada	0.0	0.5	1.8	2.0	2.3	2.4	2.3	2.3	2.0	2.2	2.4	4.0
China	5.6	7.0	7.5	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	9.0
China Salt Lake (QHSL/Minemetals Salt Lake)	3.6	5.0	5.5	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	7.0
Other	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
APC (Jordan)	2.2	2.3	2.5	2.5	2.5	2.5	2.5	2.5	2.7	2.8	2.8	3.5
SQM (Chile)	1.5	1.5	1.1	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	1.0
SCL (Chile)	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Intrepid (US)	0.5	0.3	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Carlsbad, NM (West)	0.2	0.0	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Moab, UT	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Wendover, UT	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Carlsbad, NM (HB Mine)	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
VL Mineração Ltda (Brazil)	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
EC (Russia) Usolskiy	0.1	0.6	1.1	2.0	2.2	2.2	2.4	2.4	2.4	2.7	2.9	3.7
EC (Russia) VolgaKaliy	0.0	0.0	0.0	0.1	0.2	0.4	0.7	0.9	1.2	1.4	1.4	4.6
Garlyk project/Turkmenistan	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.4
Rio Tinto (Canada)												3.3
BHP (Jansen project/Canada)									0.0	0.0	0.0	4.0
GrowMax Agri Corp (Peru)		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Western Potash Corp (solution)									0.0	0.0	0.0	2.8
Prospect Global Resources (US)		0.3	0.7	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0
Acron (Russia) Talitsky/Perm region									0.3	0.3	0.3	2.0
Slavkaliy (Belarus) Nezhinsky									0.0	0.0	0.0	2.0
MagIndustries (Mageno project/Congo)									0.0	0.0	0.0	1.2
Laos potash developments	0.5	0.7	0.9	1.1	1.1	1.1	1.4	1.6	2.1	2.3	2.5	5.5
TOTAL	62.9	69.2	74.2	72.8	74.6	75.1	66.2	74.6	78.0	80.8	81.9	126.7
Demand	60.3	65.2	66.7	63.8	69.2	70.1	61.1	68.0	72.8	74.5	74-77	76-83

2025E	Op. Capacity	%
Nutrien	14.4	18%
Belaruskali	12.0	15%
Uralkali	10.7	13%
China	8.0	10%
Mosaic	10.8	13%
K+S	7.2	9%
ICL	4.4	5%
EuroChem	3.4	4%
APC	2.8	3%
Laos potash developments	2.3	3%
Other	4.8	6%
Total	80.8	100%

Note: Years 2026 and beyond do not include possible mine depletions.

Potash Fertilizer Pricing: U.S. Gulf

(In \$ per short ton, prices as of July 3, 2026)



Source: Bloomberg Finance L.P. & J.P. Morgan estimates

Disclosures

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables, are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock’s expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts’ coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst’s coverage universe can be found on J.P. Morgan’s Research website, <https://www.jpmorganmarkets.com>.

Disclosures

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

Disclosures

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC (“JPMS”) acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries (“J.P. Morgan Private Bank”), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst’s name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst’s name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina (“BCRA”- Central Bank of Argentina) and Comisión Nacional de Valores (“CNV”- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited (“JPMSAL”) (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to “wholesale clients” (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia’s Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Disclosures

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P. Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpimipl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpimipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpimipl.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission. ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

Disclosures

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities. According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities

Disclosures

and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.